Topic	Problem numbers	Solution link
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Sequential circuits	15- 32	Sequential circuits-2
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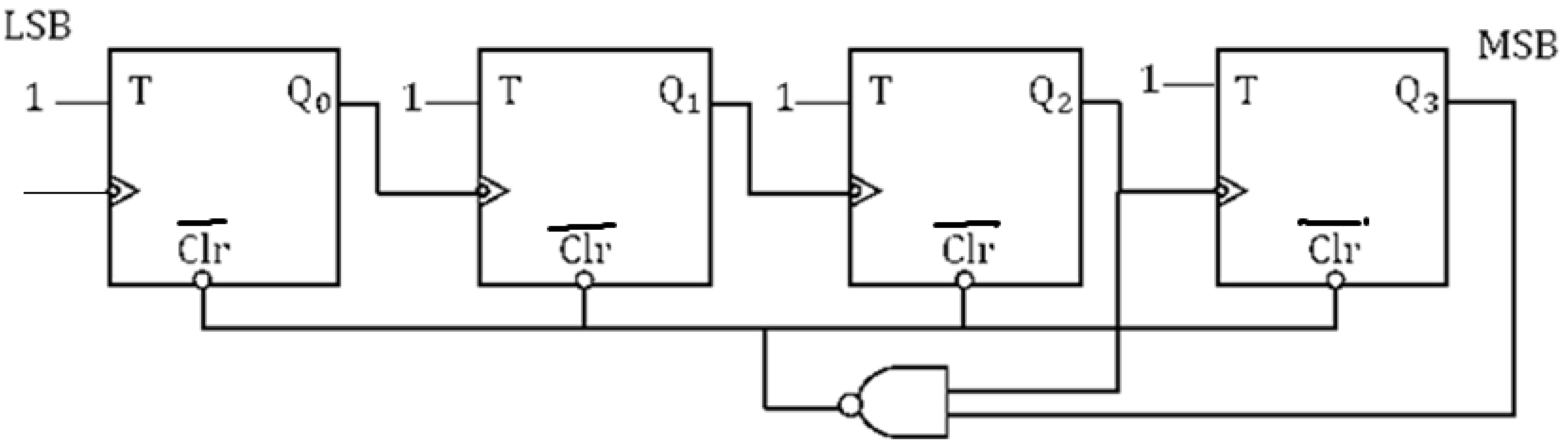
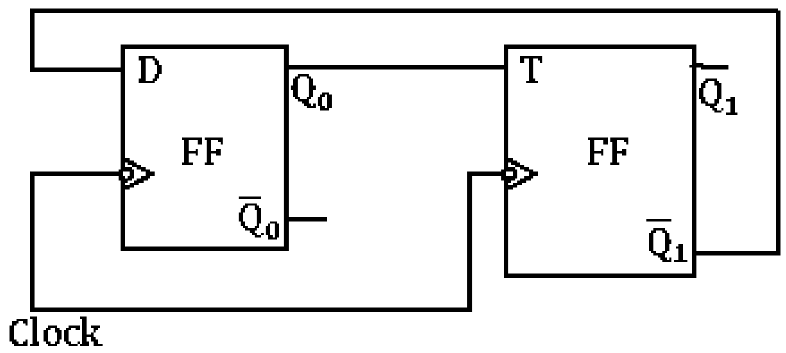
1. The mod-number of the counter shown in figure below is ________



2. Consider a combination of T and D flip flop corrected as shown below

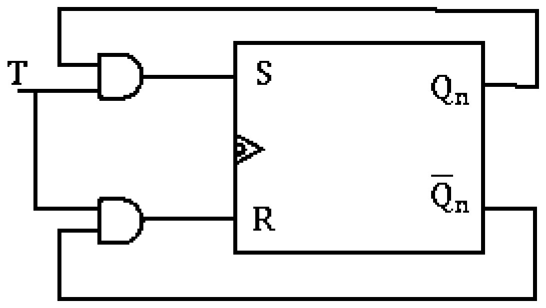


Initially Q_1 Q_0 is set to 11 (before the 1st clock) the output Q_1 Q_0 after 4 clock pulse is

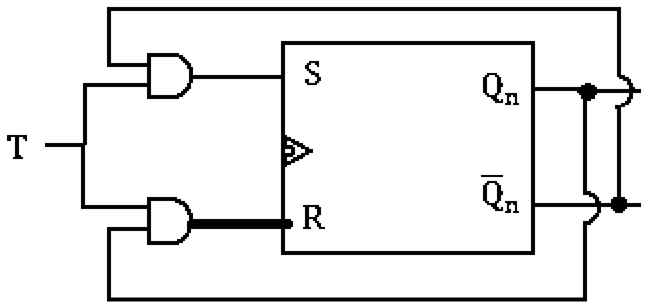
(A) 01 (B) 11 (C) 00 (D) 10

3. **A T flip flop can be implemented by S-R flip flops. Identify the correct implementations**

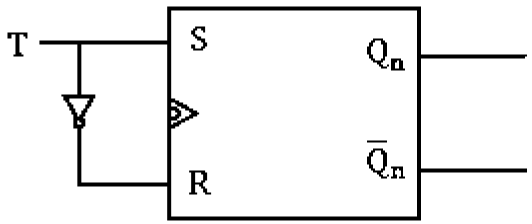
(A)



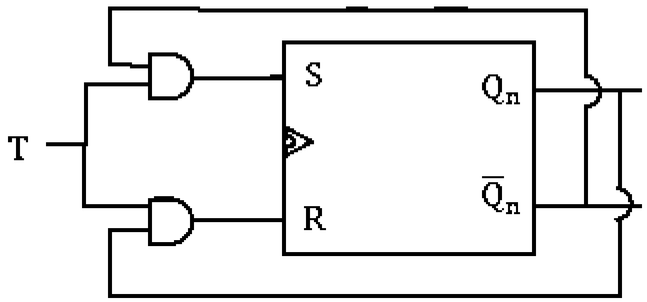
(B)



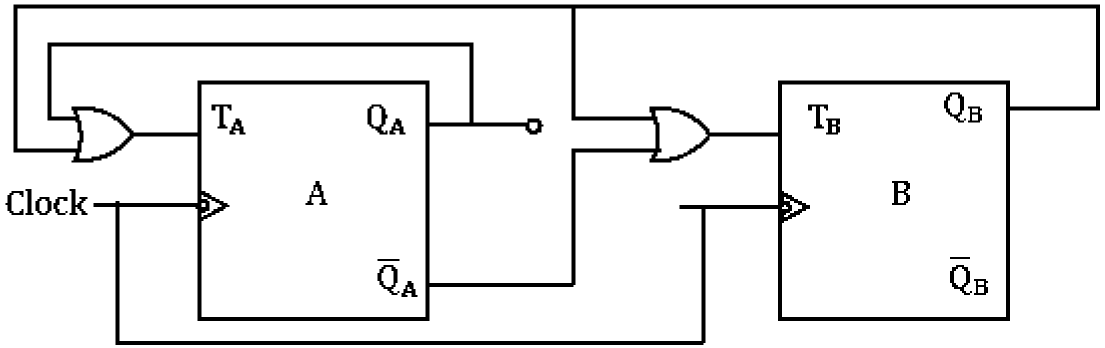
(C)



(D)

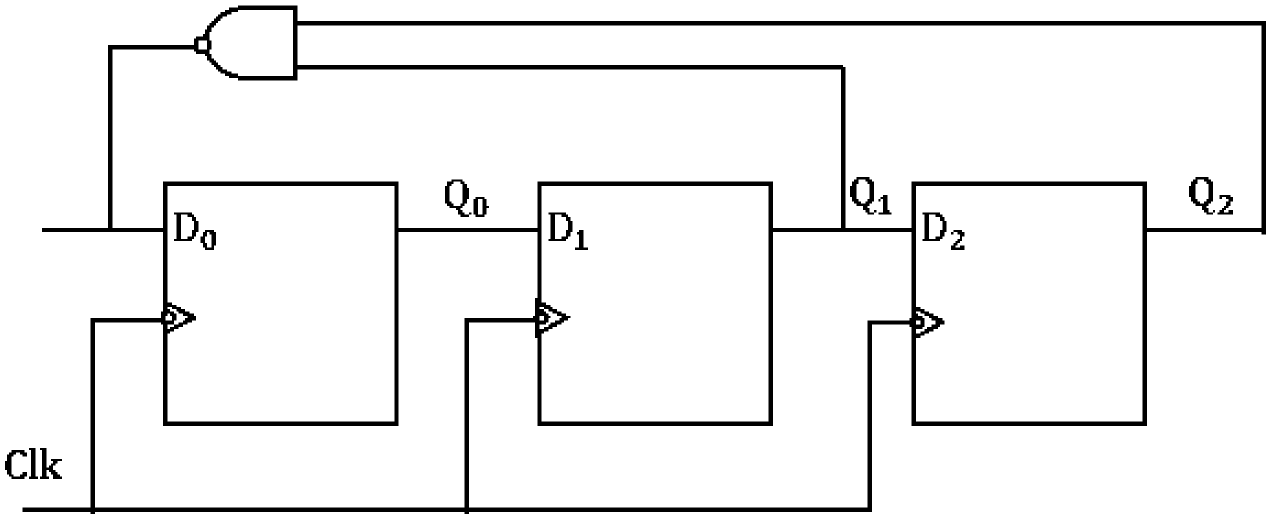


4. **Consider the circuit shown below initially both the flip flop where cleared**

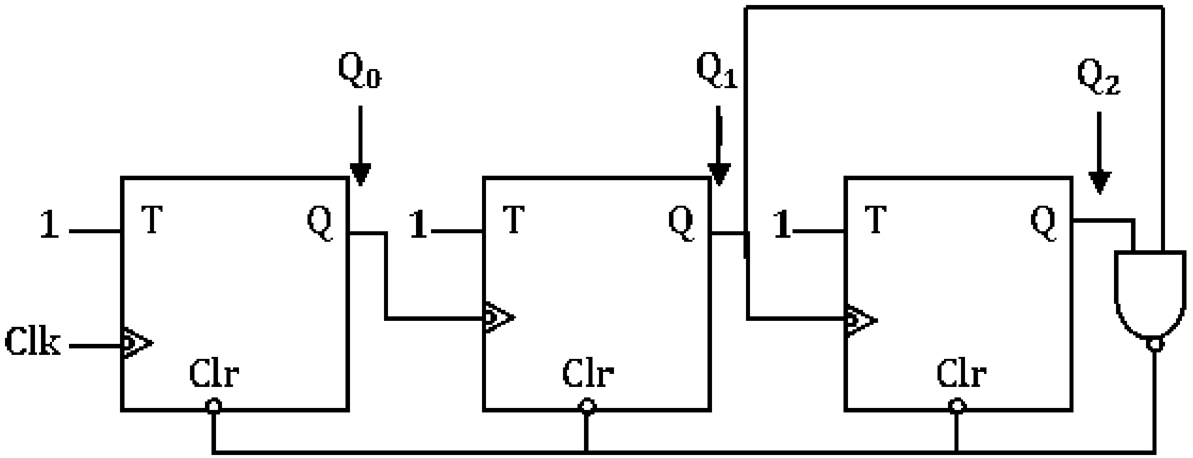


The numbers of used states in the circuit are ________

5. **In the circuit shown below, initially all flip flops are reset. The output $Q_2\ Q_1\ Q_0$ after four clock Pulse is ________**



6. **The circuit shown consists of T flip flops, each have clear option as shown in figure. The counter corresponding to this circuit is**



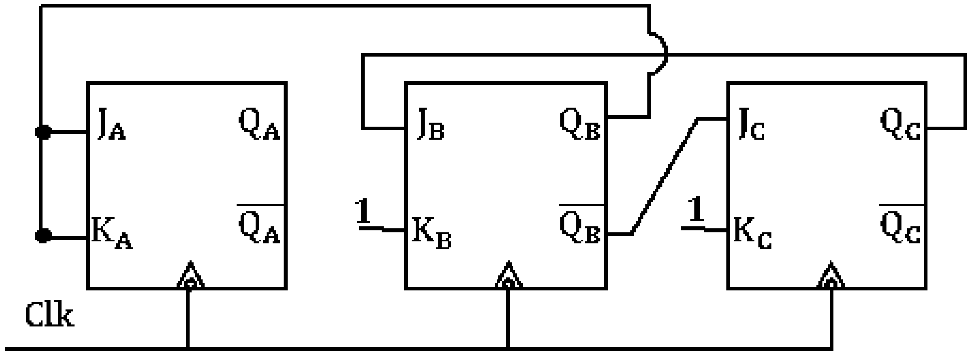
(A) Mod-5 up counter

(B) Mod-4 up counter

(C) Mod-6 down counter

(D) Mod -6 up counter

7. **Consider the circuit shown below, the sequence of Q_A from 1^{st} clock is**



(All the FF are initially relaxed)

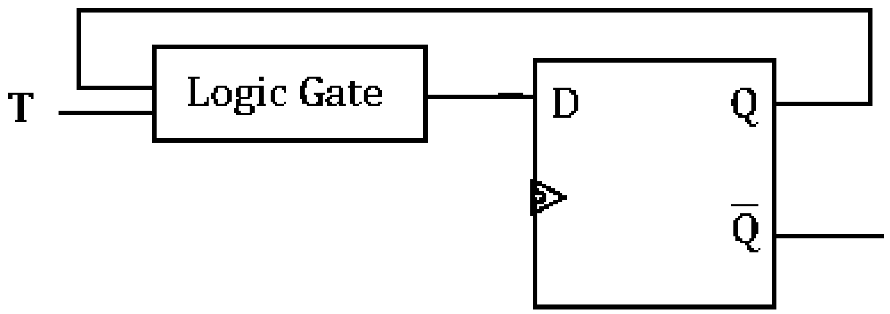
(A) 0, 0, 1, 0, 1, 0……

(B) 0, 1, 0, 1, 1, 0……

(C) 0, 0, 1, 1, 1 , 0……

(D) 1, 0, 0, 1, 1, 0……

8. A T flip flop is implemented by using D flip flop as shown in figure

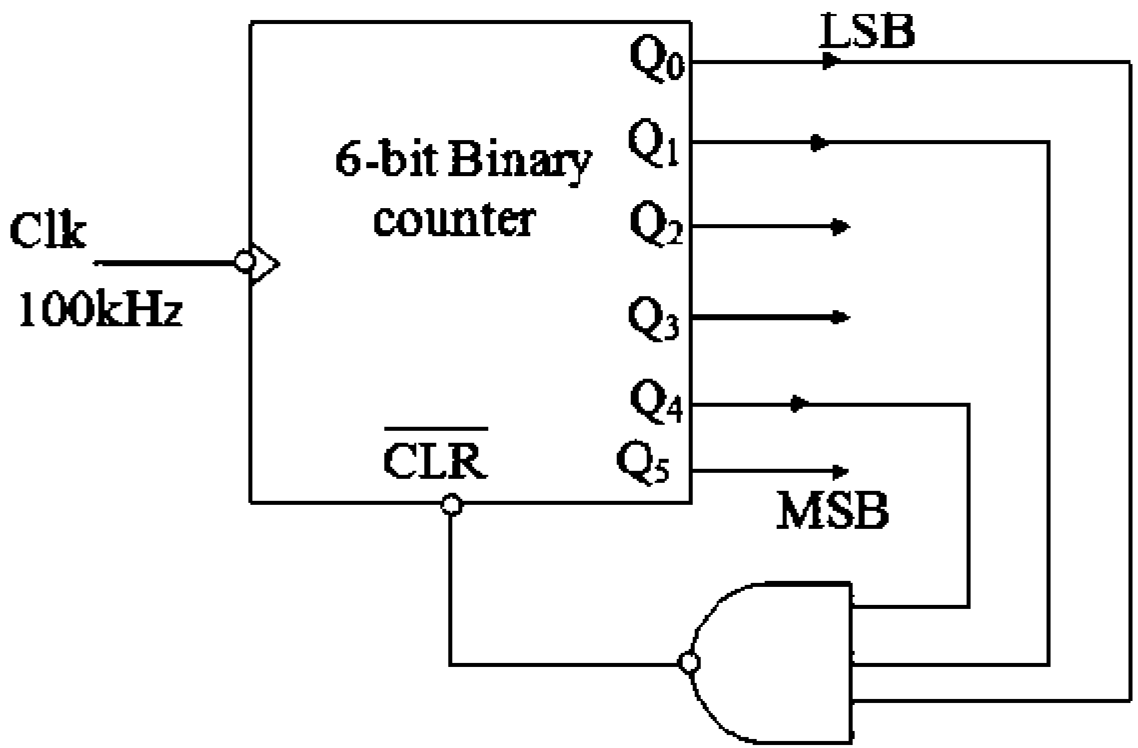


The logic gate in the given diagram is

(A) OR gate (B) AND gate

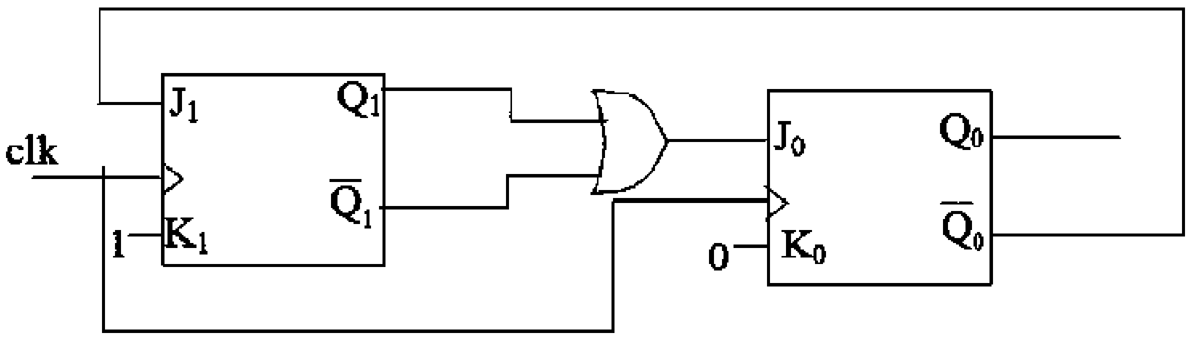
(C) EX-OR gate (D) EX-NOR gate

9. A mod K counter using Asynchronous Binary up counter with synchronous clear input is shown below



The output frequency in kHz is ______

10. **Consider the below edge triggered JK flipflops, Present output at Q_1Q_0 is 00. The output for next 3 clocks is**



(a) 11, 01, 01 (b) 11, 01, 10 (c) 11, 10, 01 (d) 11, 11, 11

11. Consider the circuit as shown in figure – 1, the sum output of a half adder is connected as clock excitation for 'Johnson' counter (rising edge triggered), the input waveforms at 'x and y' are shown in figure – 2, initially the output of counter is reset to '0'.
Find the number of state transitions at Q_1 over the duration of time 'T'.

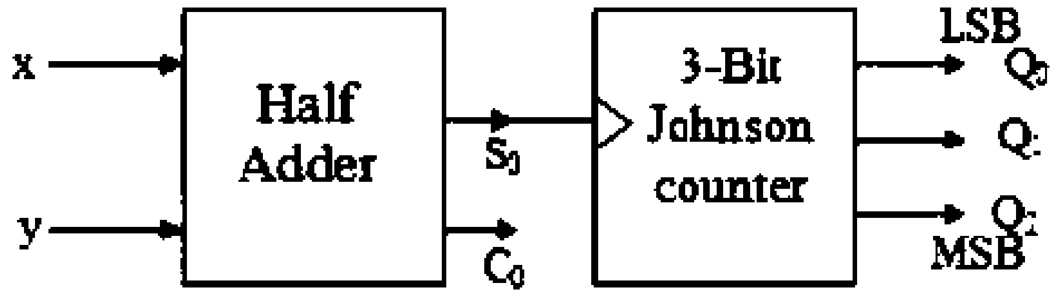


Figure 1

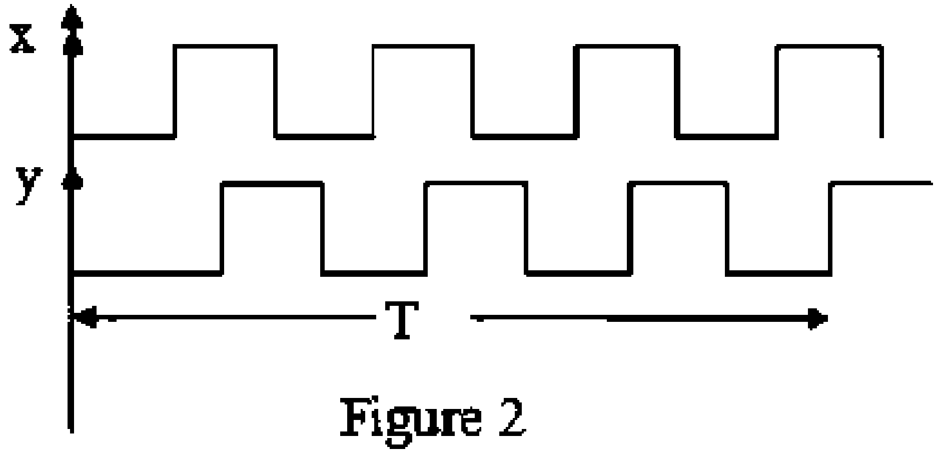
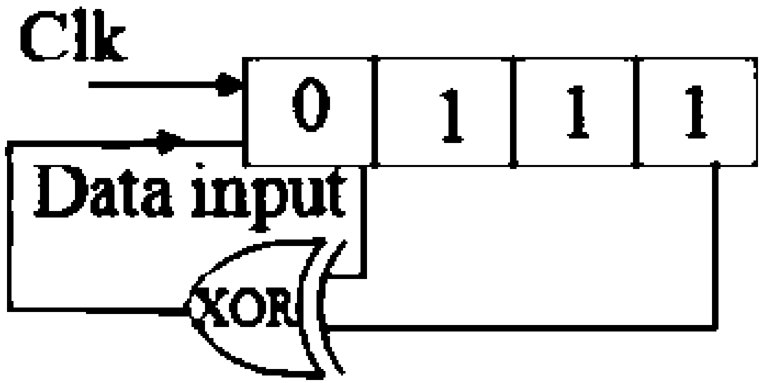


Figure 2

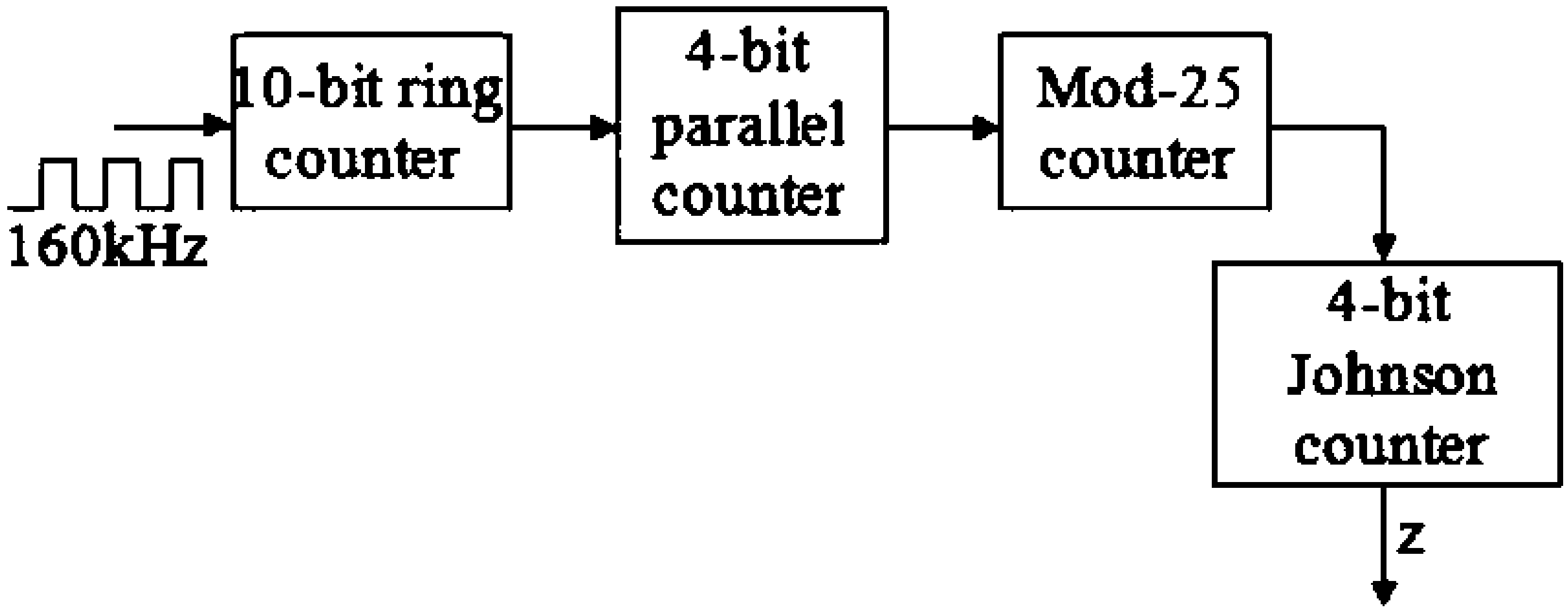
(a) 2 (b) 3 (c) 8 (d) 7

12. Initial content of the below shown 4 bit shift right register is 0111



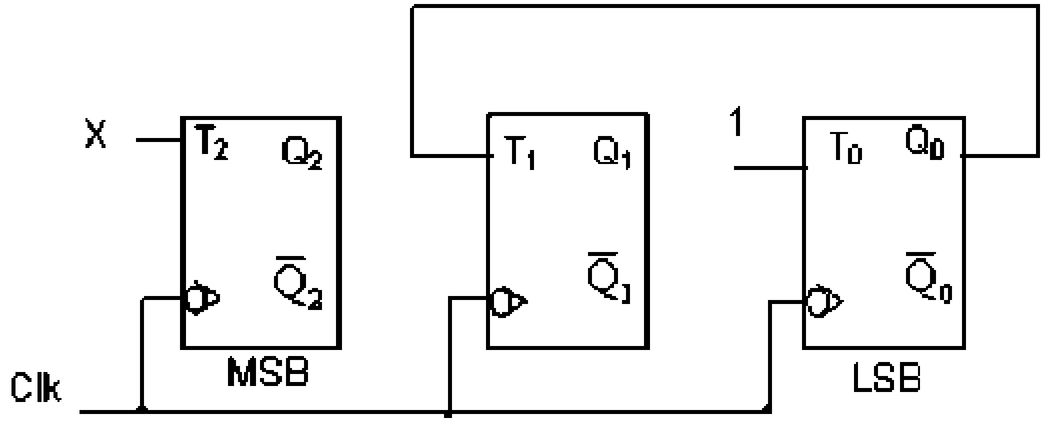
MSB and LSB are connected as input for XOR gate. Number of clock cycles required to get the same content (0111) in the register once again is ________

13.



Frequency of output (z) is ____________ Hz

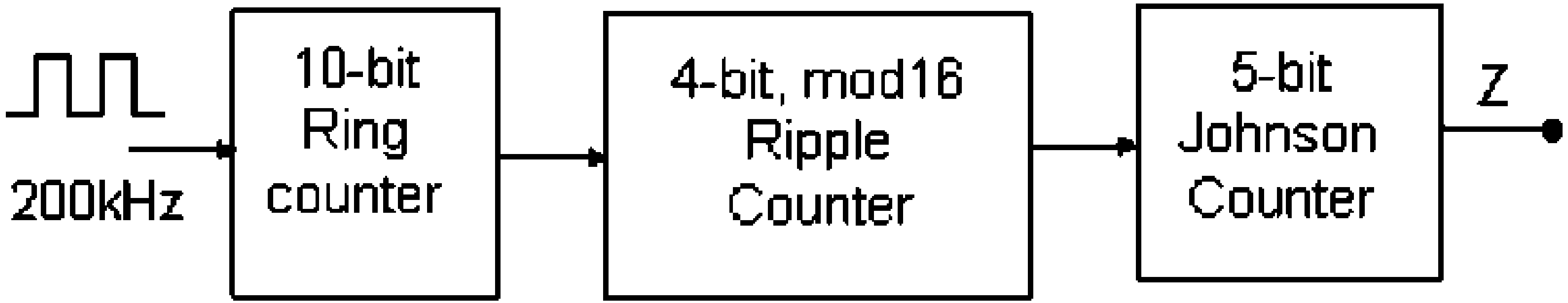
14. Consider the partial implementation of a 3-bit modulus 8 up-counter as shown below. To complete the circuit input X should be



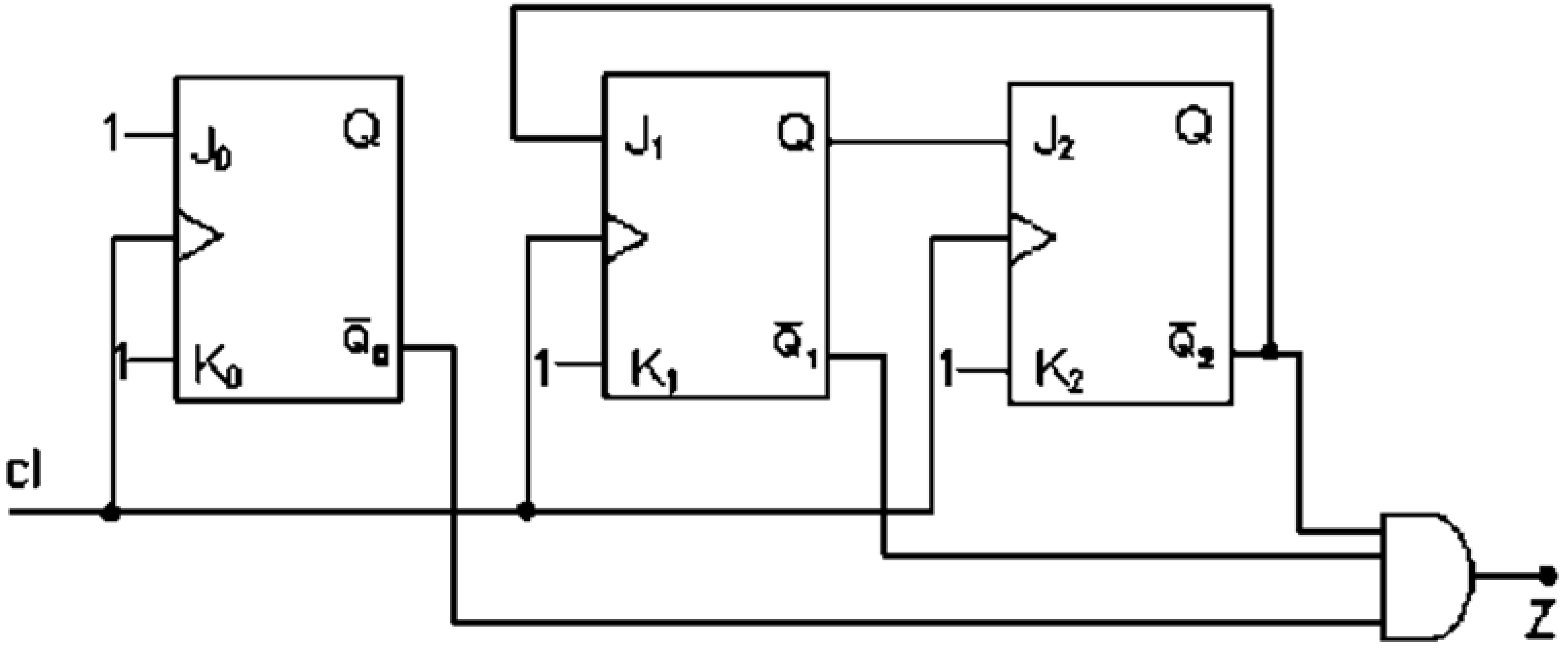
(a) $X = Q_1 + Q_2$ (b) $X = Q_1Q_2$

(c) $X = Q_0Q_1$ (d) $X = Q_0\bar{Q}_1$

15. **The frequency at the output of the following cascaded circuit is ______ Hz**



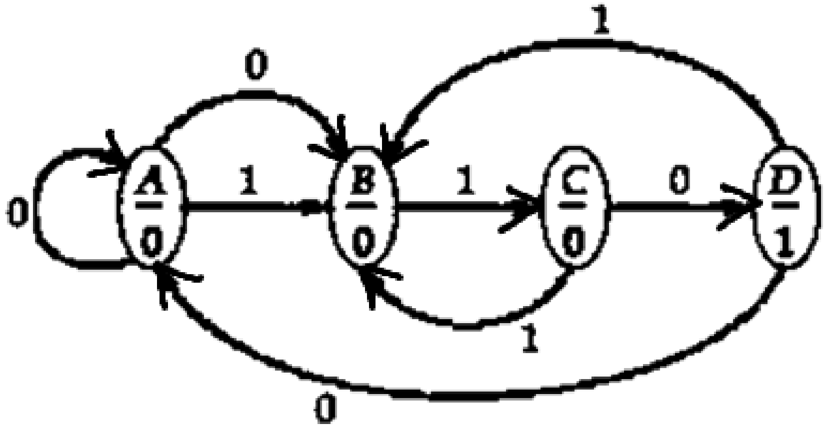
16. **Consider the sequential circuit shown below. The output Z becomes '1' after every N clock cycles. The value of N is __________[initially all flip-flops are cleared]**

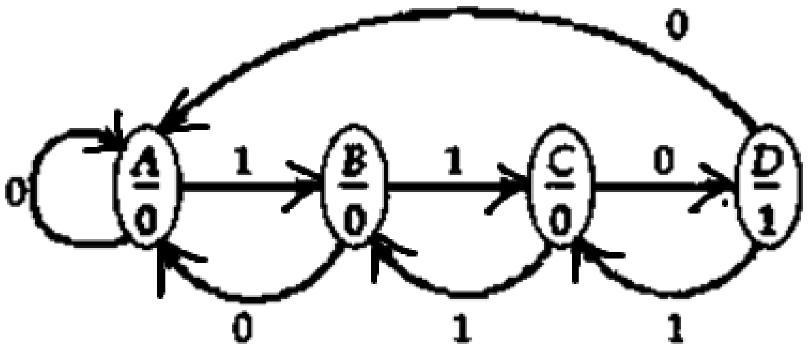


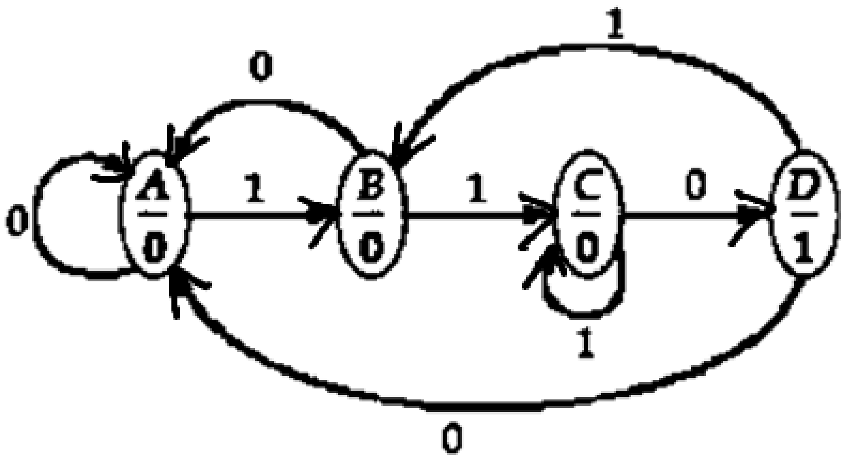
(a) 8 (b) 9 (c) 4 (d) 6

17. A 2 MHz clock signal is applied to a J-K flip flop with J = K = 1. The frequency of output signal of the flip-flop is equal to _______ kHz.

18. Out of the given state diagram of a Moore type sequence detector, which of the it represents a '110' sequence detector?

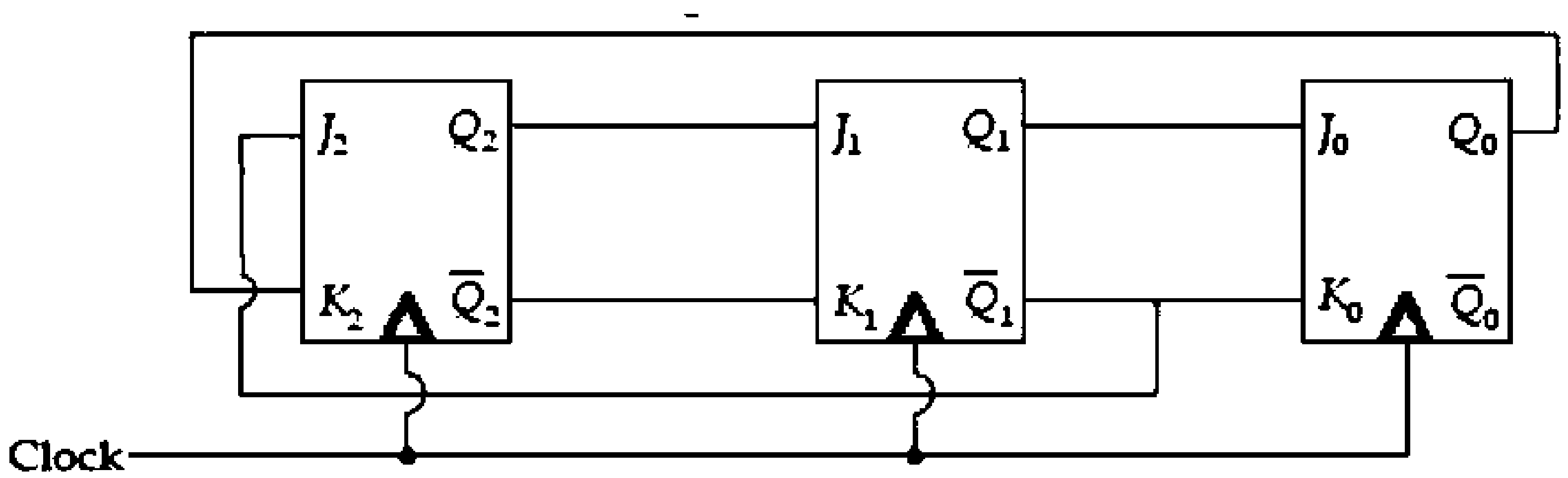
(a) 

(b) 

(c) 

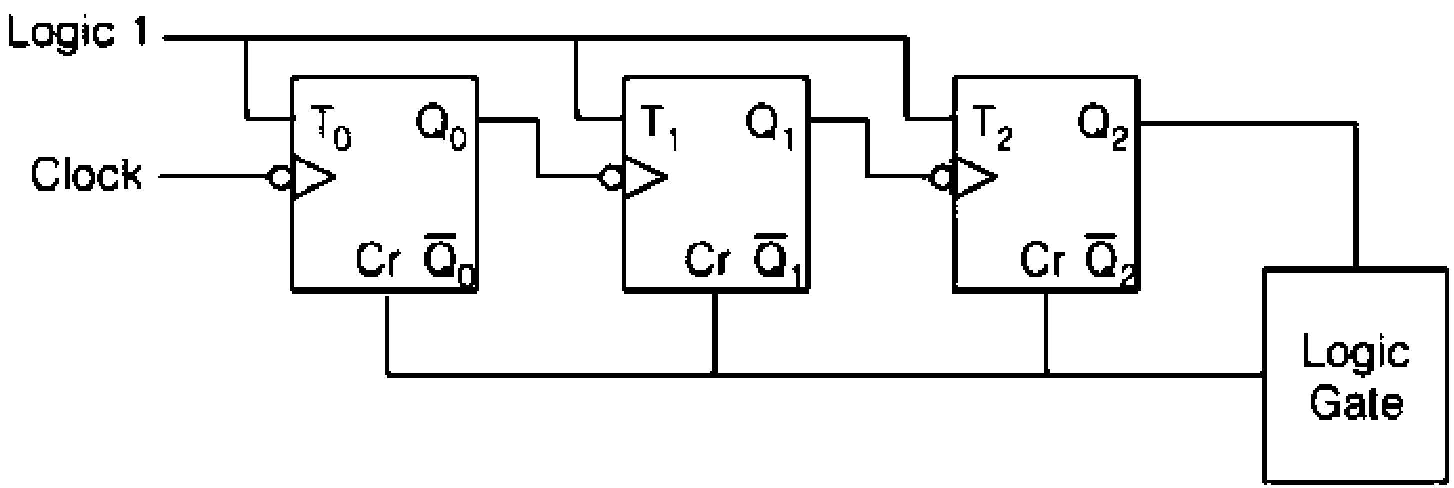
(d) None of the above

19. Consider the circuit shown in the figure below.



Then the value of $(Q_2Q_1Q_0)$ after the first clock pulse is equal to ___
(Assume all the outputs to be '0' initially)

20. **The 3-bit ripple counter (shown below) is to be designed as a MOD 4 counter**



What is the best architecture of the 'Logic gate'?

(a) a 3–bit input AND gate

(b) a 2–input AND gate

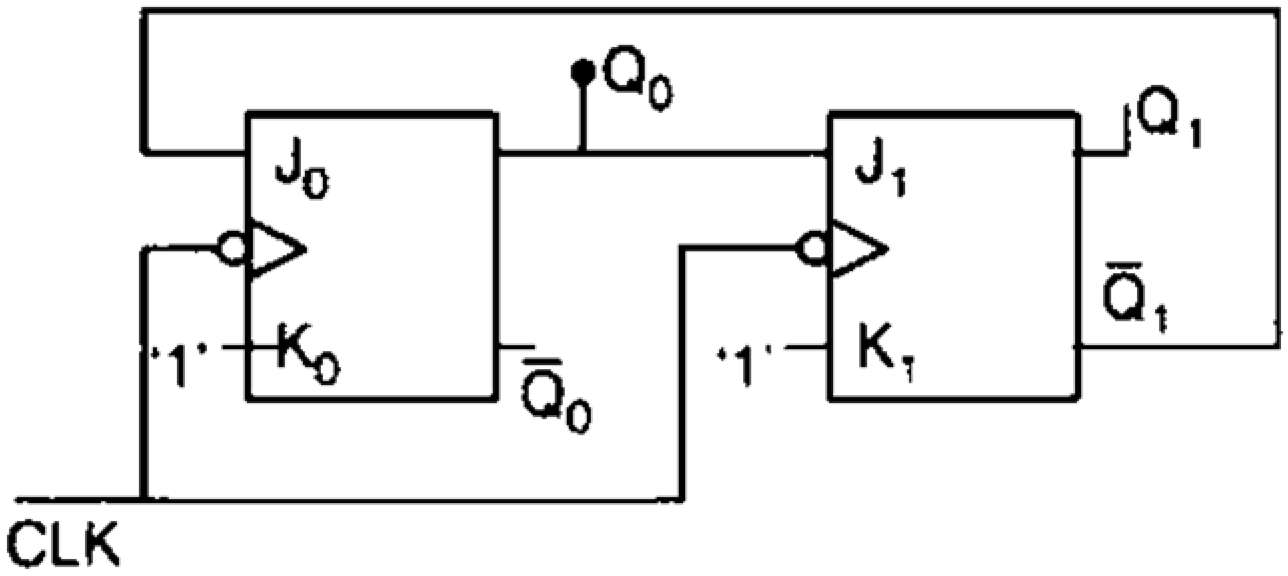
(c) a NOT gate

(d) a wire connection (no logic gate needed)

21. A 4bit synchronous UP counter is holding a state 0101. What will be the count after 27th clock pulse?

(a) 0101 (b) 0001 (c) 1111 (d) 0000

22. The following synchronous sequential circuit has _______ number of states.



23. **How many Flip-flops would be complemented in a 10-bit binary ripple up counter to reach the next count after the count 1001100111.**

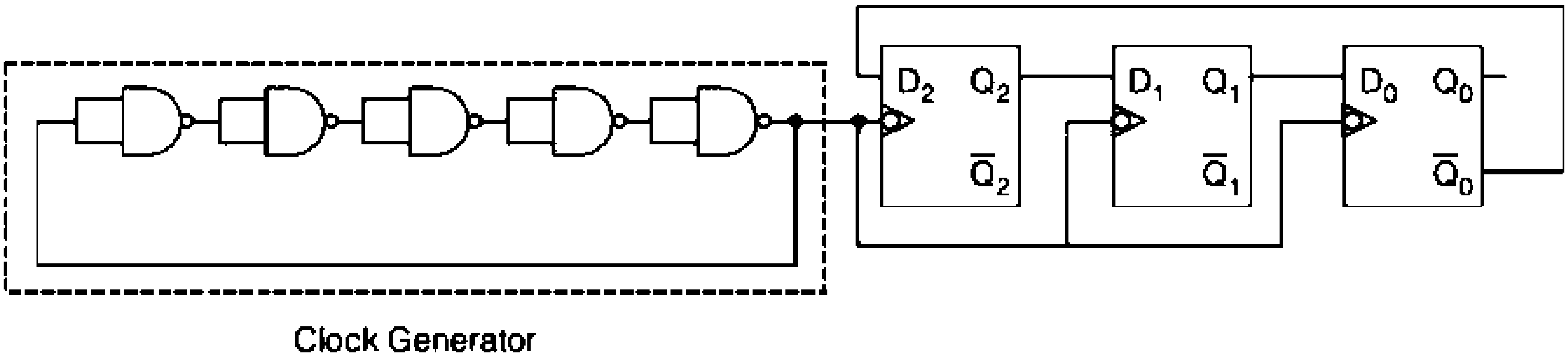
24. Match List – 1 with List – 2

List 1 (Flip-Flop)	List 2 (Characteristic Equation)
A. S – R	1. $\bar{J}Q_n + K\bar{Q}_n$
B. J – K	2. $S + \bar{R}Q_n$
C. D	3. D
D. T	4. $J\bar{Q}_n + \bar{K}Q_n$
	5. $\bar{T} \oplus \bar{Q}_n$
	6. $\bar{S} + RQ_n$
	7. $T \oplus Q_n$

Codes

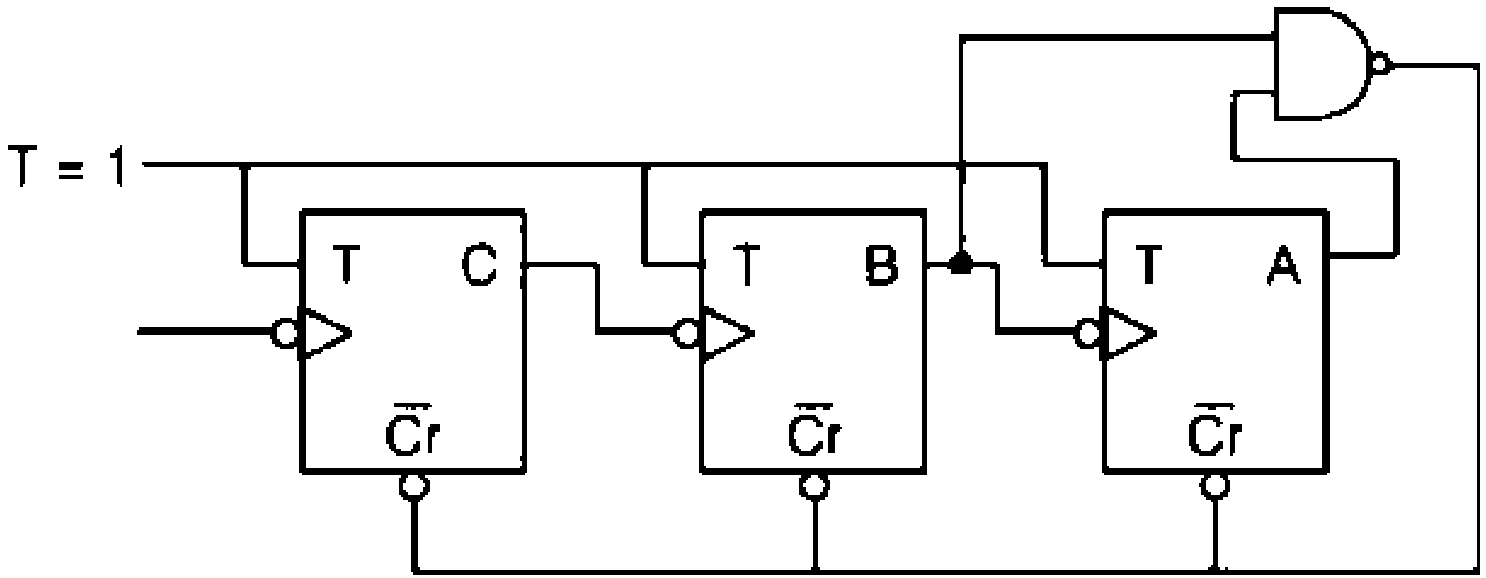
	A	B	C	D
(a)	6	1	3	7
(b)	6	4	3	5
(c)	2	1	3	5
(d)	2	4	3	7

25. Consider the digital circuit shown in below figure

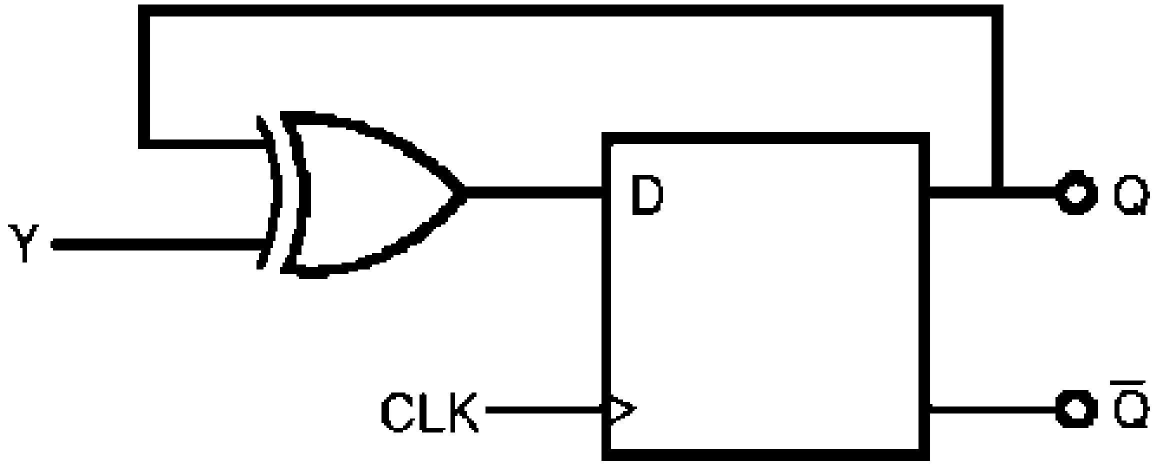


The average propagation delay of each NAND gate in the clock generator circuit is 10 ns. The frequency of the signal at Q_0 is ___ MHz.

26. **The modulus value of the below asynchronous counter is ______**



27. **The digital circuit shown in the figure works as**



(a) JK flip-flop (b) RS flip-flop
(c) T flip-flop (d) Ring counter

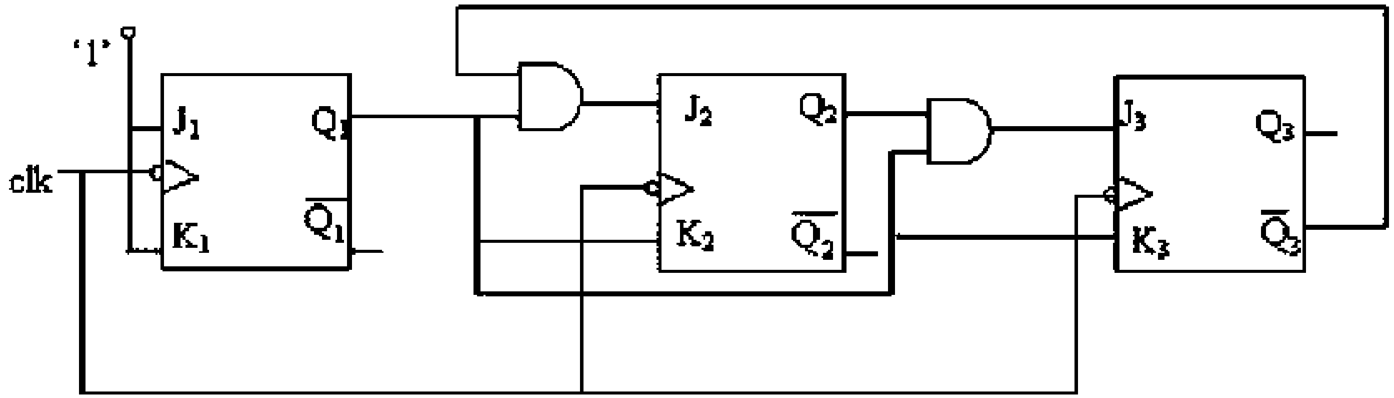
28. In a JK flip – flop, the output Q_n is 1 and it does not change when a clock pulse applied. The possible combination of J_n and K_n could be (x denotes don't care). **(IES-1992)**

(a) x and 0 (b) x and 1

(c) 0 and x (d) 1 and x

29. A binary ripple counter is required to count upto 16383_{10}.____ number of minimum flip flops are required

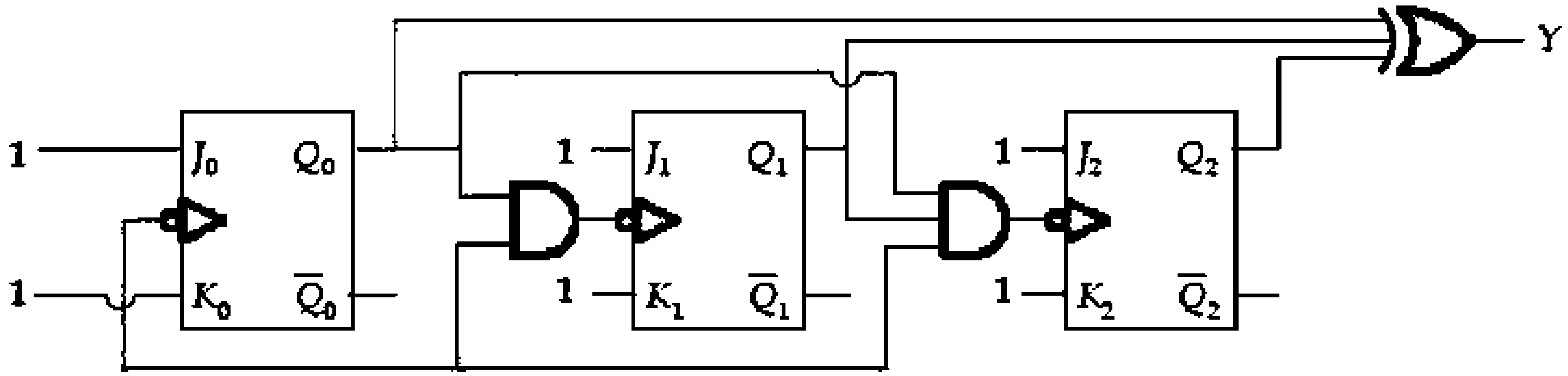
30. **The sequential circuit shown in fig is a**



(a) mod – 5 synchronous counter (b) mod – 6 synchronous counter

(c) mod – 5 Asynchronous counter (d) mod – 6 Asynchronous counter

31. **Consider the circuit shown in the figure below**



The value of output $Q_0Q_1Q_2$ and the value of output Y after the 5th clock pulse is (Assume the initialy all the flip-flop are reseted)

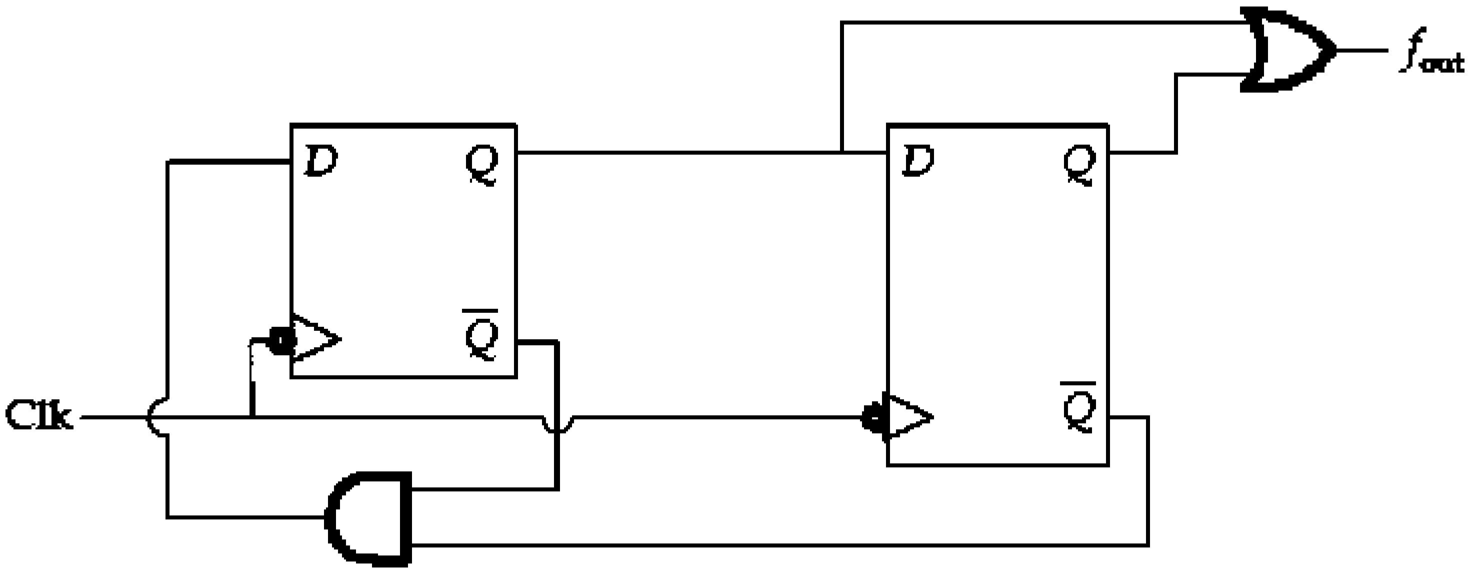
(a) Output = 001 and Y = 1
(b) Output = 101 and Y = 0
(c) Output = 100 and Y = 1
(b) Output = 010 and Y = 1

32. Consider a state table of a digital circuit given below

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	d	1	1
g	g	h	0	1
h	g	a	1	0

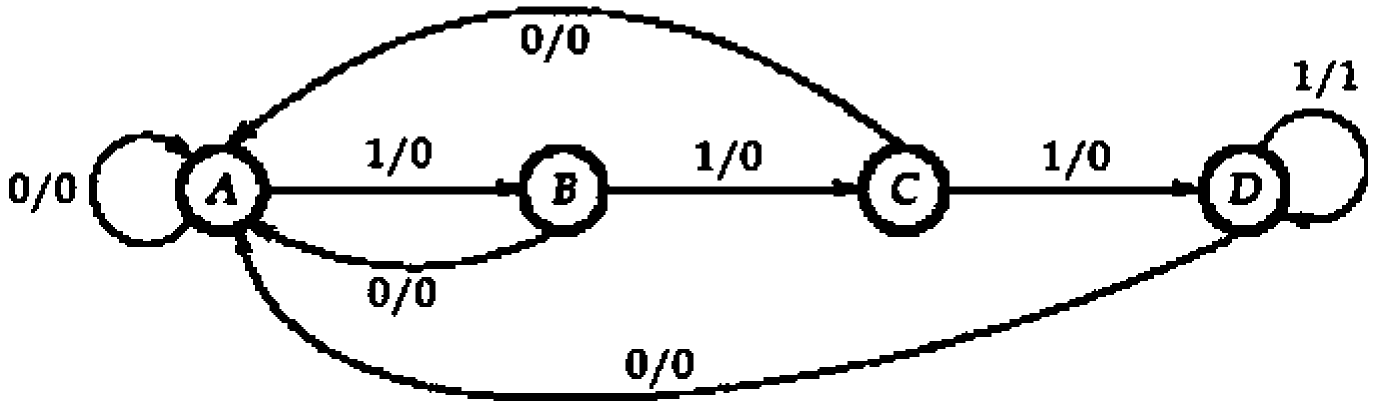
Then the total number of redundant state present is equal to _____

33. **Consider the circuit shown in the figure below**



In the input clock frequency is equal to 6 MHz, then the output signal frequency is equal to _____ kHz.

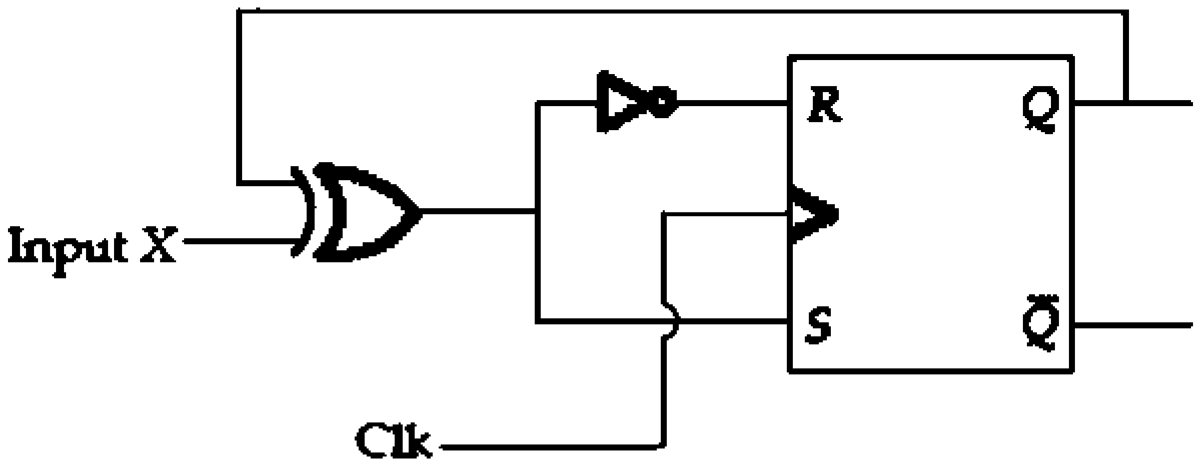
34. Consider the state diagram shown in the figure below.



The 4-bit sequence circuit is used to detect (__________)$_2$.

35. The number of clock pulses needed to change the contents of an 8-bit up-counter from $(10101011)_2$ to $(00111010)_2$ is ______

36. Consider RS flip flop having input and clock signal as shown in below figure



The characteristic equation or state equation of the circuit is

(a) $Q_{t+1} = X$

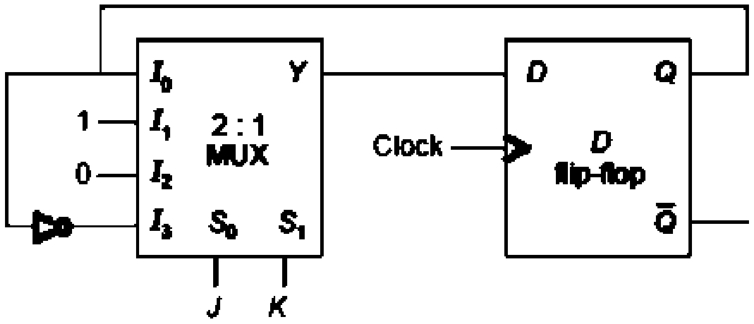
(b) $Q_{t+1} = \overline{X}$

(c) $Q_{t+1} = X \odot Q_t$

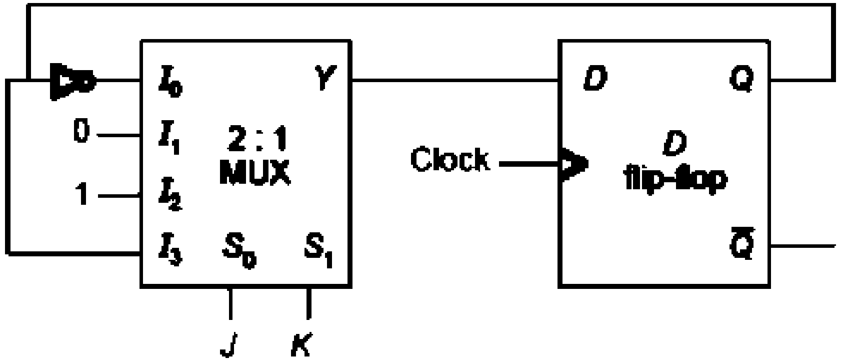
(d) $Q_{t+1} = X \oplus Q_t$

37. Which of the following circuits function as a J-K flip-flop?

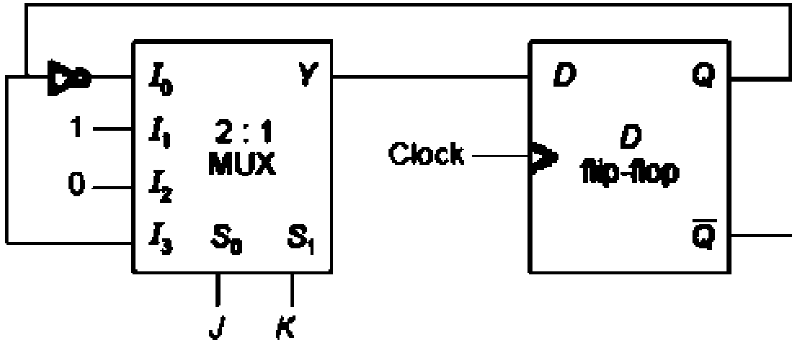
(a)



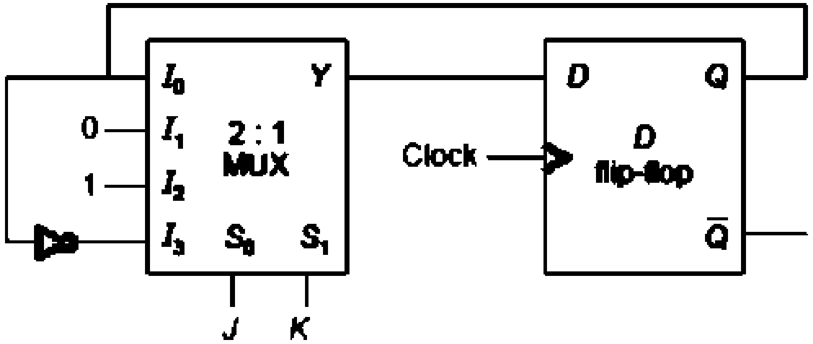
(b)



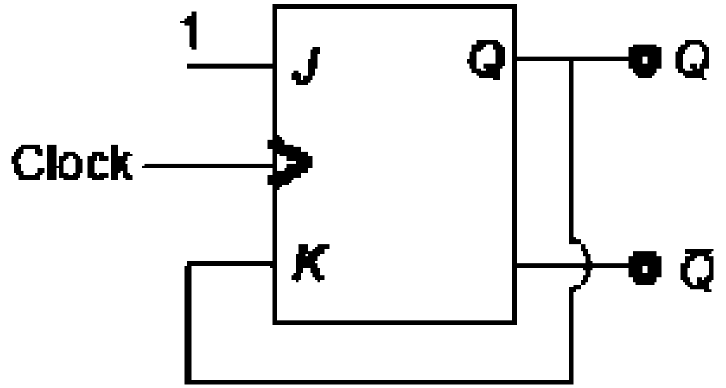
(c)



(d)



38. **Assume that the initial value of Q is 0. After applying the clock signal to the given circuit, the subsequent states of Q will be**

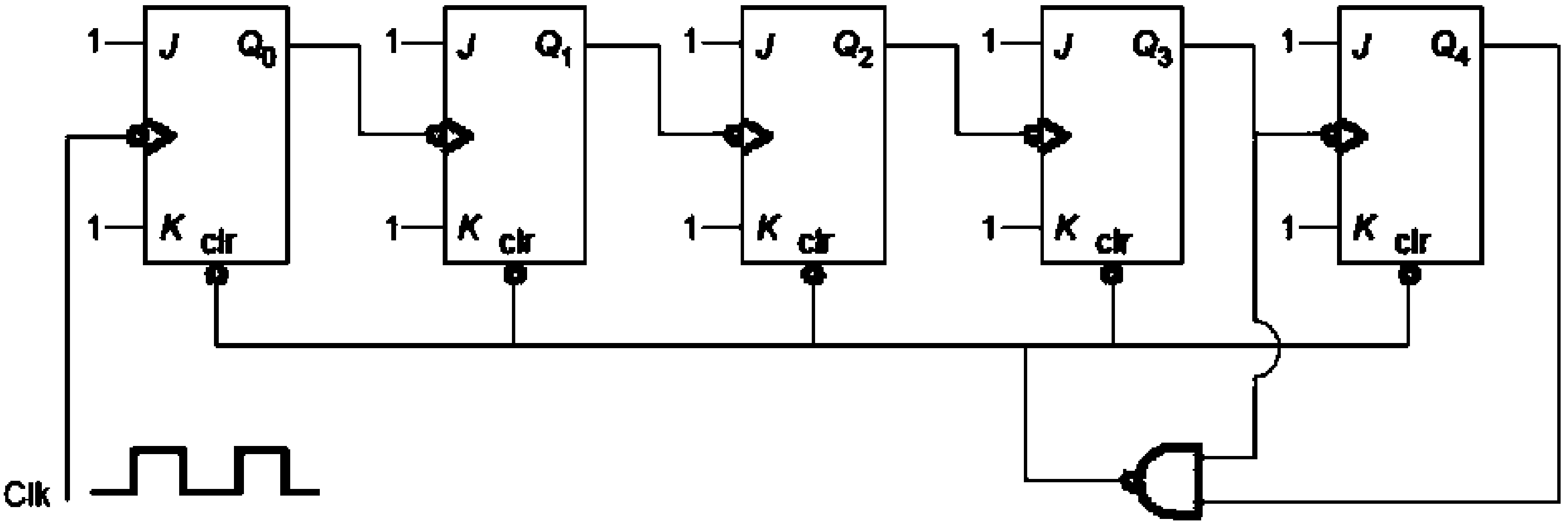


(a) 0, 1, 0, 1, 0 (b) 1, 0, 1, 0, 1

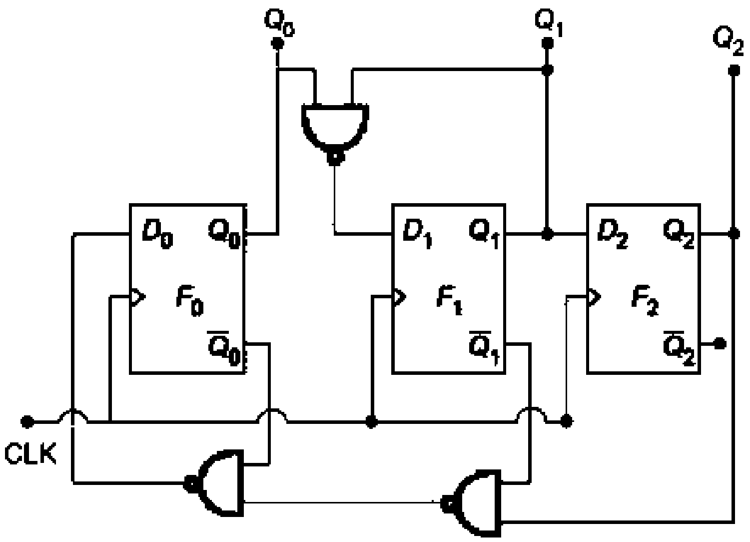
(c) 0, 0, 0, 0, 0 (d) 1, 1, 1, 1, 1

39. A modulo-16 ripple counter uses J-K flip-flops. If the propagation delay of each flip-flop is p ns and the maximum clock frequency that can be used is 5 MHz, then the value of p will be

40. The mod-number of the asynchronus counter shown in the figure is



41. **Consider the counter circuit and propagation delay table shown below.**



Device	Propagation delay
F_0	5 μ sec
F_1	4 μ sec
F_2	3 μ sec
each two-input NAND gate	3 μ sec

(Assume all the devices are ideal except for some propagation delay)
What is the maximum clock frequency that can be applied to the above counter?

(a) 125 kHz (b) 100 kHz (c) 200 kHz (d) 142.85 kHz

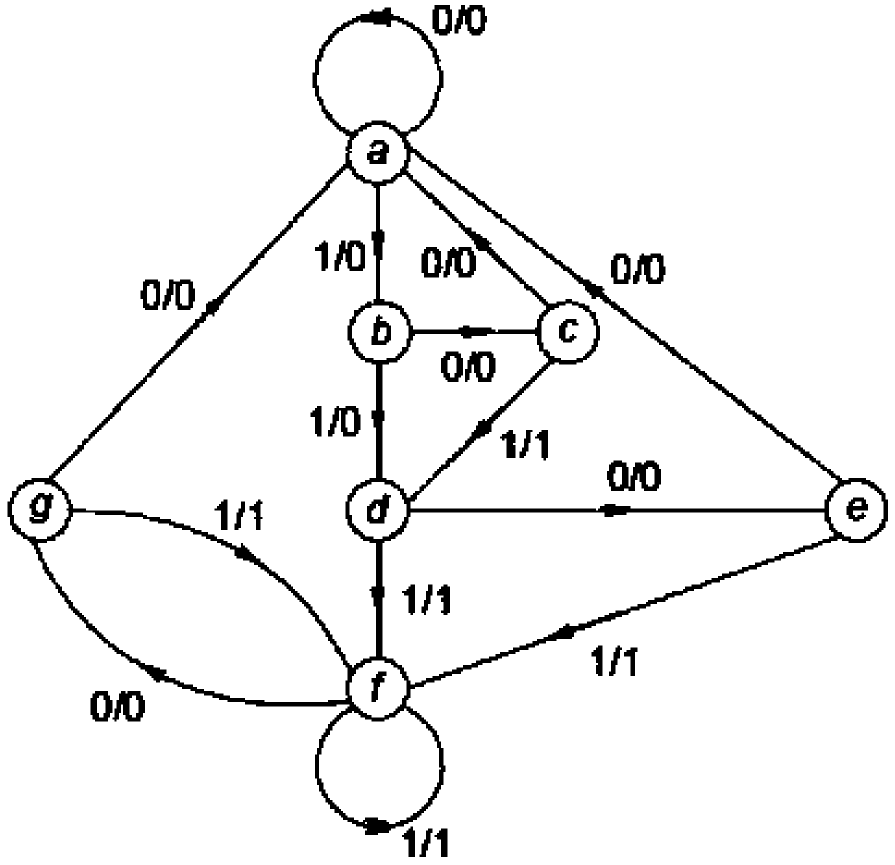
42. **Find the minimum number of flip-flops required to design a sequential circuit whose state diagram is given below?**

(a) 2

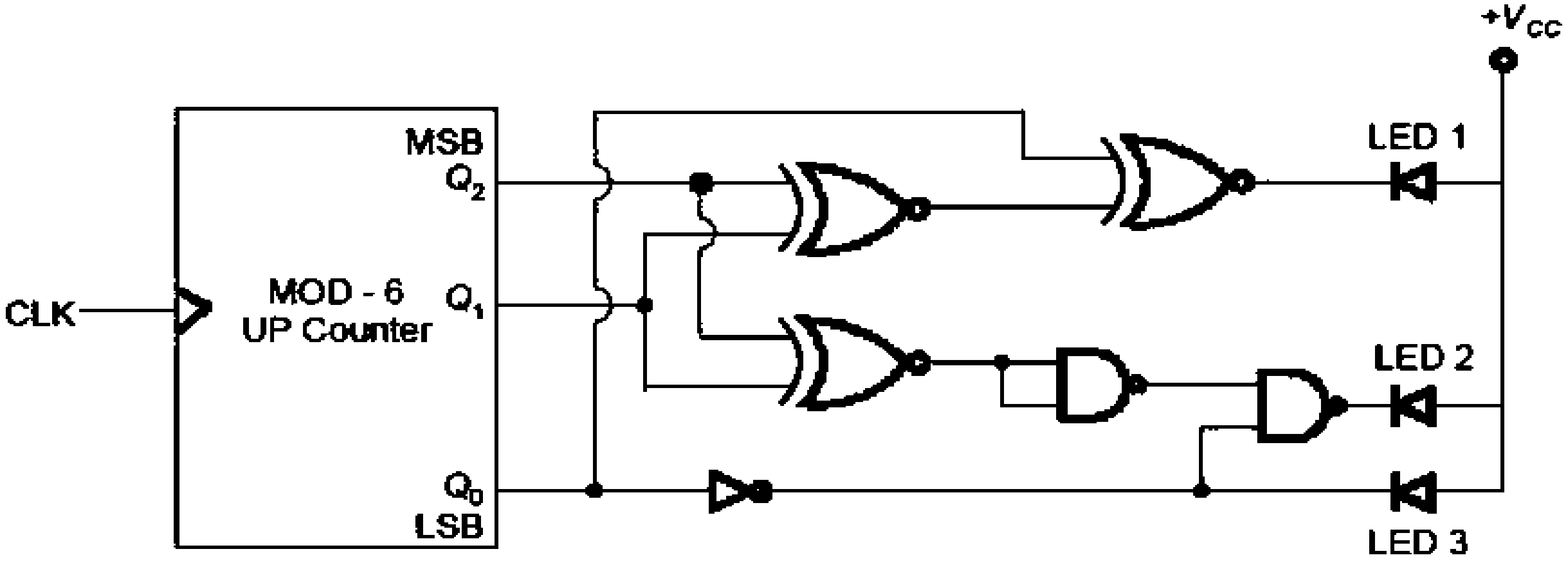
(b) 3

(c) 4

(d) 5



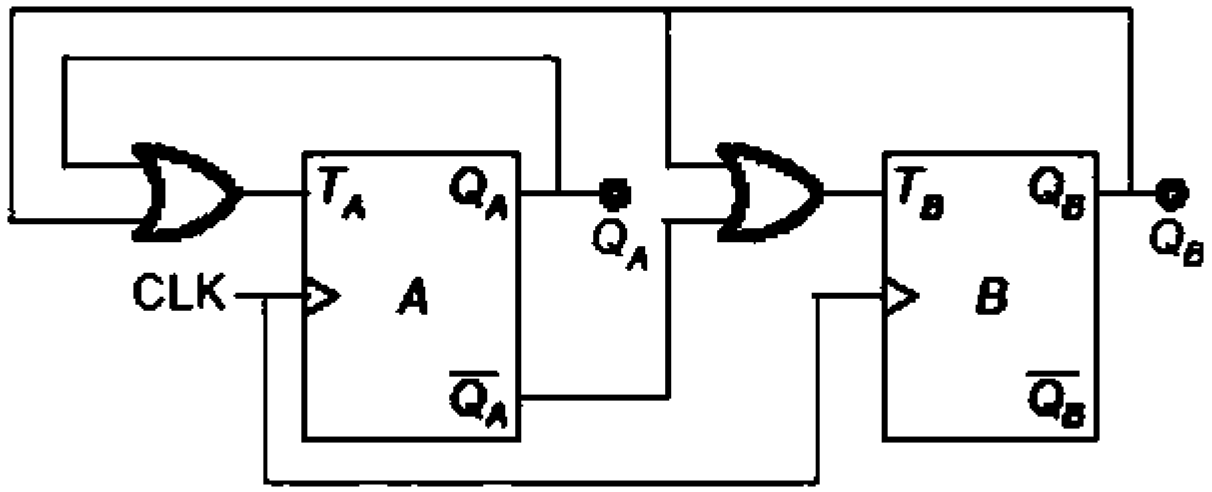
43. **Consider the following circuit**



Initially the counter is at '000'. After 9 clock pulses, the state of LEDs is (Assume + V_{CC} and 0 are used to represent logic – 1 and logic-0 respectively)

(a) only LED1 and LED 2 are ON
(b) only LED2 and LED 3 are ON
(c) only LED1 and LED 3 are ON
(d) All LEDs are ON

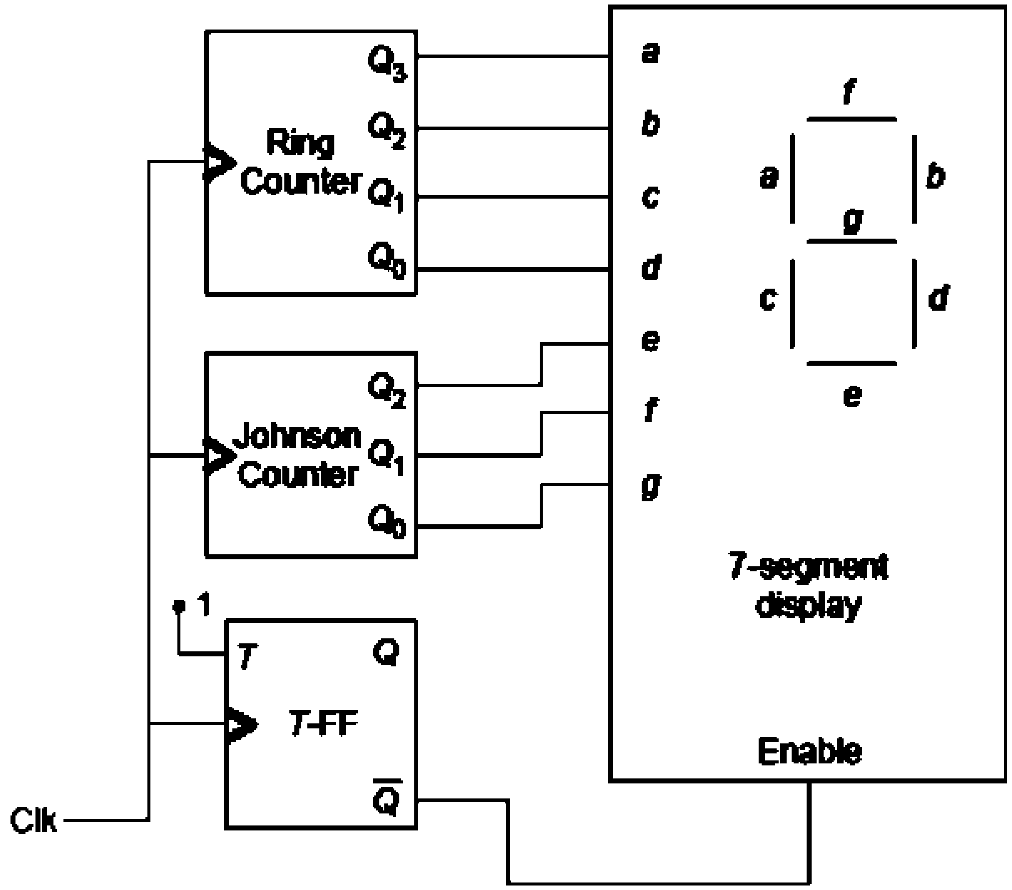
44. Consider the circuit shown below, initially both the flip-flops were cleared.



The modulus of the given counter circuit (i.e. the number of used states) is

45. A MOD-8 gray up-counter is constructed using T-flip-flops only. If the initial state of the counter is S_0 = 110, then the minimum number of clock cycles required to reach a state of (101) will be

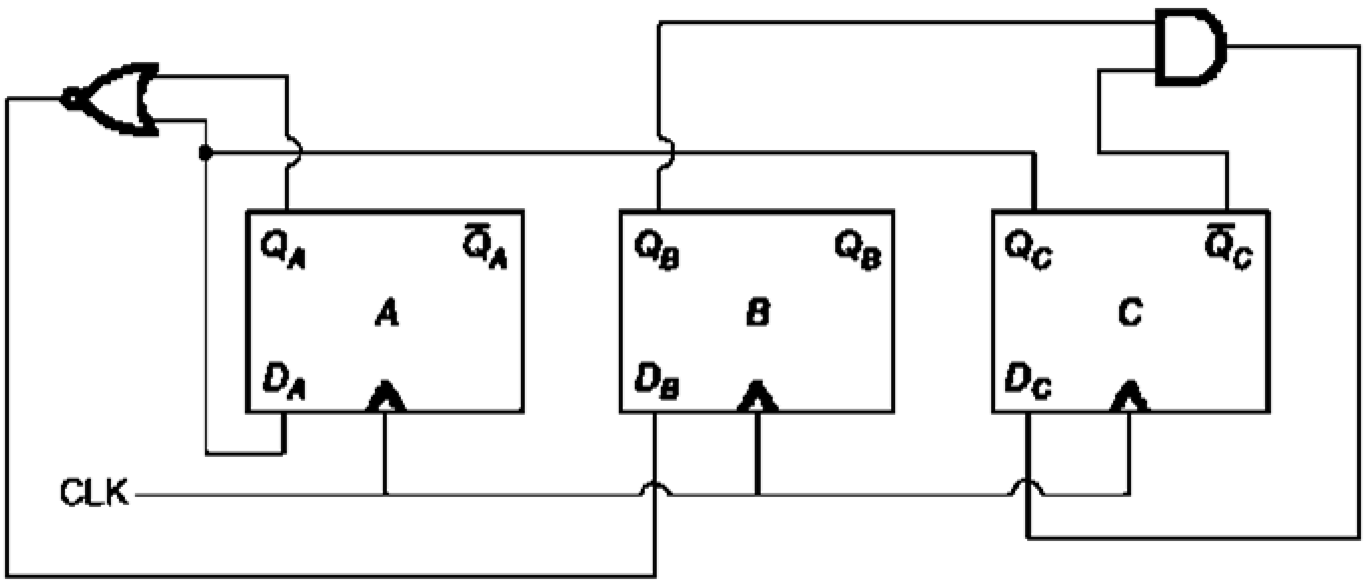
46. **Consider the circuit given below:**



After the first clock pulse, the output state of the ring counter is $(Q_3Q_2Q_1Q_0) = (1010)$ and that of Johnson counter is $(Q_2Q_1Q_0) = (101)$. If the T-flip-flip is reset before the first clock pulse is applied, then the display of the seven segment display after 6^{th} clock pulse will be

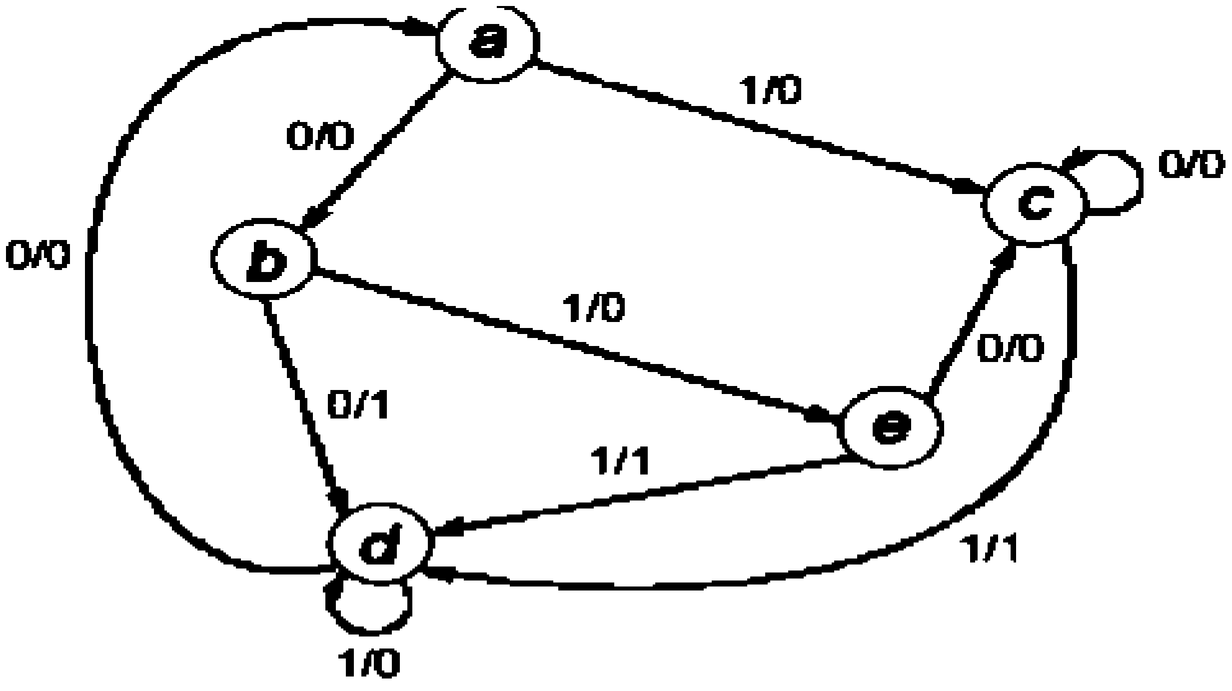
(a) 8 (b) 2 (c) 7 (d) 9

47. Consider the following circuit consisting of three D-flip flops.



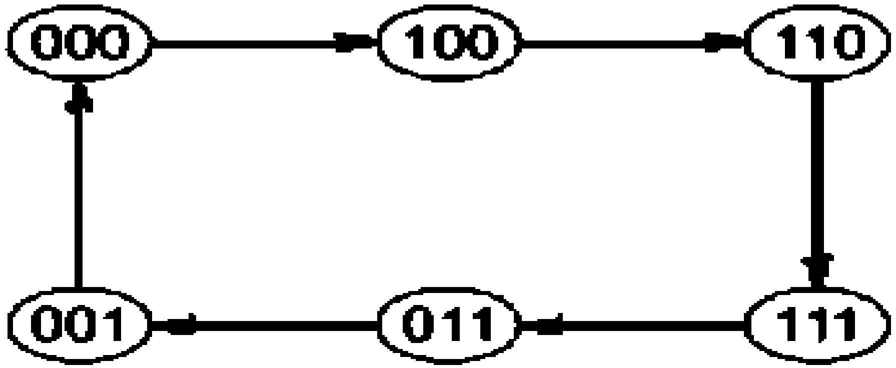
If all the flip-flops were reset to 0 at power on, then the total number of distinct output states represented by the counter ABC is equal to _____.

48. **Consider the state diagram shown in the figure below:**



The minimum number of flip-flops required to implement this state diagram is ________.

49. A counter is constructed whose output count is given by the sequence diagram shown in the figure below.



The minimum number of flip-flops required to construct such type of counter is

(a) 6 (b) 3 (c) 2 (d) 5

50 **A ripple counter is made with three positive edge triggered flip-flops. If the output of previous lower significant bit flip-flop is used as a triggering clock pulse of the next higher significant bit flip-flop, then the resultant counter is a**

(a) MOD 3 up counter (b) MOD 3 down counter

(c) MOD 8 up counter (d) MOD 8 down counter

51. A new flip-flop is designed with inputs A and B and output of the flip-flop being Q. The characteristic table of the flip-flop is shown below.

A	B	Q	Q^+
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

If Q^+ represents the next state output of the flip-flop, then the expression for Q^+ will be

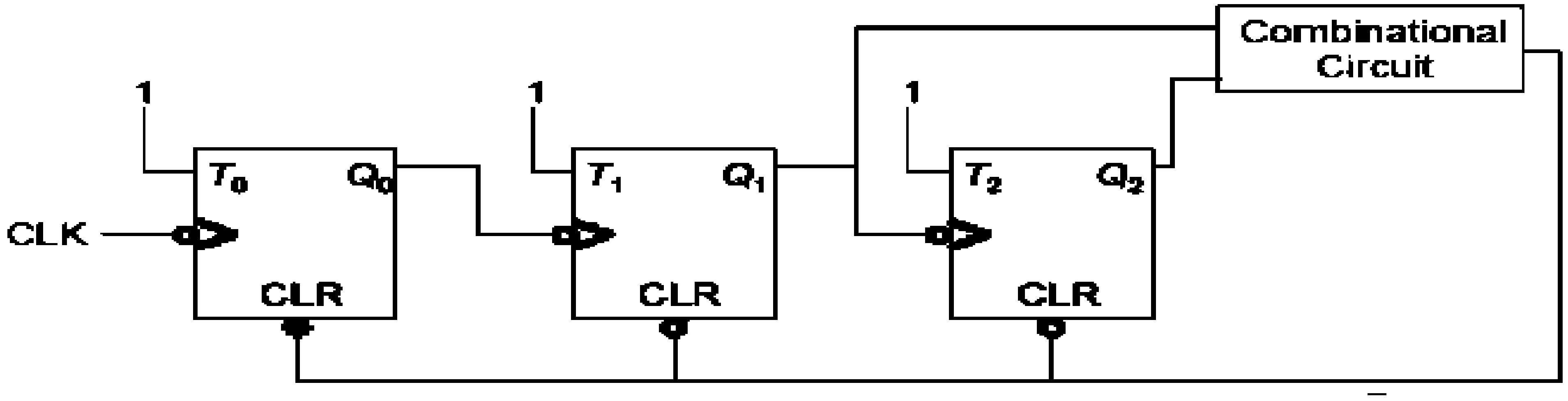
(a) $\bar{A}B + BQ$

(b) $A\bar{B} + B\bar{Q}$

(c) $A\bar{B} + AQ$

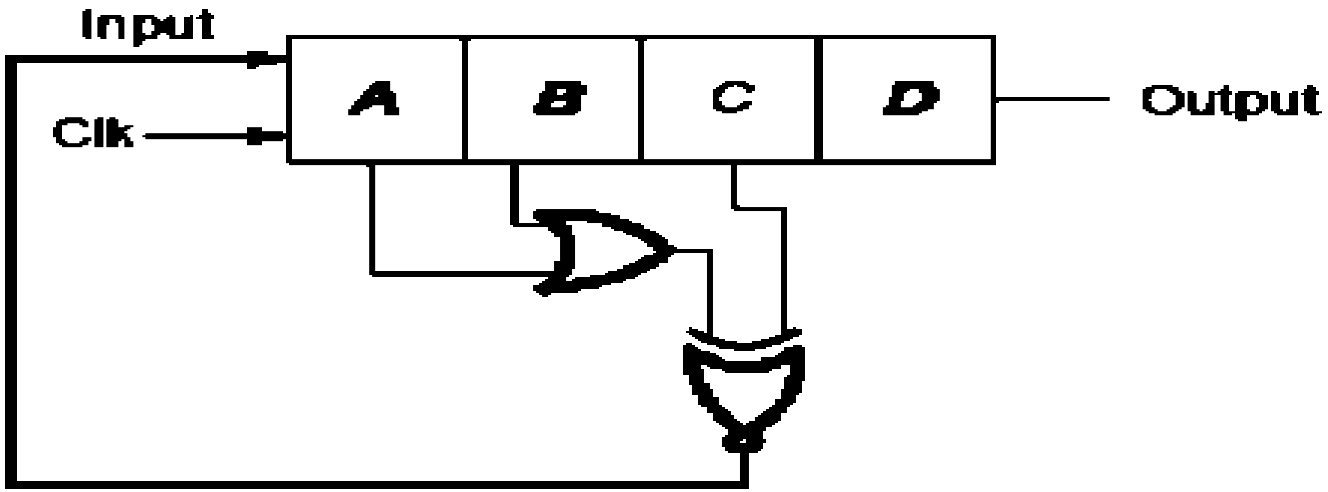
(d) $A\bar{B} + A\bar{Q}$

52. A MOD 6 ripple up-counter is to be designed using three flip-flops. The flip-flops can be reset to their initial condition by providing an active low external trigger to the CLR input. If the counter starts counting the sequence from $(Q_2Q_1Q_0) = (000)$, then the combinational circuit that is shown in the figure below can be constructed by using



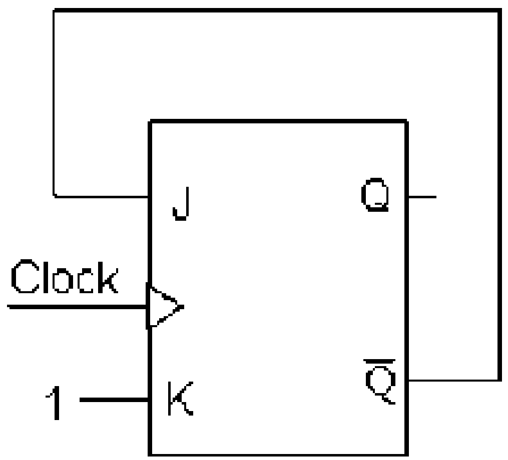
(a) an OR gate followed by a NOT gate
(b) an EX-OR gate followed by a buffer gate
(c) an AND gate followed by a NOT gate
(d) an EX-OR gate followed by a NOT gate

53. A four bit serial in parallel out shift register is constructed as shown in the figure below. The register is a right shift register i.e. A → B, B → C, C → D and D is connected to the output. If the initial state (ABCD) of the shift register is (1000), then the minimum number of clock cycles after which the state (ABCD) of the shift register will be again equal to (1000) is ________ .



(a) 6 (b) 5 (c) 4 (d) 3

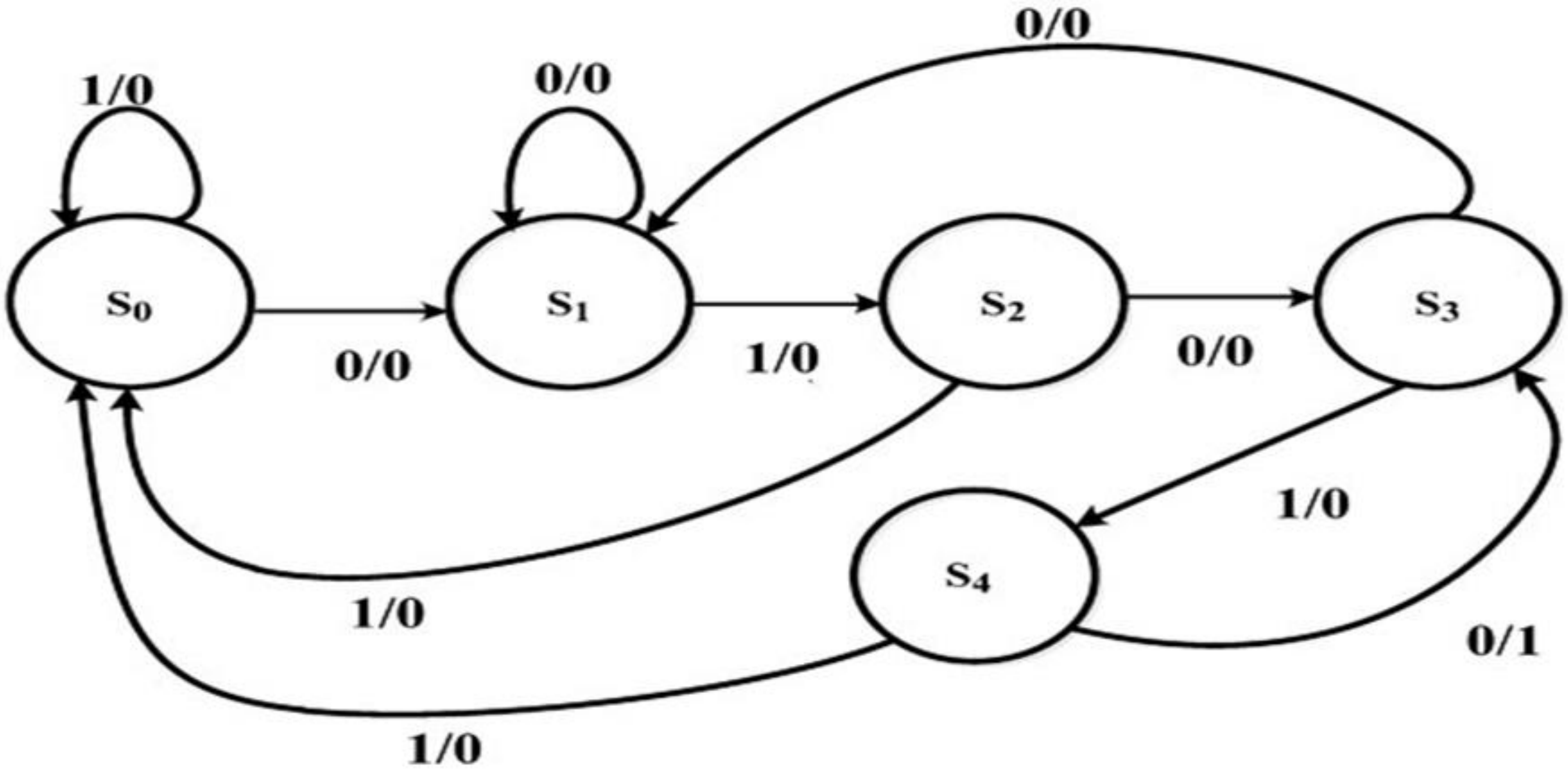
54. JK Flip-Flop circuit shown below



Assuming the Flip-Flop was initially cleared. The counting sequence at the 'Q' output will be

(a) 010000 (b) 011001 (c) 010010 (d)010101

55. The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1, then



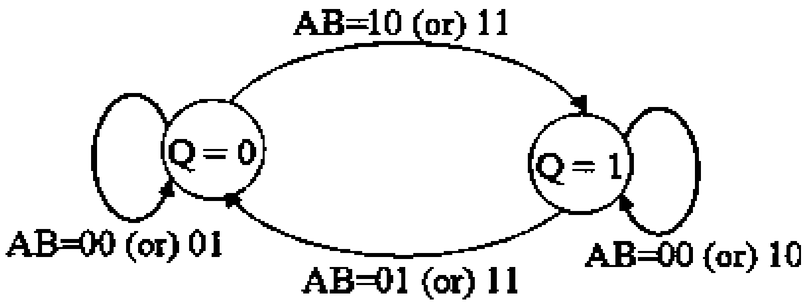
(A) the sequence 01010 is detected.

(B) the sequence 01011 is detected.

(C) the sequence 01110 is detected.

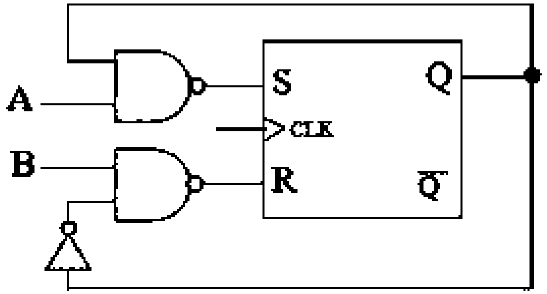
(D) the sequence 01001 is detected.

56. To eliminate the forbidden state of the SR flip-flop, we are generating a special flip-flop called as AB flip-flop whose state transition diagram is shown below

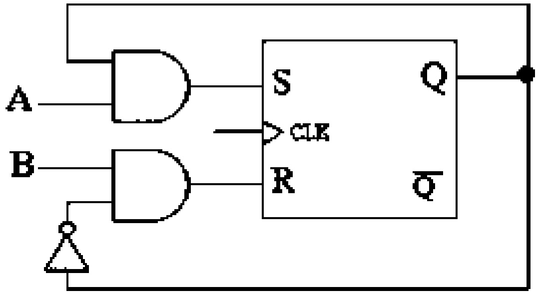


Such AB flip-flop logic diagram constructed from SR flip-flop is given by

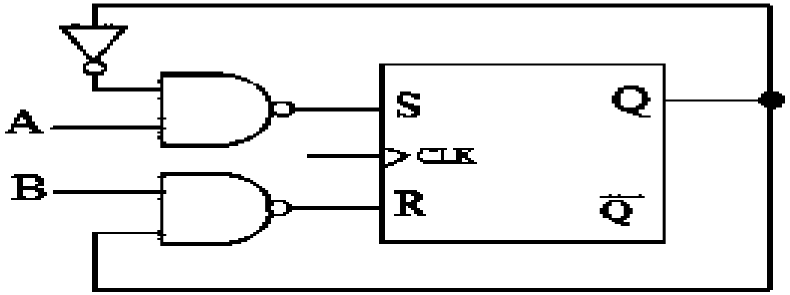
(a)



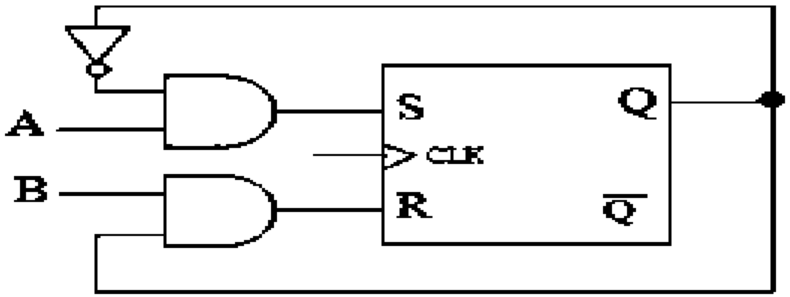
(b)



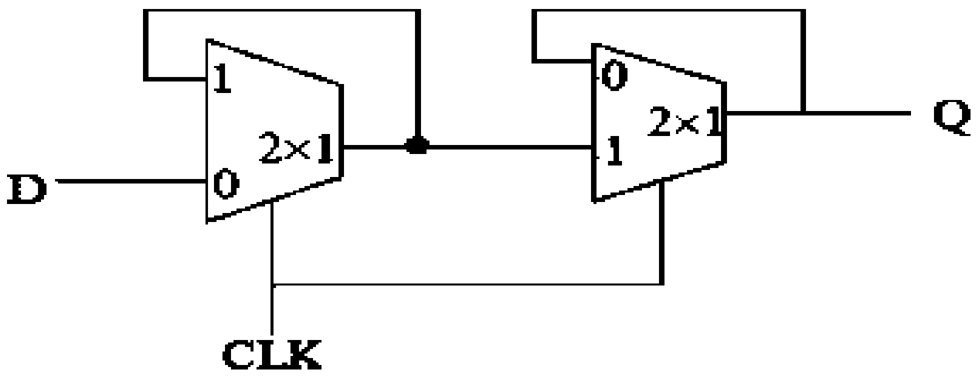
(c)



(d)



57.



Considering level triggered clock 'CLK' is used as selection input to both the multiplexers, then the above circuit represents which one of the following circuit operations

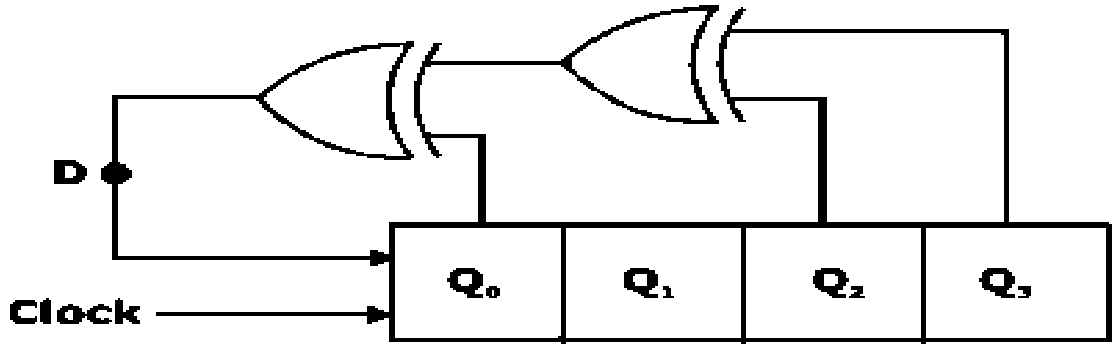
(a) Positive edge-triggered D flip-flop whose input is D and output is Q

(b) Negative edge-triggered D flip-flop whose input is D and output is Q

(c) Positive D – latch

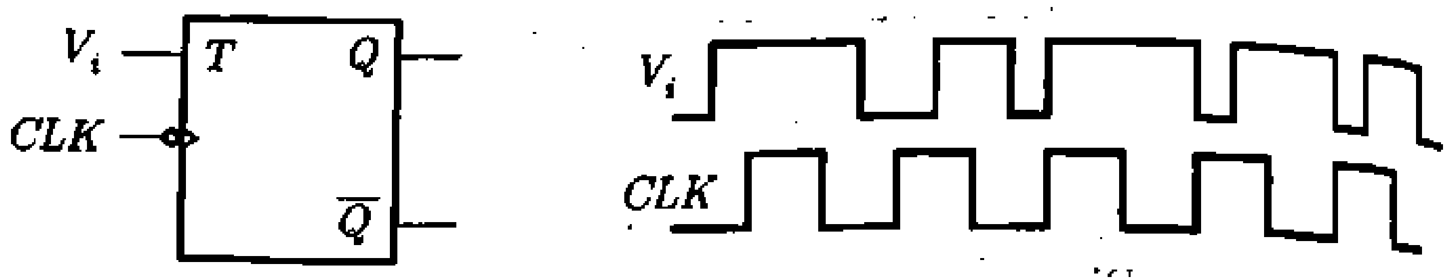
(d) T flip-flop constructed using two 2 x 1 multiplexers whose input is D and output is Q

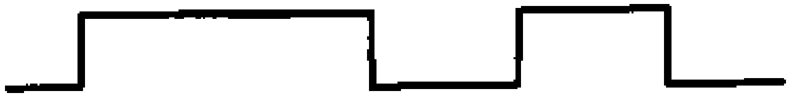
58. A 4-bit right shift register has initialized the value 1000 for (Q_3 Q_2 Q_1 Q_0). The D input is derived from Q_3, Q_2 and Q_0 through two XOR gates as shown in figure below. Number of clock pulses required to get pattern 1000 are

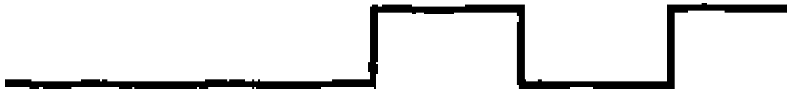


(a)6 (b) 5 (c) 7 (d) 8

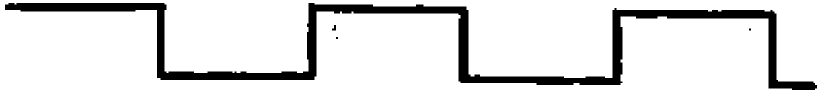
59. **The input signal V_i shown below is applied to the FF in given figure when initially in 0 state. Assume all timing constraints are satisfied.**



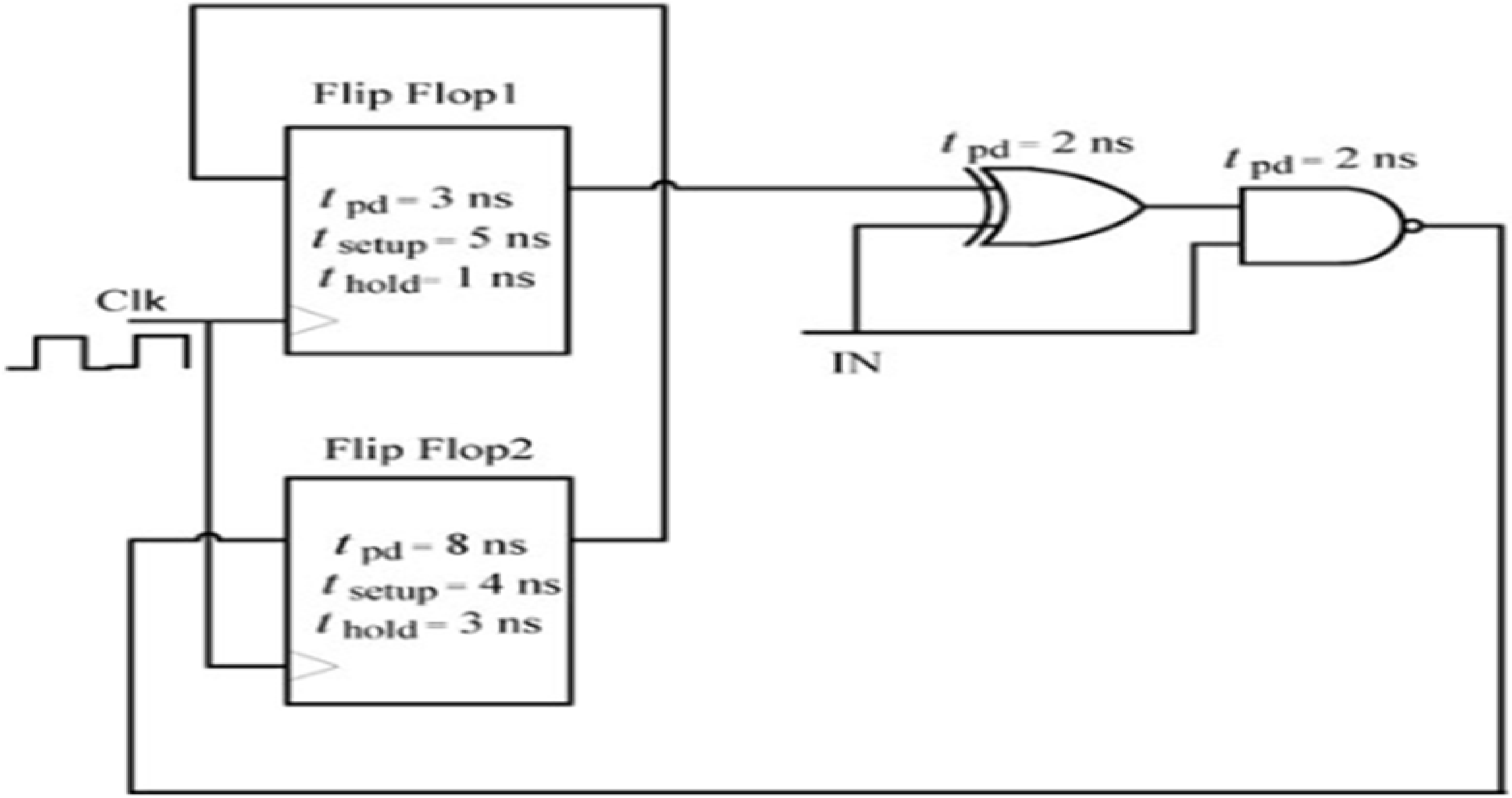
(a) 

(b) 

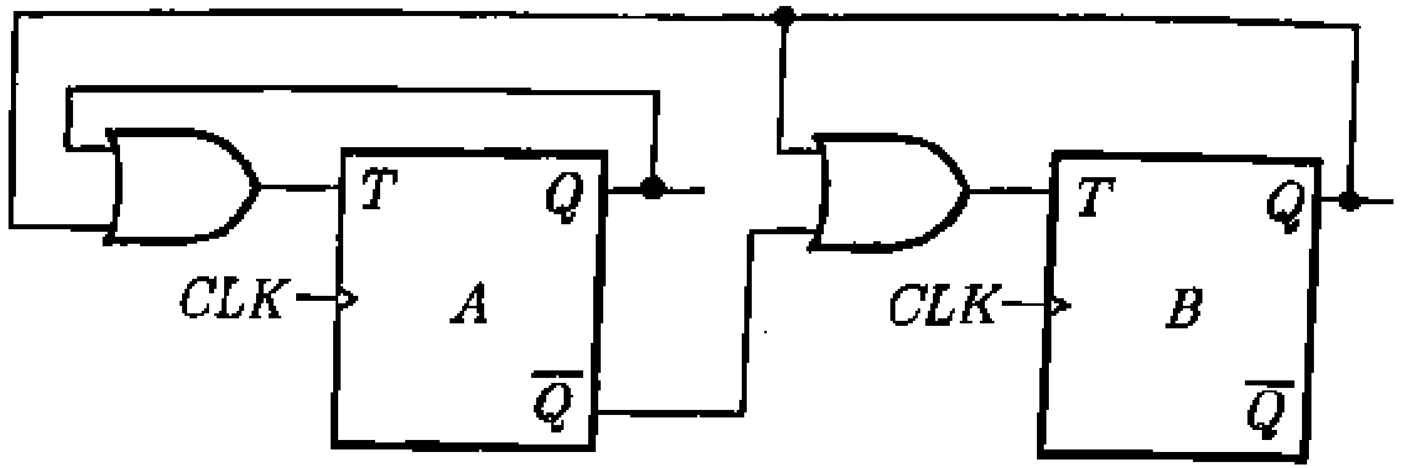
(c) 

(d) 

60. For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency **(rounded off to the nearest integer)**, at which the given circuit can operate reliably, is __________ MHz.



61. **The circuit shown in figure below is**



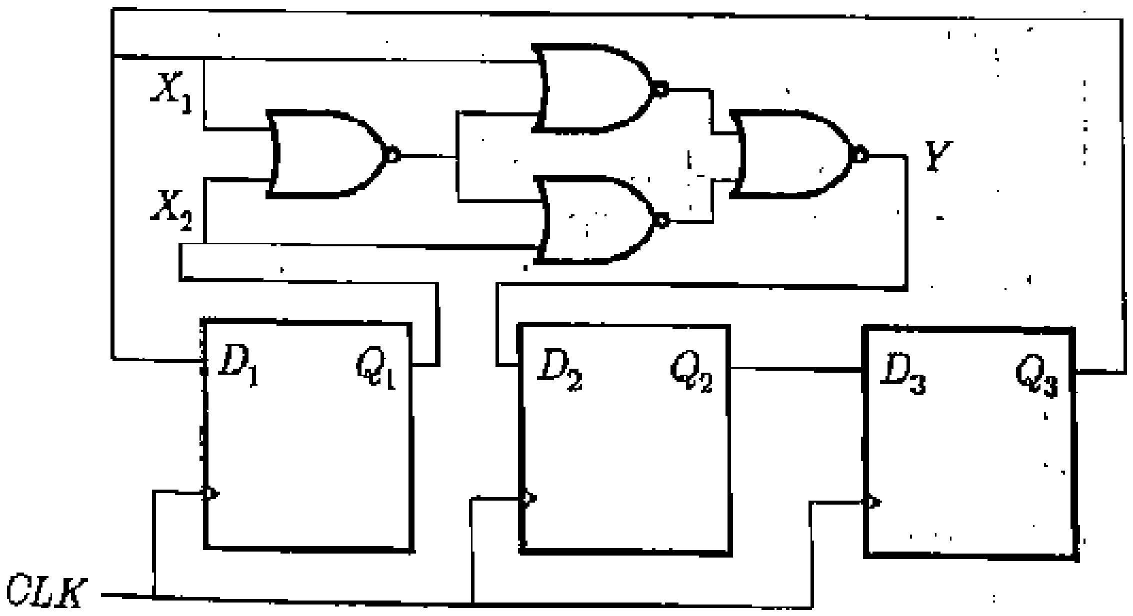
(a) a MOD-2 counter

(b) a MOD-3 counter

(c) generate sequence 00, 10, 01, 00, ...

(d) generate sequence 00, 10, 00, 10, 00, ...

Consider the circuit shown in following figure



62. **The correct input output relationship between Y and (X_1, X_2) is**

(a) $Y = X_1 + X_2$
(b) $Y = X_1X_2$
(c) $Y = X_1 \oplus X_2$
(d) $Y = \overline{X_1 \oplus X_2}$

63. **The D flip-flop are initialized to $Q_1\ Q_2\ Q_3 = 000$. After 1 clock cycle, $Q_1\ Q_2\ Q_3$ is equal to**

(a) 011
(b) 010
(c) 100
(d) 101

64. A sequence detector is designed to detect precisely 3 digital inputs, with overlapping sequences detectable. For the sequence (1,0,1) and input data (1,1,0,1,0,0,1,1,0,1,0,1,1,0), what is the output of this detector?

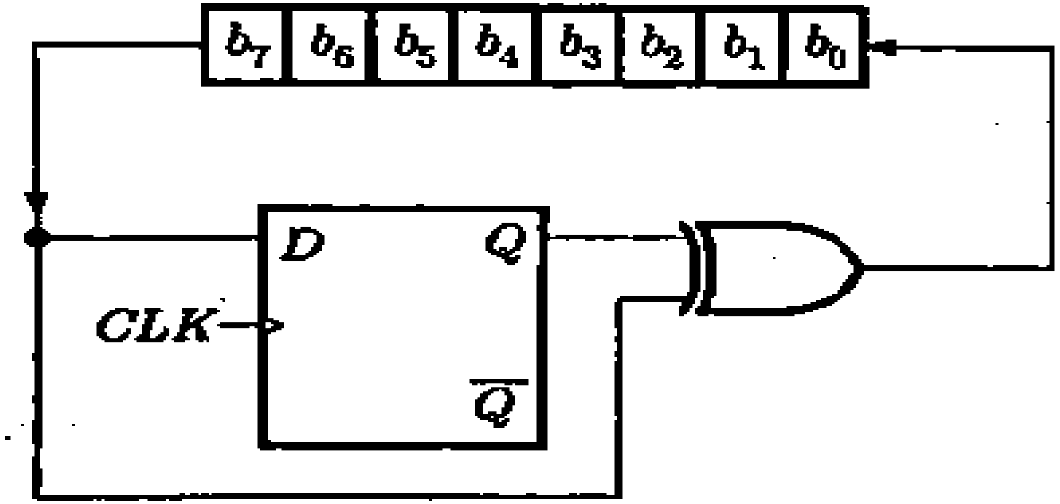
(A) 0,0 1,1,0,0,0,0,1,1,0,1,0,0

(B) 0,0 0,1,0,0,0,0,0,1,0,1,0,0

(C) 0,0 0,1,0,0,0,0,0,1,0,1,1,0

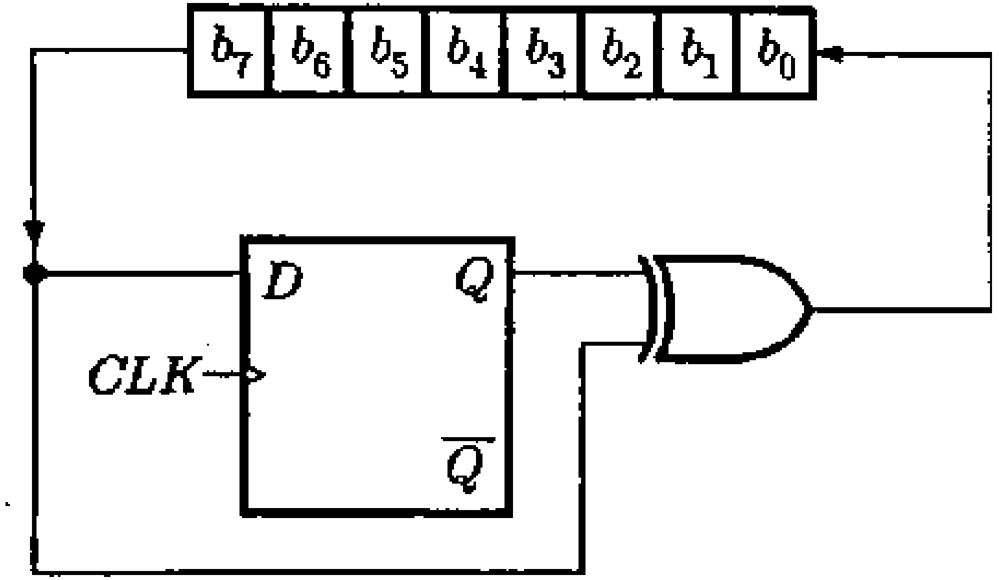
(D) 0,0 0,1,0,0,0,0,0,0,1,0,0,0

The 8-bit left shift register and D flip-flop shown in figure below is synchronized with same clock. The D flip-flop is initially cleared.



65. The circuit acts as
(a) Binary to 2's complement converter
(b) Binary to Gray code converter
(c) Binary to 1st complement converter
(d) Binary to Excess-3 code converter

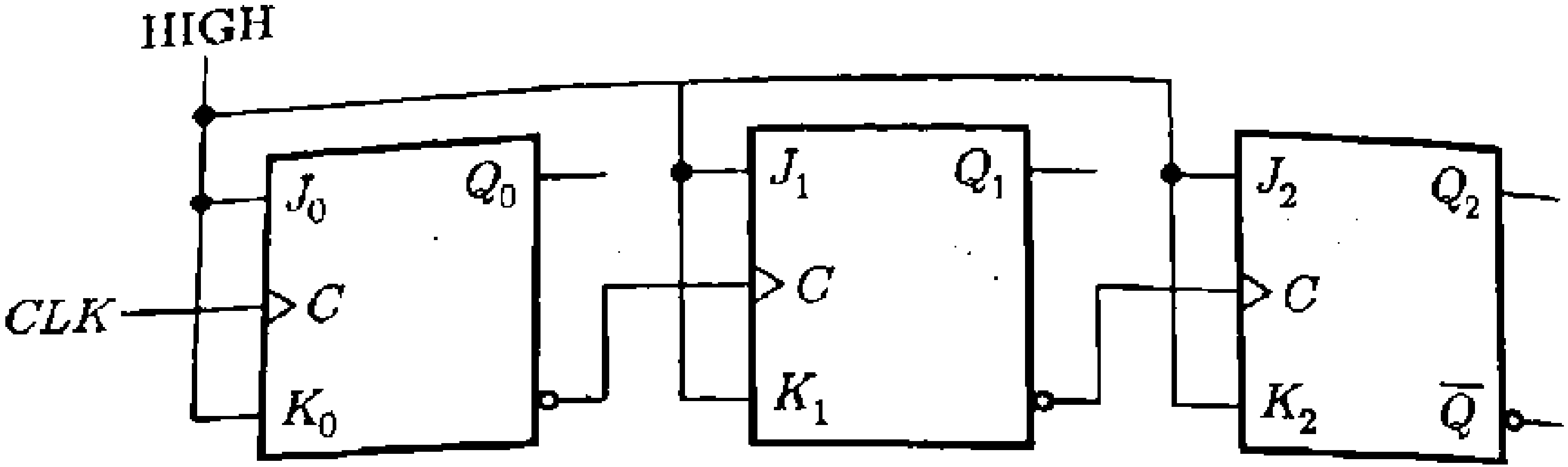
The 8-bit left shift register and D flip-flop shown in figure below is synchronized with same clock. The D flip-flop is initially cleared.



66. If initially register contains byte B7, then after 4 clock pulse contents of register will be

(a) 73 (b) 72 (c) 7E (d) 74

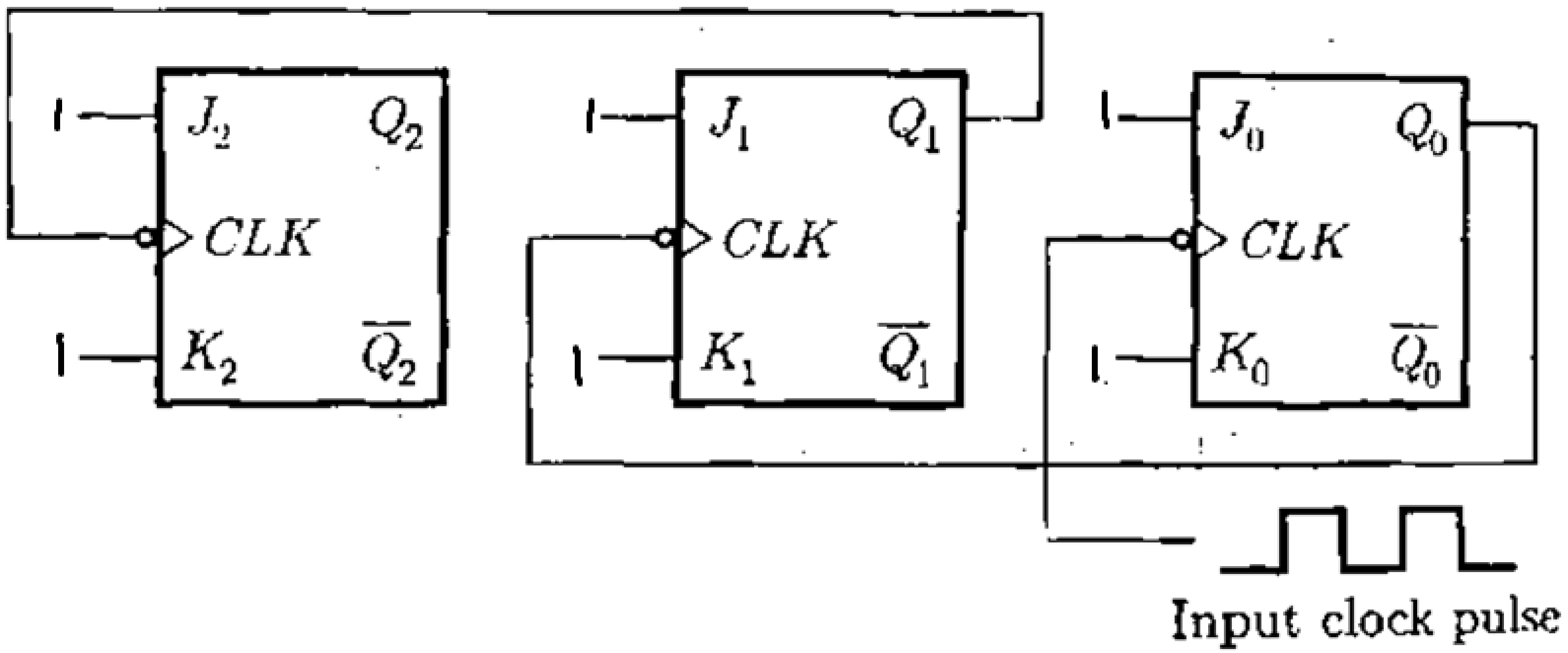
67. In the given counter, each flip-flop has a propagation delay of 8 ns.



The worst-case (longest) delay time from a clock pulse to the arrival of the counter in a given state is

(a) 8 n sec (b) 16 n sec (c) 24 n sec (d) 10 n sec

68. **Consider the following counter**



If counter starts at 000, what will be the count after 13 clock pulses?

(a) 100 (b) 101 (c) 110 (d) 111

69. In a master-slave JK flip-flop

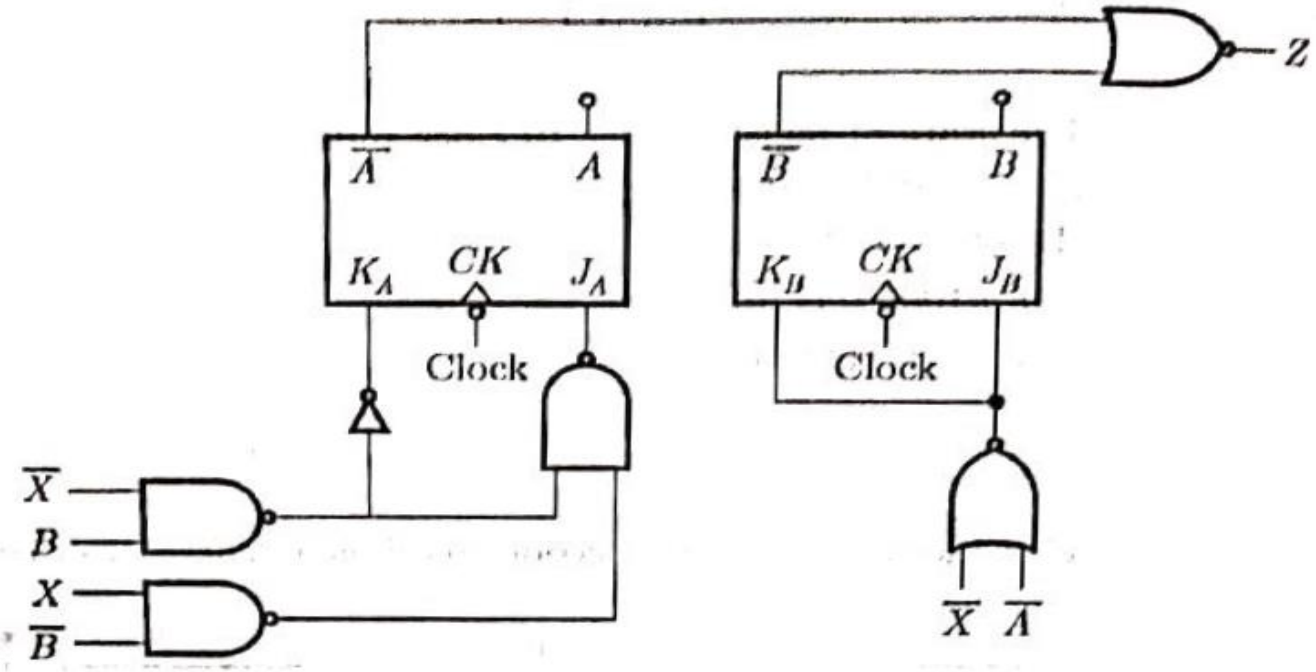
(a) both master and slave are positive-edge-triggered

(b) both master and slave are negative-edge-triggered

(c) master is positive- level -triggered and slave is negative- level triggered

(d) master is negative-edge-triggered and slave is positive-edge-triggered

Consider the following sequential circuit



70. **For the given sequential circuit, the next state equations for flip-flop A and B are**

(a) $A^+ = A(B' + X) + A'(BX' + B'X)$ and $B+ = AB'X + B(A' + X')$

(b) $A^+ = A(B'X) + A'(BX' + B'X)$ and $B^+ = A(B' + X) + B(A' + X')$

(c) $A^+ = A(B'X) + A'(BX')$ and $B^+ = A(B'X) + B(A'X')$

(d) $A^+ = A(B' + X) + A'(BX' + B'X)$ and $B^+ = A'X + B'X'A'$

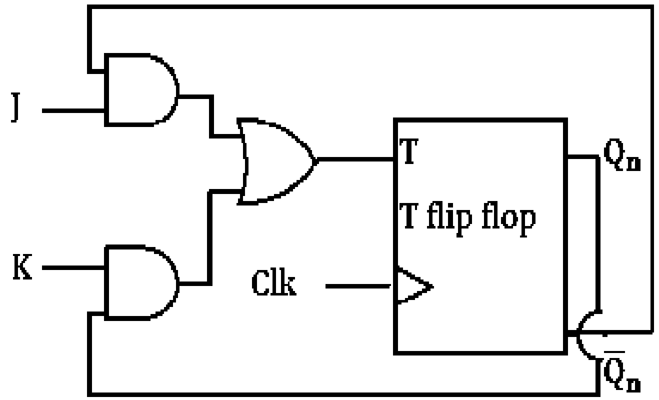
71. The minimum number of D flip-flops needed to design a mod-258 counter is

(A) 9 (B) 8 (C) 512 (D) 258

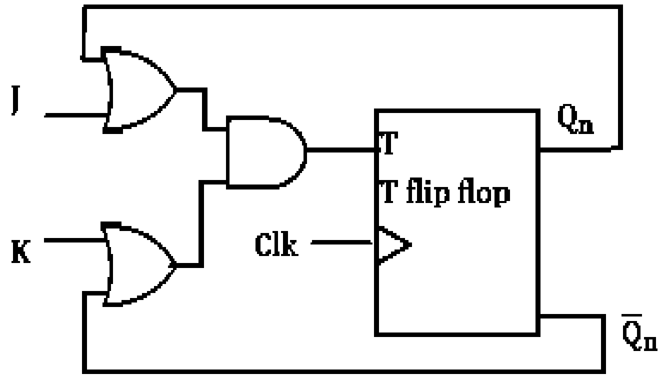
[2011, 2 Marks]

72. A JK flip flop can be implemented by T flip-flops. Identify the correct implementation **GATE (EE-2014)**

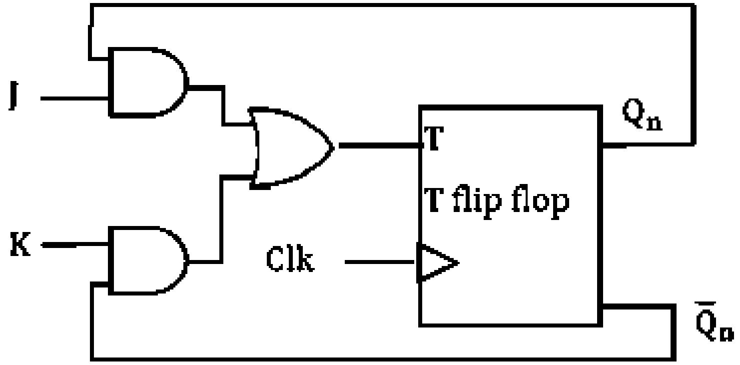
(A)



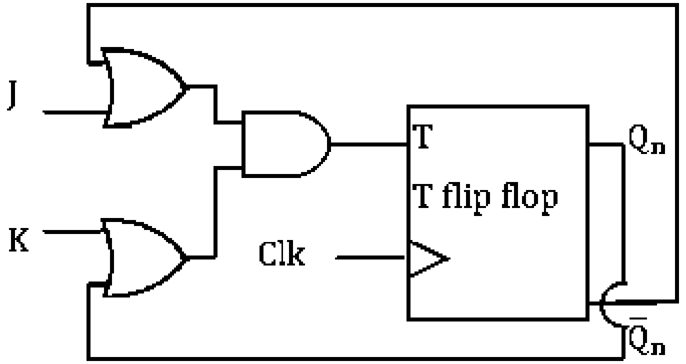
(B)



(C)



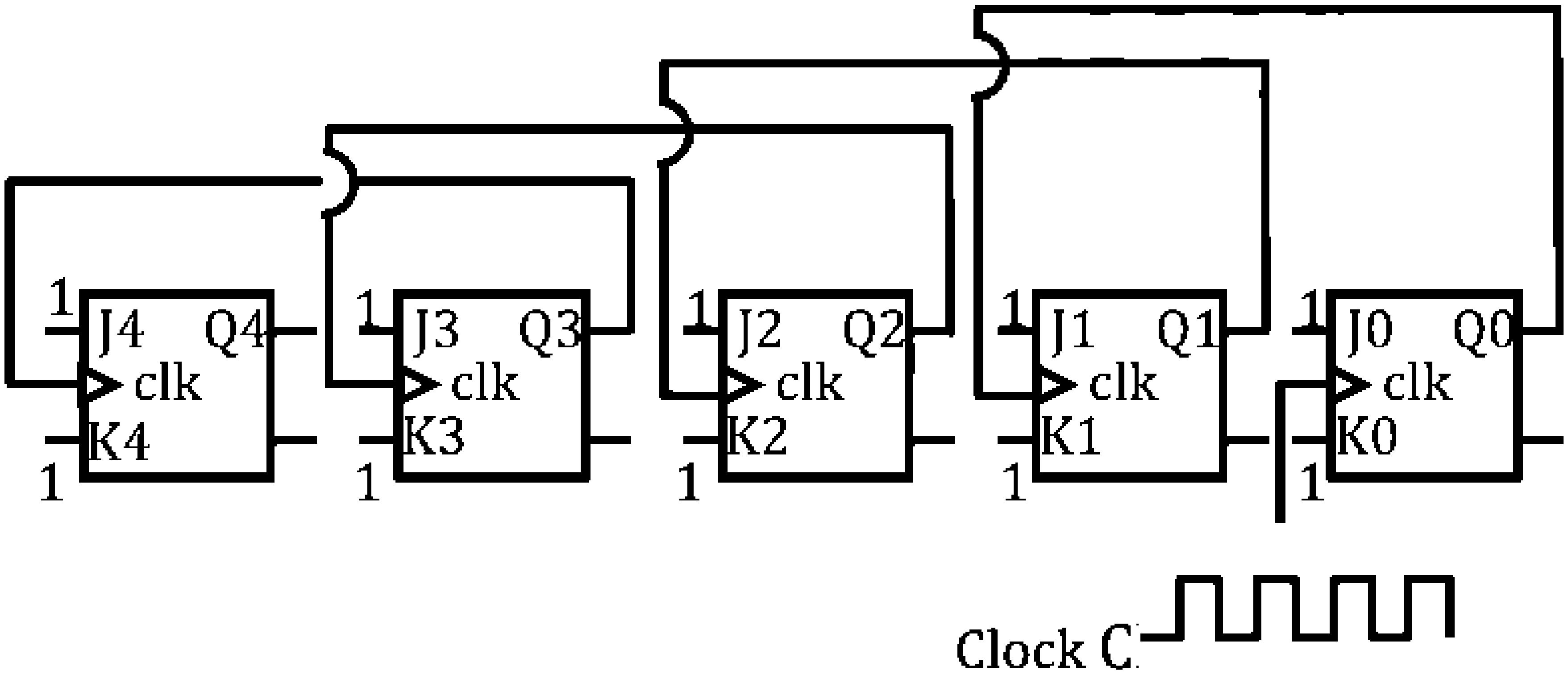
(D)



73. **If the input to a toggle mode T flip flop is a 100 MHz signal, the final output of three T flip flops in a cascade is** ESE (2017)

(a) 1000 MHz (b) 520 MHz (c) 333 MHz (d) 12.5 MHz

74. Five JK flip-flops are cascaded to form the circuit shown in Figure. Clock pulses at a frequency of 1 MHz are applied as shown. The frequency (in kHz) of the waveform at Q3 is _____. GATE (ECE-2014)



75. **A 3-bit ripple counter is constructed using three T flip-flops to do the binary counting. The three flip-flops have T-inputs fixed at**

ESE (2016)

(a) 0, 0 and 1

(b) 1, 0 and 1

(c) 0, 1 and 1

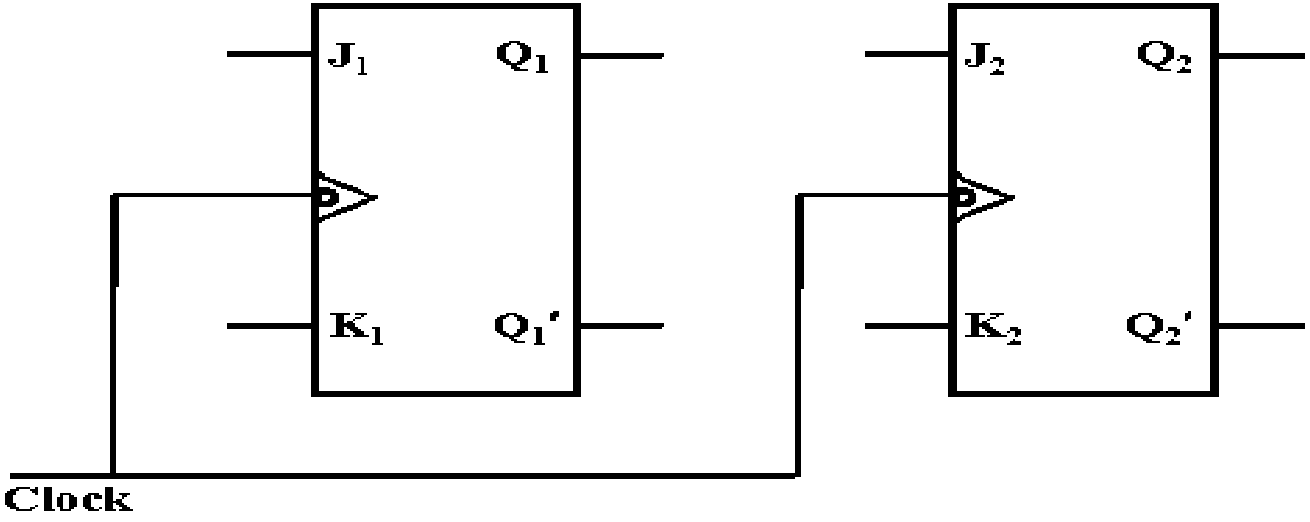
(d) 1, 1 and 1

76. Two D flip-flops are connected as a synchronous counter that goes through the following $Q_B Q_A$ sequence $00 \rightarrow 11 \rightarrow 01 \rightarrow 10 \rightarrow 00 \rightarrow \ldots$ The connections to the inputs D_A and D_B are

GATE (EC-2011)

(a) $D_A = Q_B$, $D_B = Q_A$
(b) $D_A = \overline{Q}_A$, $D_B = \overline{Q}_B$
(c) $D_A = (Q_A \overline{Q}_B + \overline{Q}_A Q_B) D_B = Q_A$
(d) $D_A = (Q_A Q_B + \overline{Q}_A \overline{Q}_B)$, $D_B = \overline{Q}_B$

77. A synchronous counter using two J – K flip flops that goes through the sequence of states; $Q_1Q_2 = 00 \rightarrow 10 \rightarrow 01 \rightarrow 11 \rightarrow 00\ldots$ is required. To achieve this, the inputs to the flop flops are **GATE (IN-2016)**



(a) $J_1 = Q_2,\ K_1 = 0;\ J_2 = Q_1',\ K_2 = Q_1$

(b) $J_1 = 1,\ K_1 = 1;\ J_2 = Q_1,\ K_2 = Q_1$

(c) $J_1 = Q_2,\ K_1 = Q_2';\ J_2 = 1',\ K_2 = 1$

(d) $J_1 = Q_2',\ K_1 = Q_2;\ J_2 = Q_1,\ K_2 = Q_1'$

78. The next state table of a 2-bit saturating up-counter is given below

Q1	Q0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuits using T flip-flop. The expression for T_1 and T_0 are **GATE (CS-2017)**

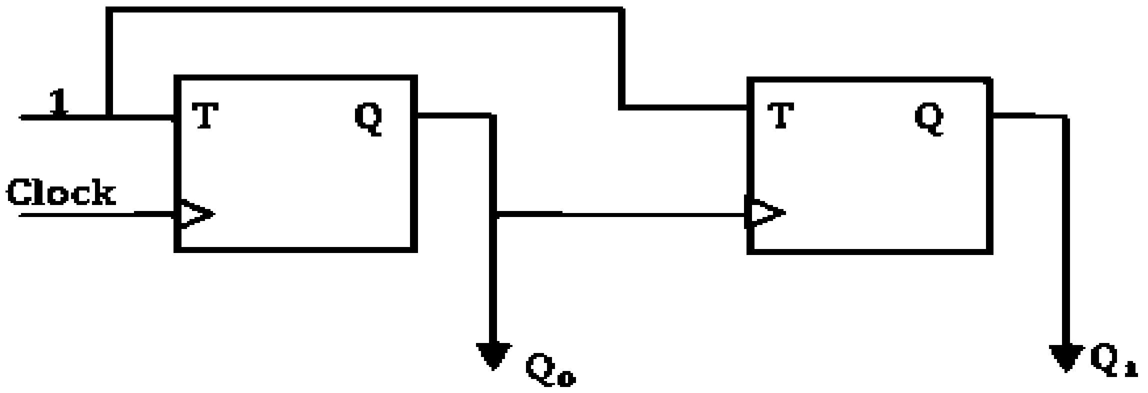
(a) $T_1 = Q_1Q_0$, $T_0 = \overline{Q}_1\overline{Q}_0$

(b) $T_1 = \overline{Q}_1Q_0$, $T_0 = \overline{Q}_1\overline{Q}_0$

(c) $T_1 = Q_1+Q_0$, $T_0 = \overline{Q}_1+\overline{Q}_0$

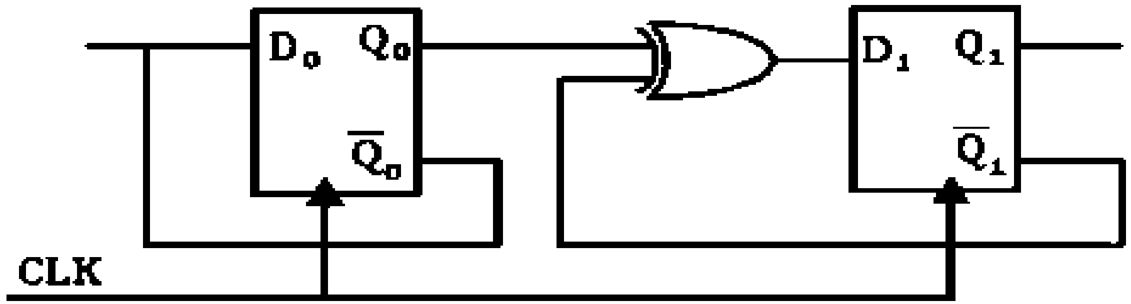
(d) $T_1 = \overline{Q}_1Q_0$, $T_0 = \overline{Q}_1 + \overline{Q}_0$

79. In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00, what are the next four values of Q_1Q_0? GATE (CS-2010)



(a) 11, 10, 01, 00
(b) 10, 00, 01, 11
(c) 10, 11, 01, 00
(d) 11, 10, 00, 01

80. **Consider the following circuit**



The flip-flops are positive edge triggered D FFs. Each state is designated as a two bit string Q_0Q_1. Let the initial state be 00. The state transition sequence is **GATE (CS-2005)**

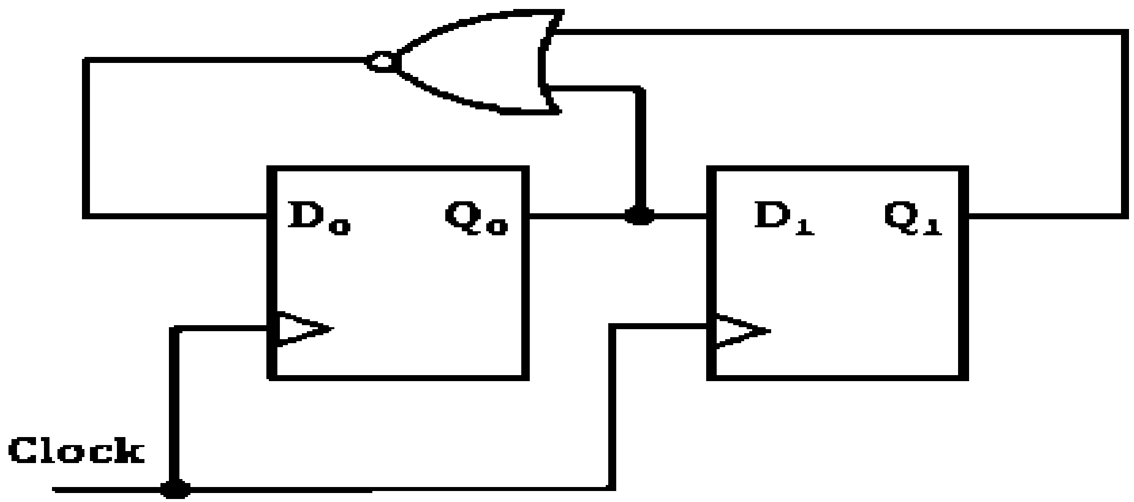
(A) 00→11→01

(B) 00→11

(C) 00→10→01→11

(D) 00→11→01→10

81. **For the circuit shown, the counter state (Q_1Q_0) follows the sequence**



GATE (EC-2007)

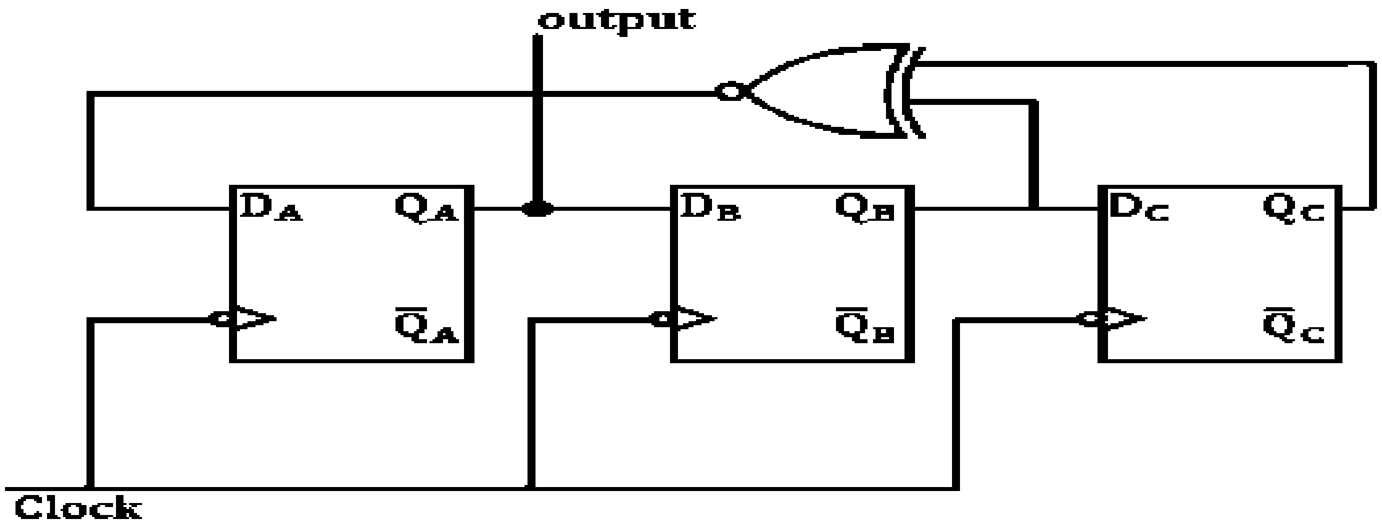
(a) 00, 01, 10, 11, 00 … … …

(b) 00, 01, 10, 10, 01 … … …

(c) 00, 01, 11, 00, 01 … … …

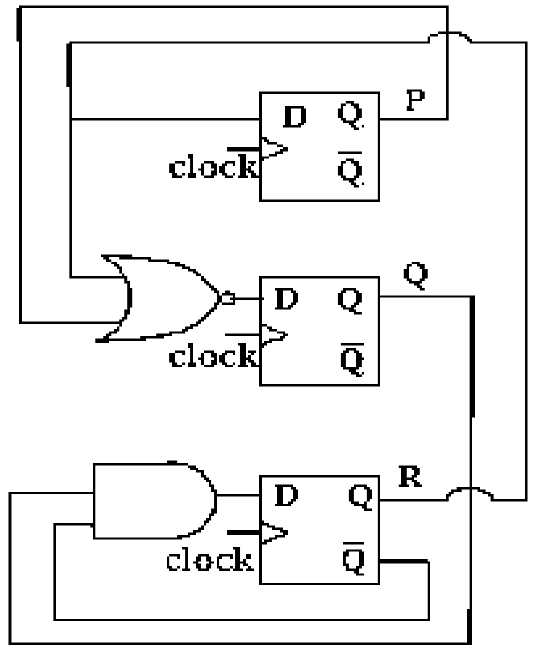
(d) 00, 10, 11, 00, 10 … … …

82. Assuming that all flips-flops are in reset condition initially, the count sequence observed at Q_A in the circuit shown is GATE (EC-2010)



(a) 0010111 ...
(b) 0101111 ...
(c) 0001011 ...
(d) 0110100 ...

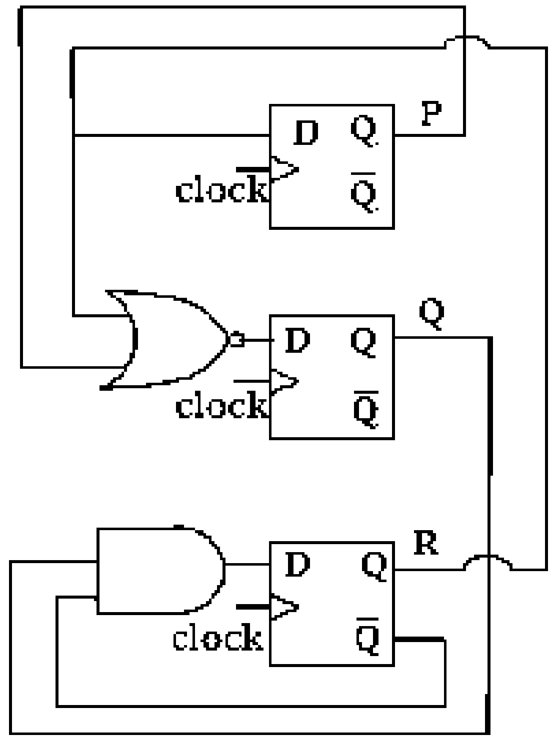
Consider the following circuit involving three D-type flip-flop used in a certain type of counter configuration. **GATE (CS-2011)**



83. If some instance prior to the occurrence of the clock edge, P, Q and R have a value 0, 1 and 0 respectively, what shall be the value of PQR after the clock edge?

(a) 000 (b) 001 (c) 010 (d) 011

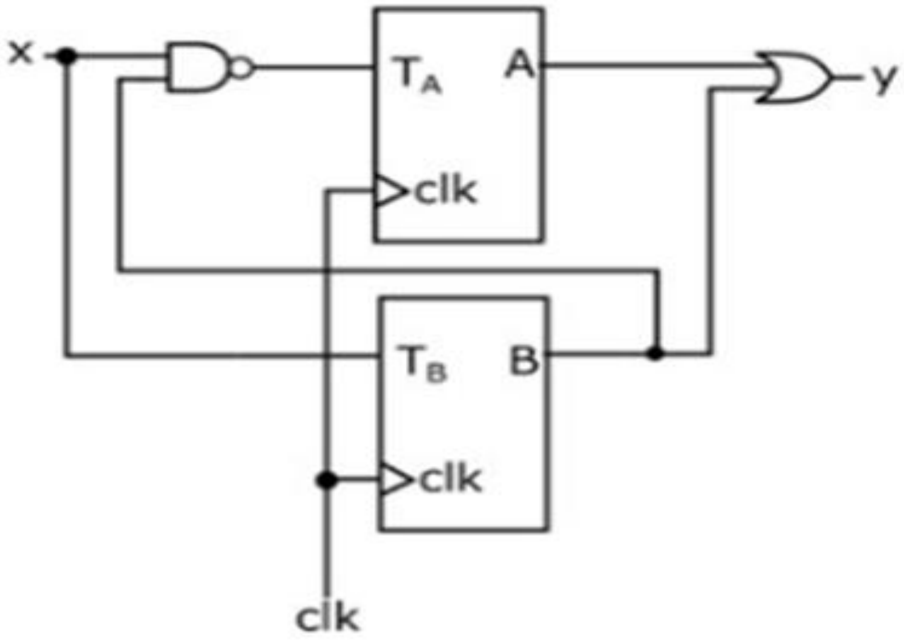
Consider the following circuit involving three D-type flip-flop used in a certain type of counter configuration. **GATE (CS-2011)**



84. If all the flip-flops were reset to 0 at power on, what is the total number of distinct output (states) represented by PQR generated by the counter?

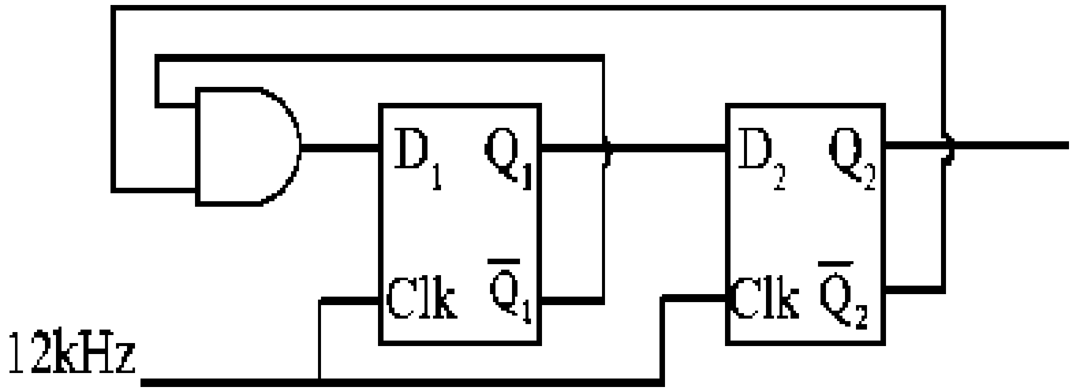
(a) 3 (b) 4 (c) 5 (d) 6

85. Two T-flip flops are interconnected as shown in the figure. The present state of the flip flops are: A = 1, B = 1. The input x is given as 1, 0, 1 in the next three clock cycles. The decimal equivalent of $(ABy)_2$ with A being the MSB and y being the LSB, after the 3rd clock cycle is ___



86. If the circuit shown, the clock frequency, i.e. the frequency of the CLK signal, is 12 kHZ. The frequency of the signal at Q_2 is ____ kHz.

GATE (EC-2019)



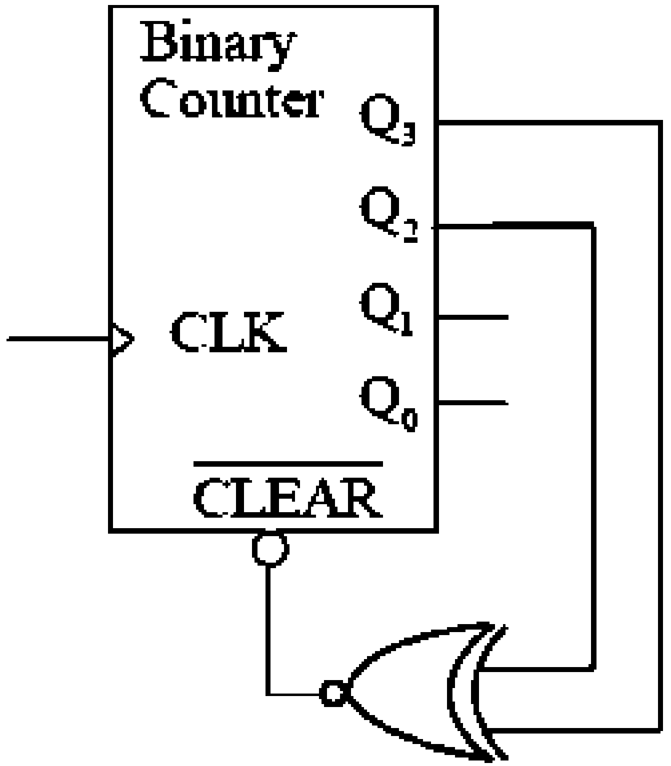
87. The figure shows a binary counter with synchronous clear input. With the decoding logic shown, the counter works as a GATE (EC-2015)

(a) mod-2 counter

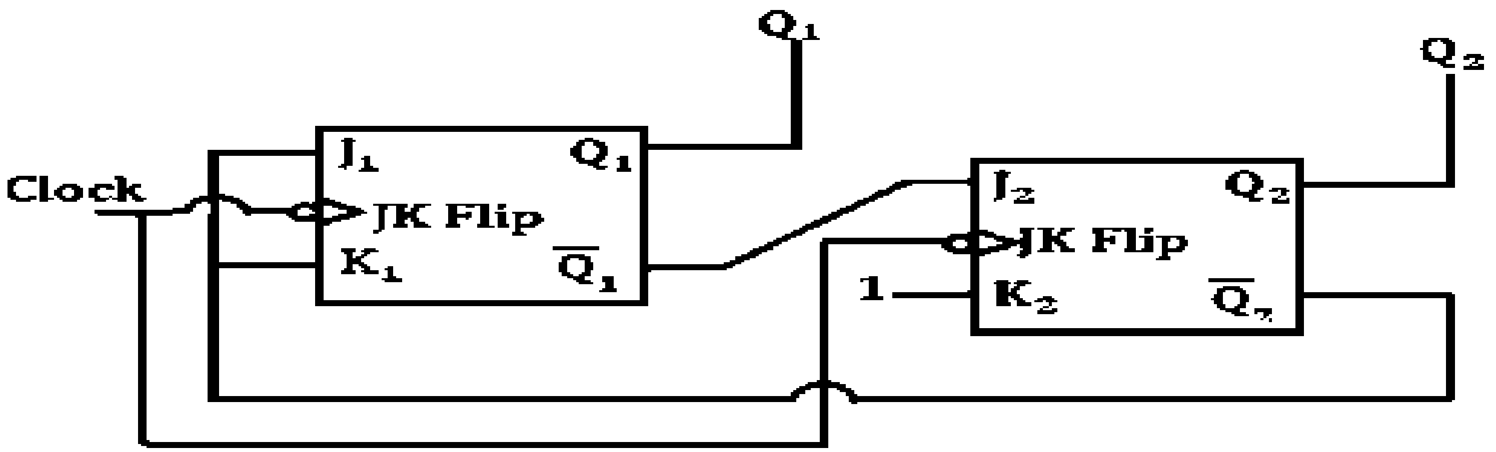
(b) mod-4 counter

(c) mod-5 counter

(d) mod-6 counter

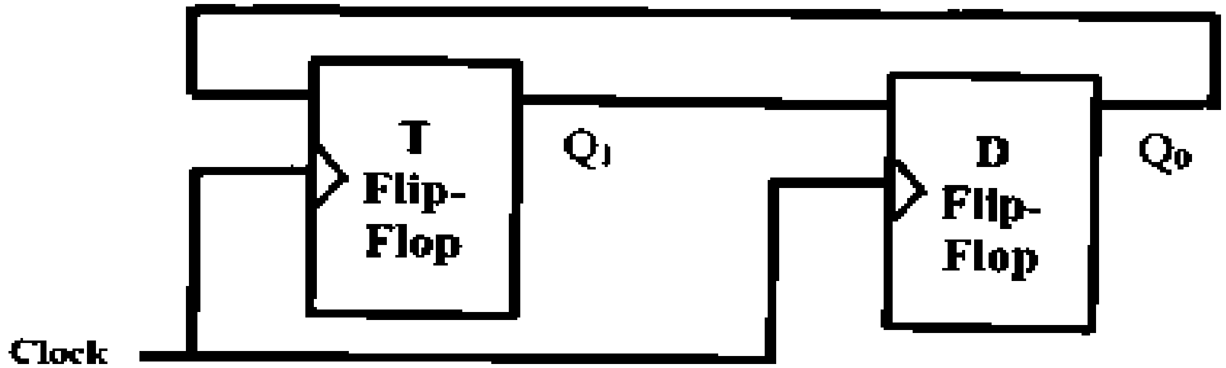


88. **What are the counting states (Q_1, Q_2) for the counter shown in the figure below?** **GATE (EC-2009)**



(a) 11, 10, 00, 11, 10, ...
(b) 01, 10, 11, 00, 01, ...
(c) 00, 11, 01, 10, 00, ...
(d) 01, 10, 00, 01, 10

89. **Consider a combination of T and D flip-flops connected as shown below. The output of the flip-flop is connected to the input of the flop-flop and the output of the T flip-flop is connected to the input of the D flip-flop.**



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs **GATE (CS-2017)**

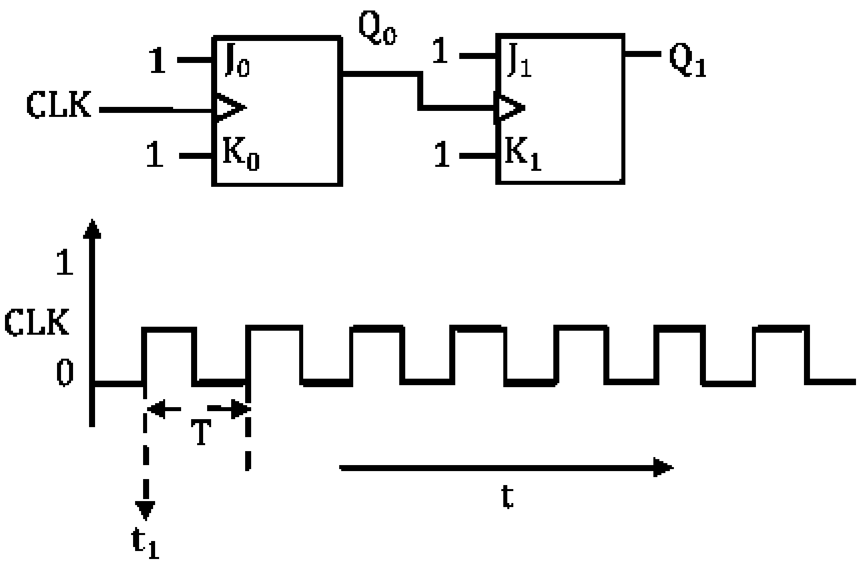
(a) Q_1 Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively

(b) Q_1 Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively

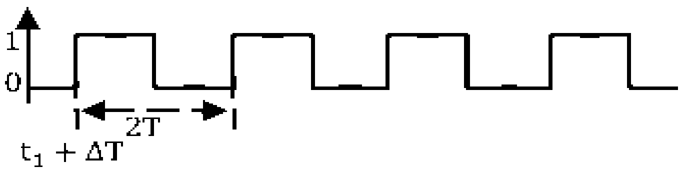
(c) Q_1 Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively

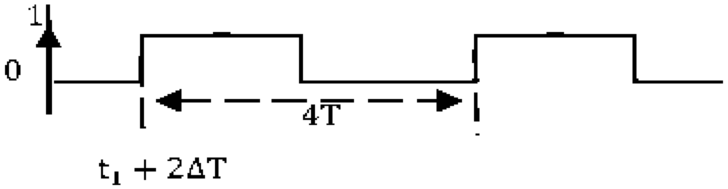
(d) Q_1 Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively

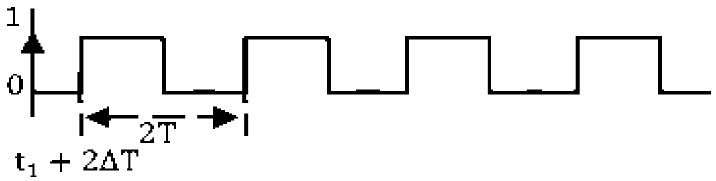
90. **For each of the positive edge – triggered J-K flip flop used in the following figure, the propagation delay is ΔT.**

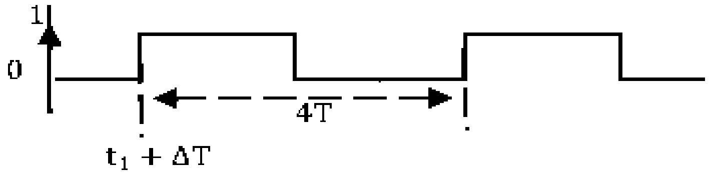


Which of the following waveforms correctly represents the output at Q_1 ? **GATE (ECE-2017)**

(A)


(B)


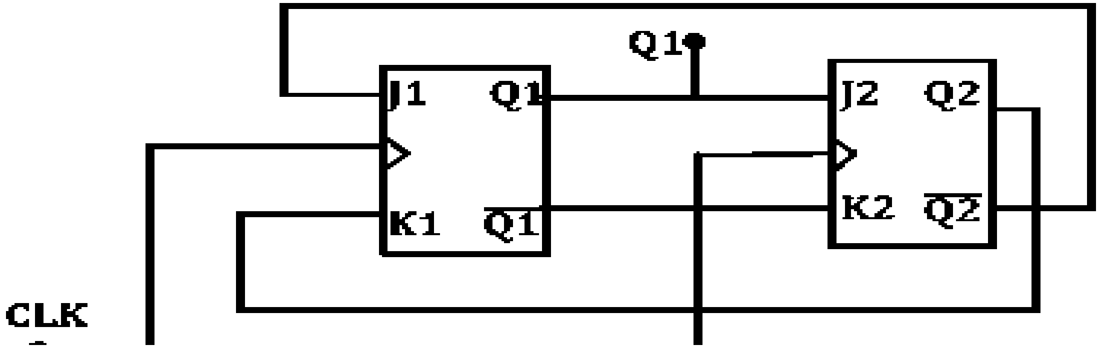
(C)


(D)


91. How many pulses are needed to change the contents of a 8-bit upcounter from 10101100 to 00100111 (rightmost bit is the LSB)?

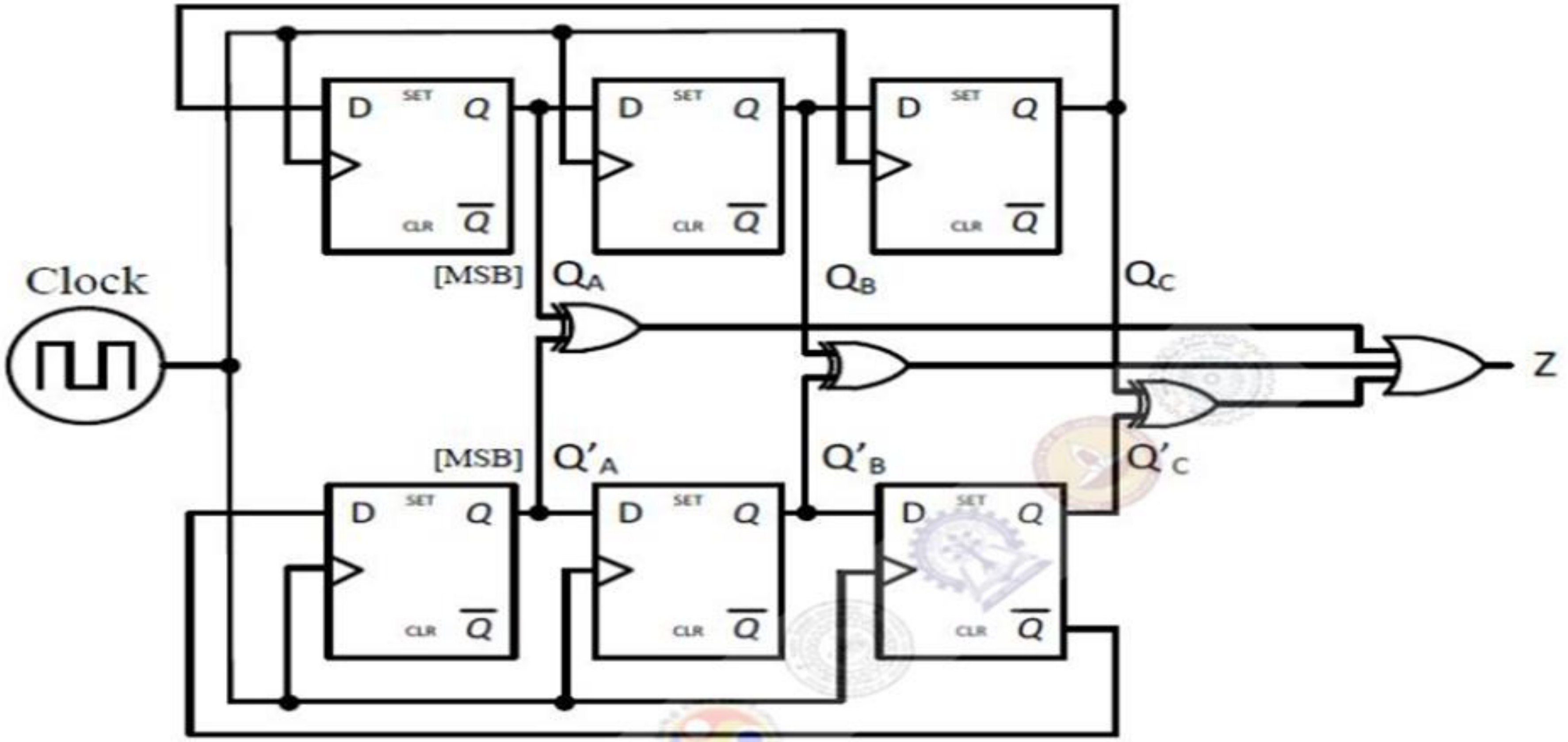
(a) 134 (b) 133 (c) 124 (d) 123

92. **The outputs of the two flip-flops Q_1, Q_2 in the figure shown are initialized to 0, 0. The sequence generated at Q_1 upon application of clock signal is** **GATE (EC - 2014)**



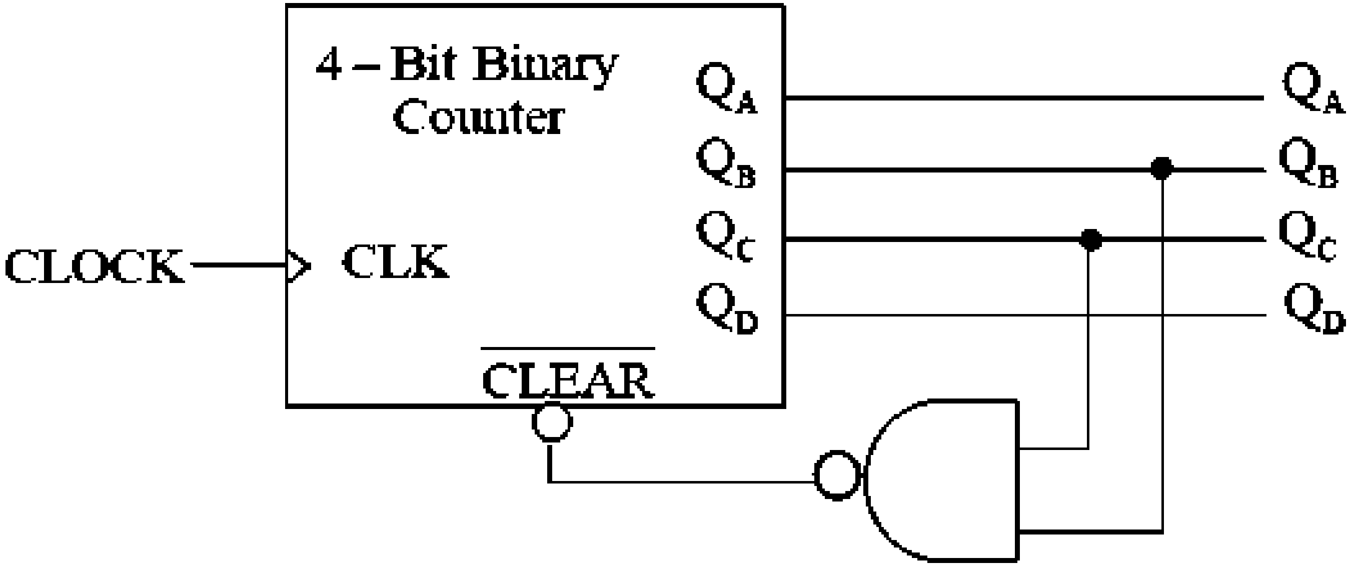
(a) 01110 ... (b) 01010 ... (c) 00110 ... (d) 01100 ...

93 For the synchronous sequential circuit shown below, the output Z is zero for the initial conditions $Q_A Q_B Q_C = Q'_A Q'_B Q'_C = 100$.



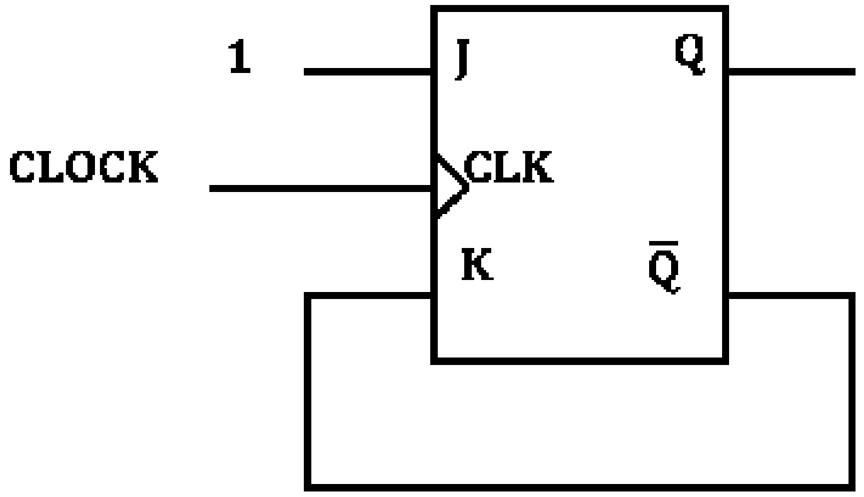
The minimum number of clock cycles after which the output Z would again become zero is

94. A mod-n counter using a synchronous binary up-counter with synchronous clear input is shown in the figure. The value of n is _____



GATE (EC-2015)

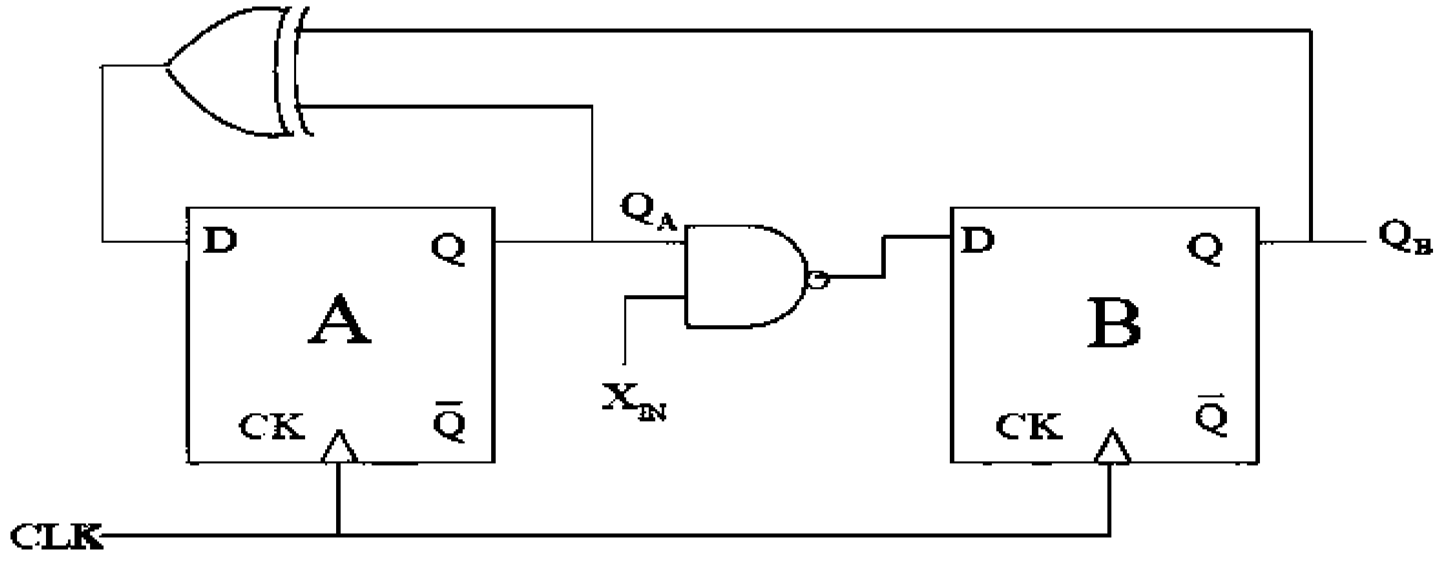
95. **In the figure shown, the initial state of Q is 0. The output is observed after the application of each clock pulse. The output sequence at Q is**



GATE (IN-2009)

(a) 0 0 0 0 ... (b) 1 0 1 0 ... (c) 1 1 1 1 ... (d) 1 0 0 0 ....

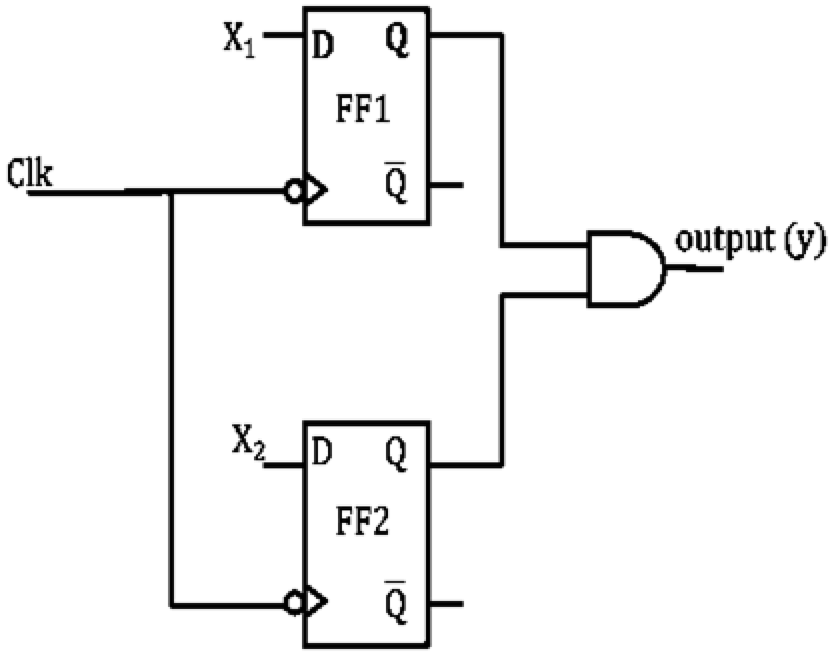
96. A finite state machine (FSM) is implemented using the D flip-flops A and B, and logic gates, as shown in the figure below. The four possible states of the FSM are $Q_A Q_B = 00$, 01, 10 & 11

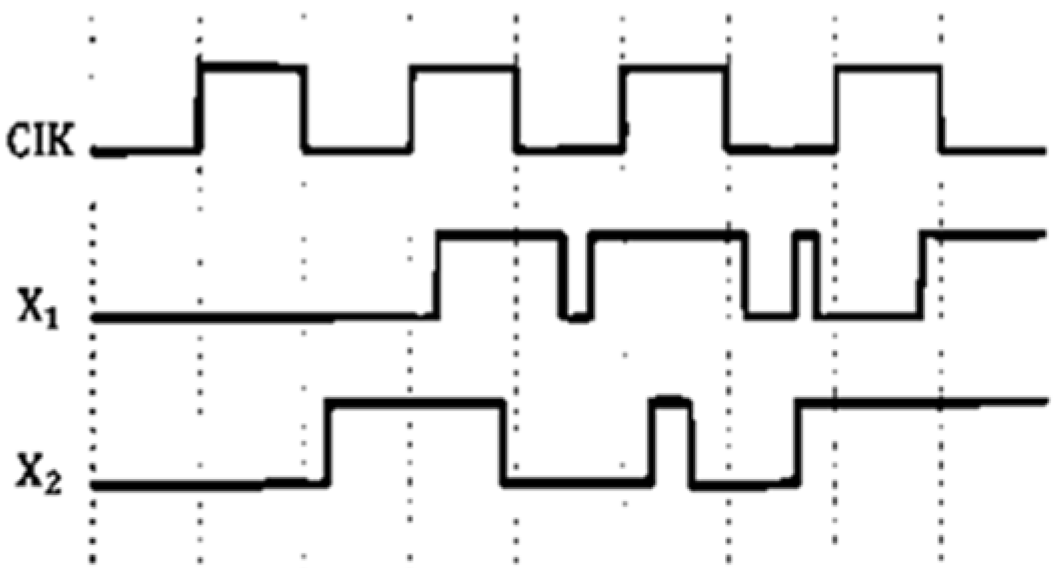


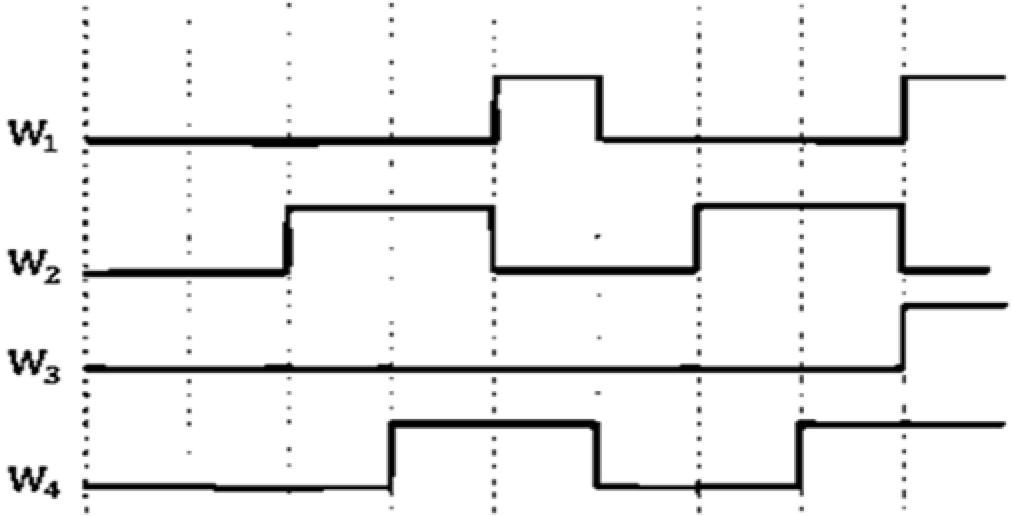
Assume that X_{IN} is held at a constant logic level throughout the operation of the FSM. When the FSM is initialized to the state $Q_A Q_B = 00$ and clocked, after a few clock cycles, it starts cycling through

(a) all of the four possible states if $X_{IN} = 1$
(b) three of the four possible states if $X_{IN} = 0$
(c) only two of the four possible states if $X_{IN} = 1$
(d) only two of the four possible states if $X_{IN} = 0$

97. **In the circuit shown, choose the correct timing diagram of the output (y) from the given waveforms W_1, W_2, W_3 and W_4.** **GATE (ECE-2014)**

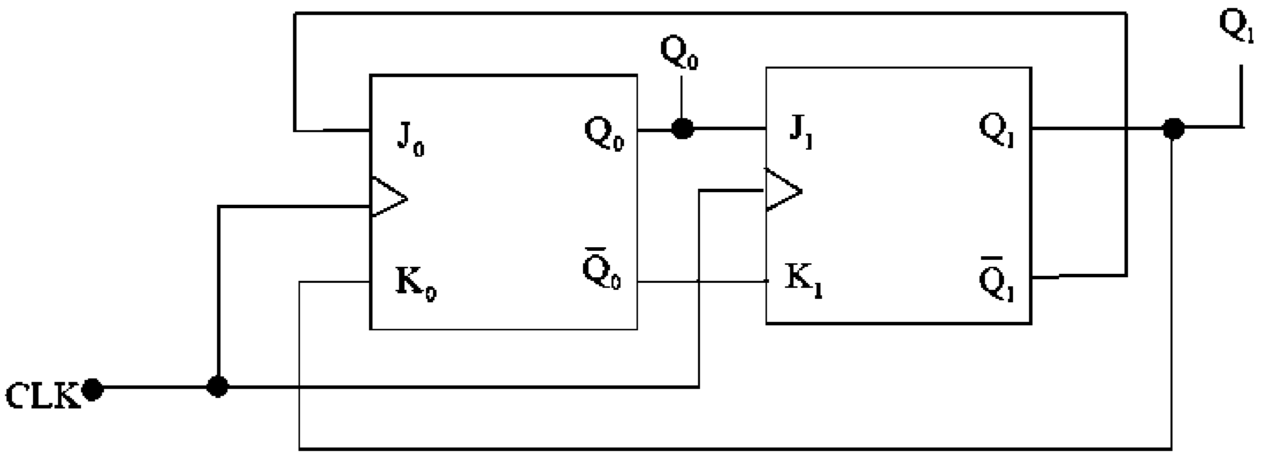






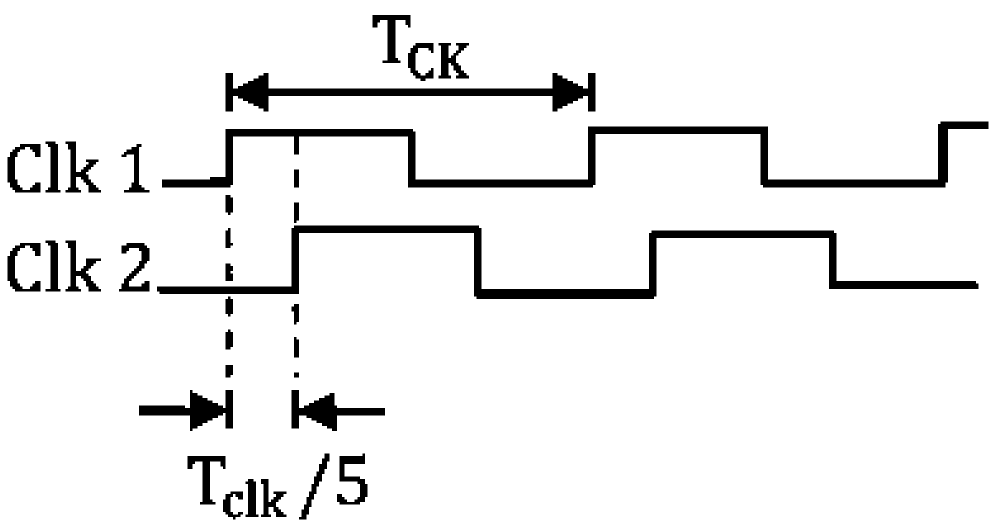
(A) W_1 (B) W_2 (C) W_3 (D) W_4

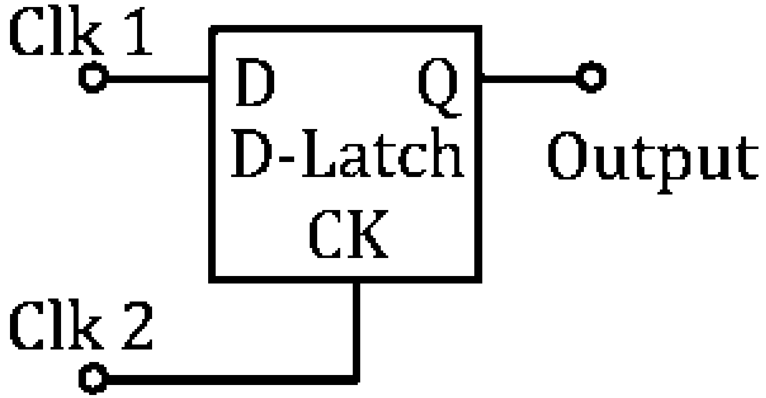
98. **In the following sequential circuit, the initial state (before the first clock pulse) of the circuit is $Q_1Q_0 = 00$. The state (Q_1Q_0), immediately after the 333rd clock pulse is** **GATE (EE-2015)**



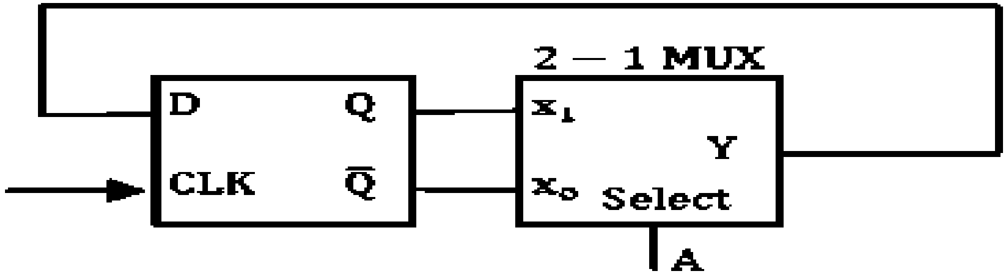
(a) 00 (b) 01 (c) 10 (d) 11

99. Consider the D-Latch shown in the figure, which is transparent when its clock input CK is high and has zero propagation delay. In the figure, the clock signal CLK1 has a 50% duty cycle and CLK2 is a one-fifth period delayed version of CLK1. The duty cycle at the output of the latch in percentage is________ GATE (ECE-2017)



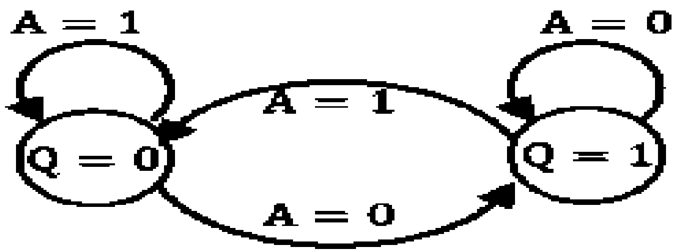


100. The state transition diagram for the logic circuit shown is

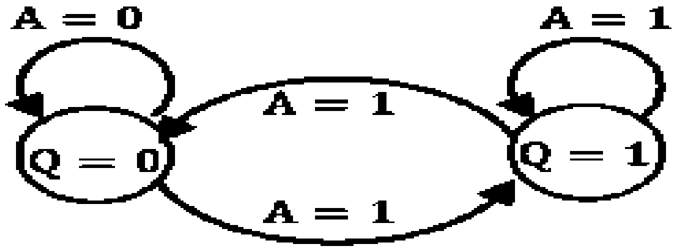


GATE (IN/EC-2013)

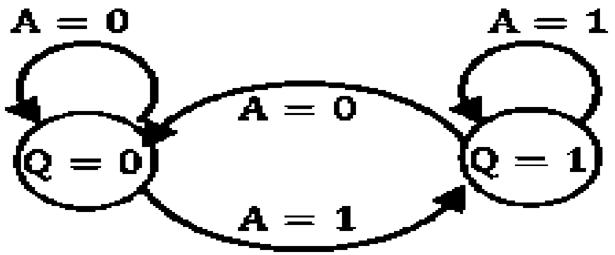
(A)



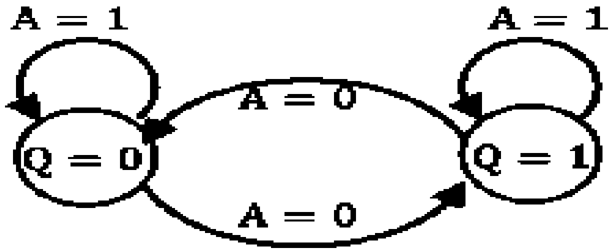
(B)



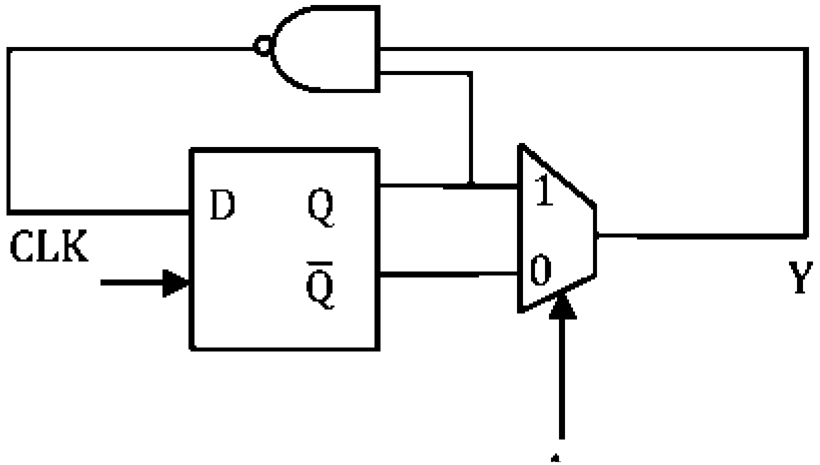
(C)



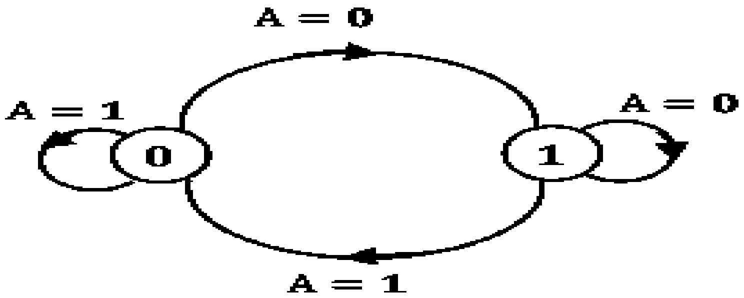
(D)



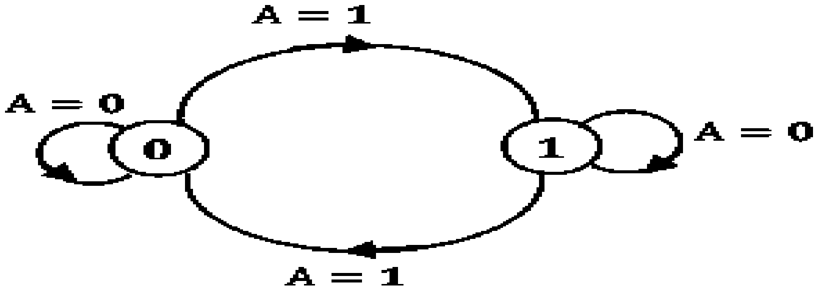
101. For the given circuit state diagram is ________. When A=1 or 0 [0 and 1 are the output states] GATE (CS-2019)



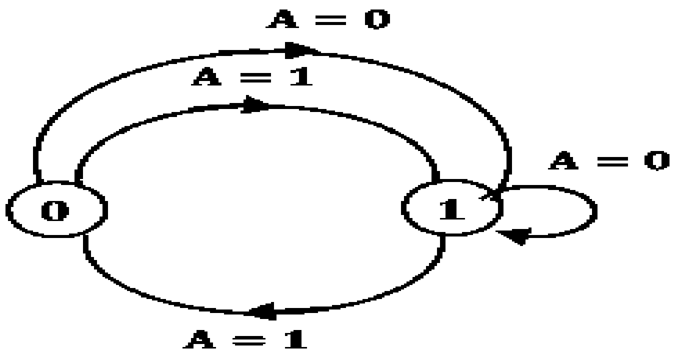
(A)



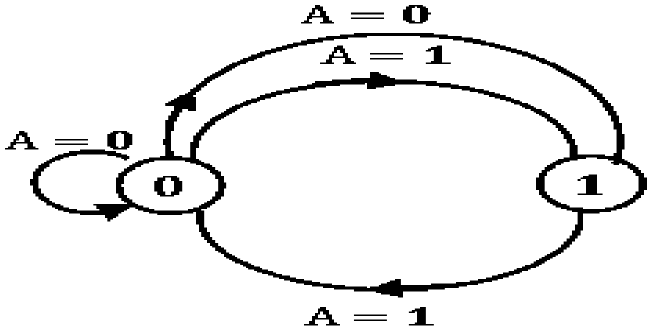
(B)



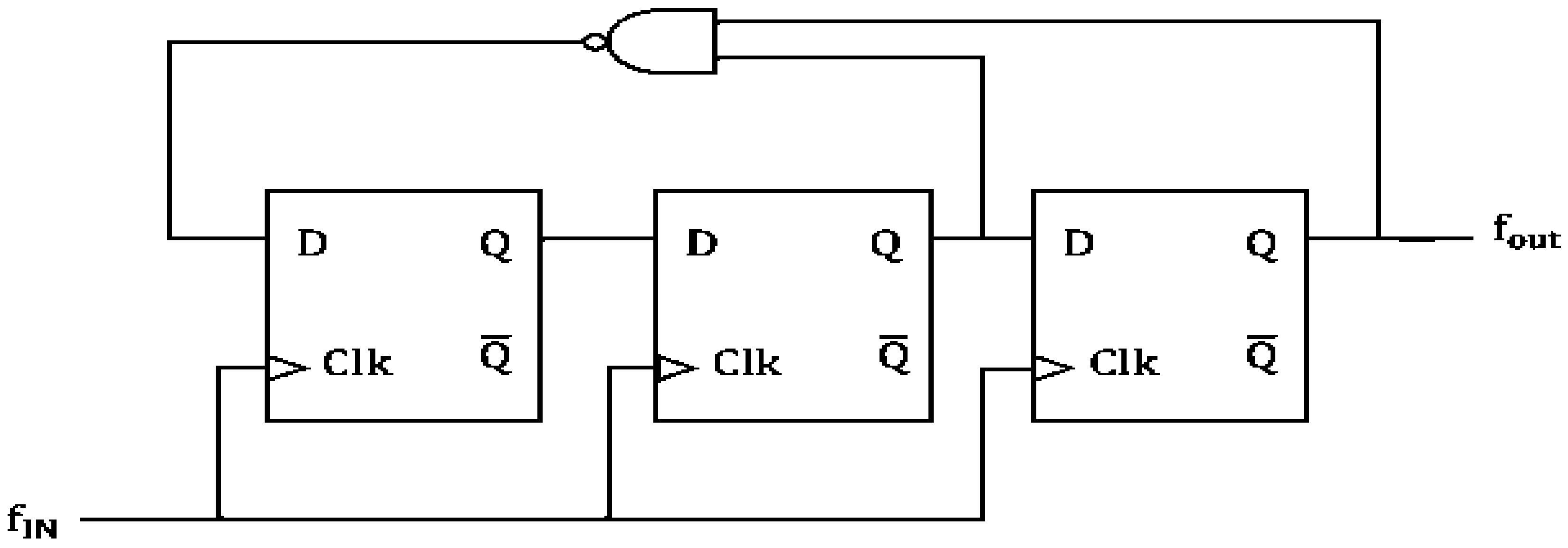
(C)



(D)



102. What one of the following statement is true about the digital circuit shown in the figure? GATE (EC-2018)



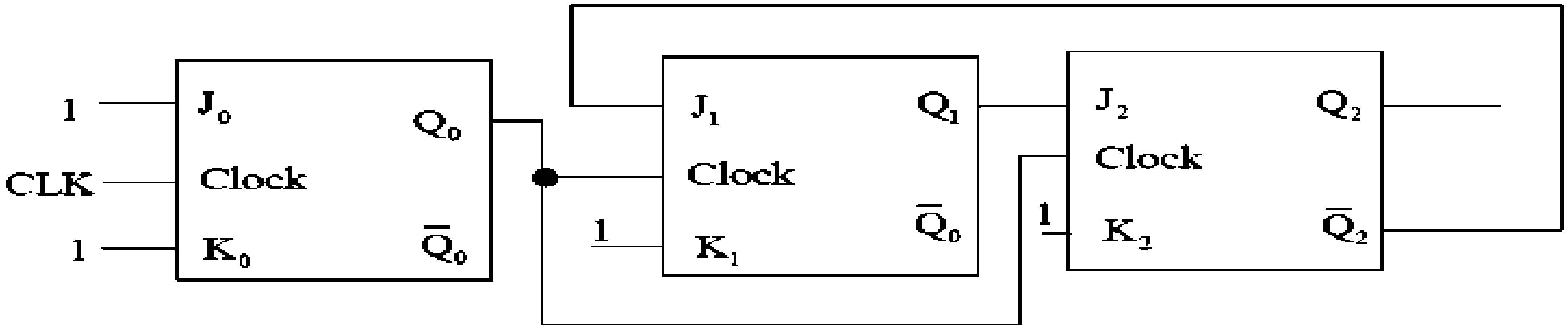
(A) It can be used for dividing the input frequency by 3

(B) It can be used for dividing the input frequency by 5

(C) It can be used for dividing the input frequency by 7

(D) It cannot be reliably used as a frequency divider due to disjoint internal cycles

103. The figure shown a digital circuit constructed using negative edge triggered J-K flip flops. Assume a starting state of $Q_2Q_1Q_0$ = 000. This state $Q_2Q_1Q_0$ = 000 will repeat after ____________ number of cycles of the clock CLK. **GATE (EE-2015)**



104. We want to design a synchronous counter that counts the sequence 0–1–0–2–0–3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is ______. **GATE (CS-2016)**

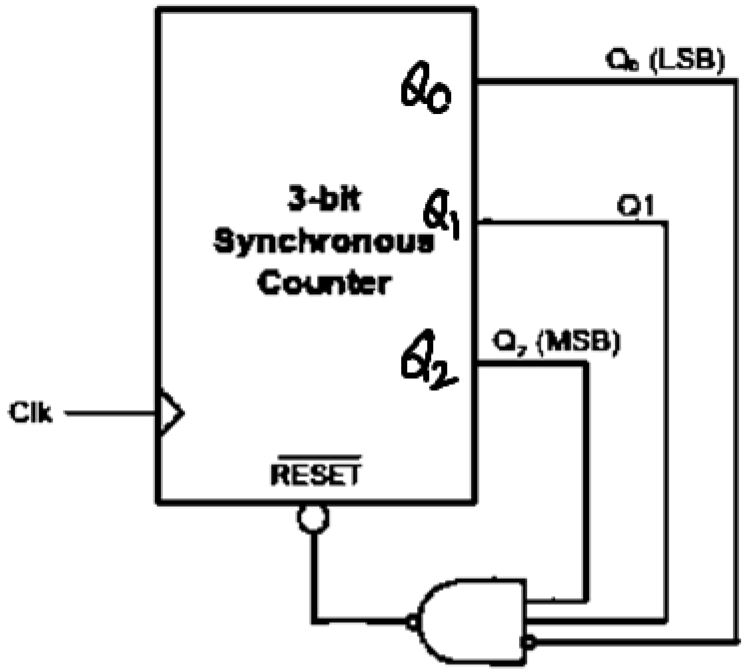
105. For the circuit shown in the figure, the delay of the bubbled NAND gate is 2 ns and that of the counter is assumed to be zero. If the clock (Clk) frequency is 1 GHz, then the counter behaves as a **GATE (ECE-2016)**

(a) mod-5 counter

(b) mod-6 counter

(c) mod-7 counter

(d) mod-8 counter



106. Master-Slave flip-flop is also called

(a) Pulse triggered flip-flop

(b) Latch

(c) Level triggered flip-flop

(d) Buffer

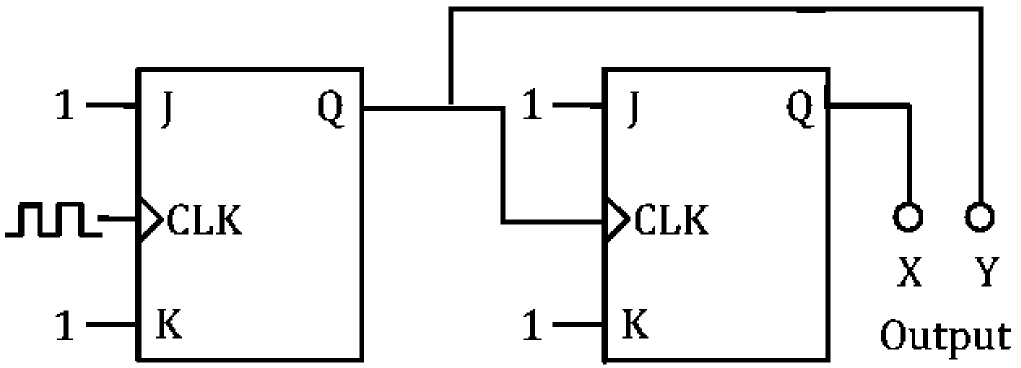
107. The circuit shown in the figure below uses ideal positive edge-triggered synchronous J-K flip flops with outputs X and Y. If the initial state of the output is $X = 0$ *and* $Y = 0$ just before the arrival of the first clock pulse, the state of the output just before the arrival of the second clock pulse is **GATE (IN-2019)**

(a) X = 0, Y = 0

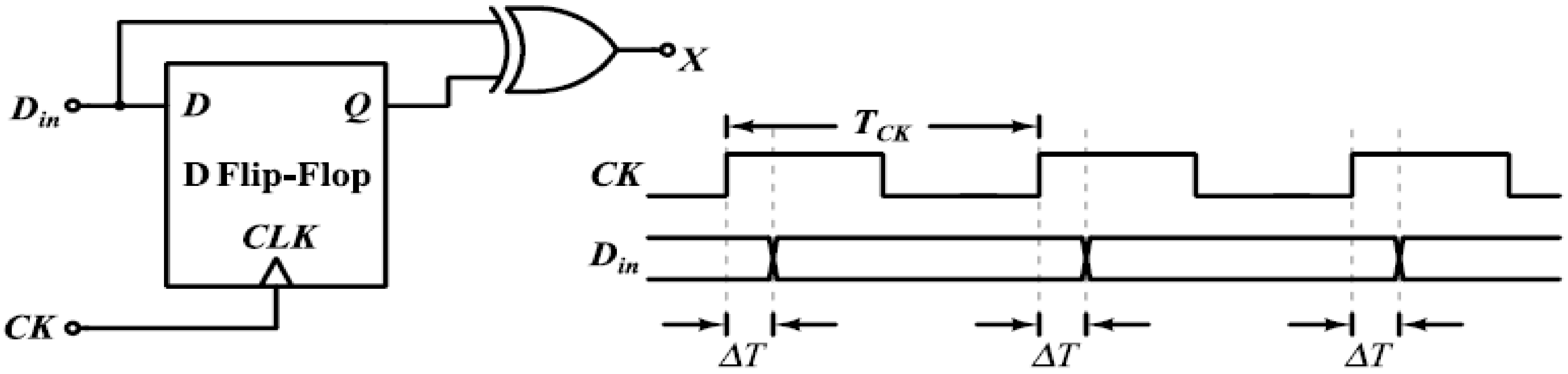
(b) X = 0, Y = 1

(c) X = 1, Y = 0

(d) X = 1, Y = 1



108. In the circuit shown below, a positive edge-triggered D Flip-Flop is used for sampling input data D_{in} using clock CK. The XOR gate outputs 3.3 volts for logic HIGH and 0 volts for logic LOW levels. The data bit and clock periods are equal and the value of $\Delta T / T_{CK} = 0.15$, where the parameters ΔT and T_{CK} are shown in the figure. Assume that the Flip-Flop and the XOR gate are ideal.



If the probability of input data bit (D_{in}) transition in each clock period is 0.3, the average value (in volts, accurate to two decimal places) of the voltage at node X, is ________.

109. Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is **GATE (2015)**

(a) 0, 1, 3, 7, 15, 14, 12, 8, 0

(b) 0, 1, 3, 5, 7, 9, 11, 13, 15, 0

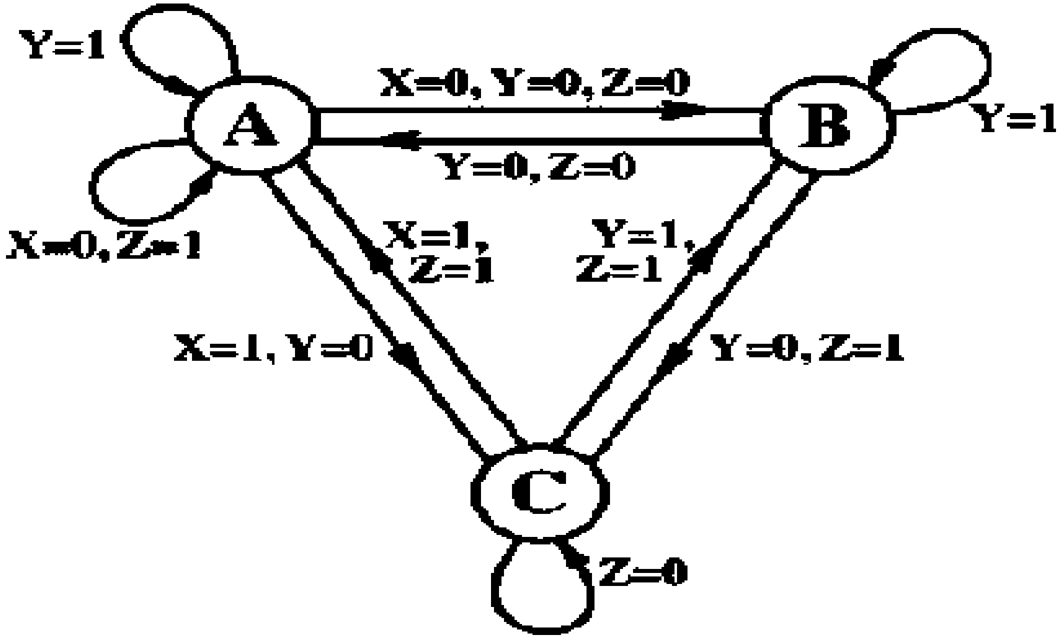
(c) 0, 2, 4, 6, 8, 10, 12, 14, 0

(d) 0, 8, 12, 14, 15, 7, 3, 1, 0

110. The minimum number of D flip-flops needed to design a mod-258 counter is **GATE (CS-2011)**

(a) 9 (b) 8 (c) 512 (d) 258

111. The state transition diagram for a finite state machine with states A, B and C, and binary inputs X,Y and Z is shown in the figure.



Which one of the following statements is correct GATE (ECE-2016)

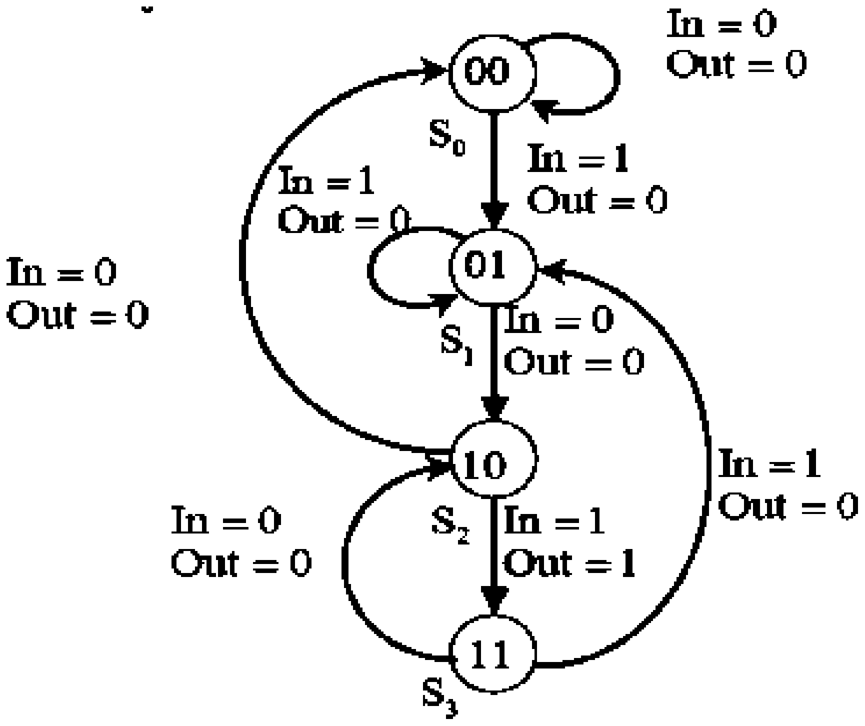
(a) Transitions from State A are ambiguously

(b) Transitions from State B are ambiguously

(c) Transitions from State C are ambiguously

(d) All of the state transitions are defined unambiguously.

112. The state diagram of a finite state machine (FSM) designed to detect an overlapping sequence of three bits is shown in the figure. The FSM has an input 'In' and an output 'Out'. The initial state of the FSM is S_0.

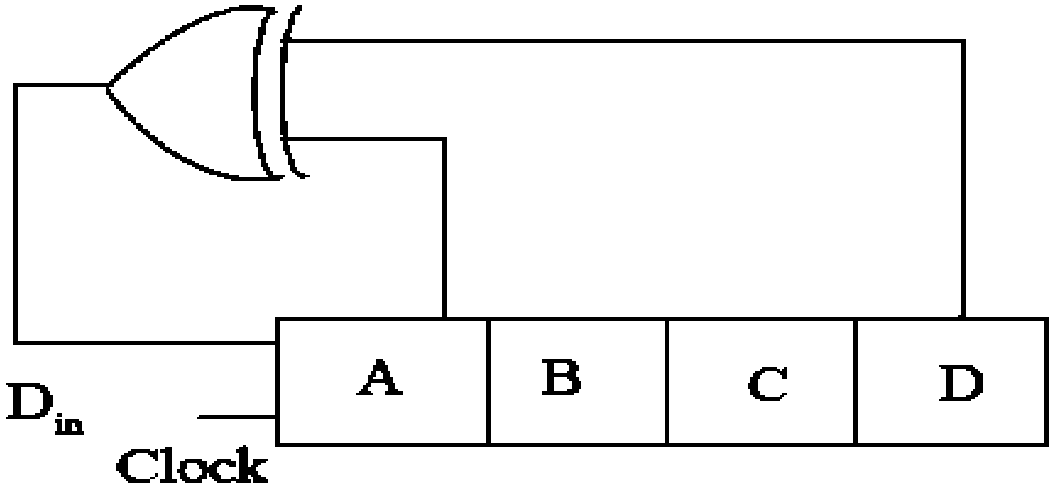


If the input sequence is 10101101001101, starting with the left-most bit, then the number times 'Out' will be 1 is ________ GATE (EC-2017)

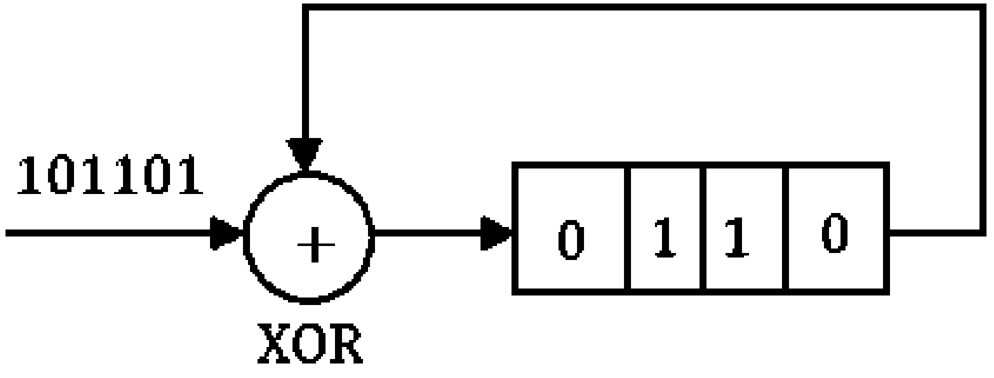
113. A traffic signal cycles from GREEN to YELLOW, YELLOW to RED and RED to GREEN. In each cycle. GREEN is turned on for 70 seconds. YELLOW is turned on for 5 seconds and the RED is turned on far 75 seconds. This traffic light has to be implemented using a finite state machine (FSM). The only input to this. FSM is a clock of 5 second period. The minimum number of flip-flops required TO implement this FSM is ________.

GATE (EC-2018)

114. A 4-bit shift register circuit configured for right-shift operation, i.e, $D_{in} \to A \to B \to C \to D$ is shown. If the present state of the shift register is ABCD = 1101, the number of clock cycles required to reach the state ABCD = 1111 is ________. **GATE (EC-2017)**



115. **What is the final value stored in the linear feedback shift register if the input is 101101?** GATE (CS-2007)

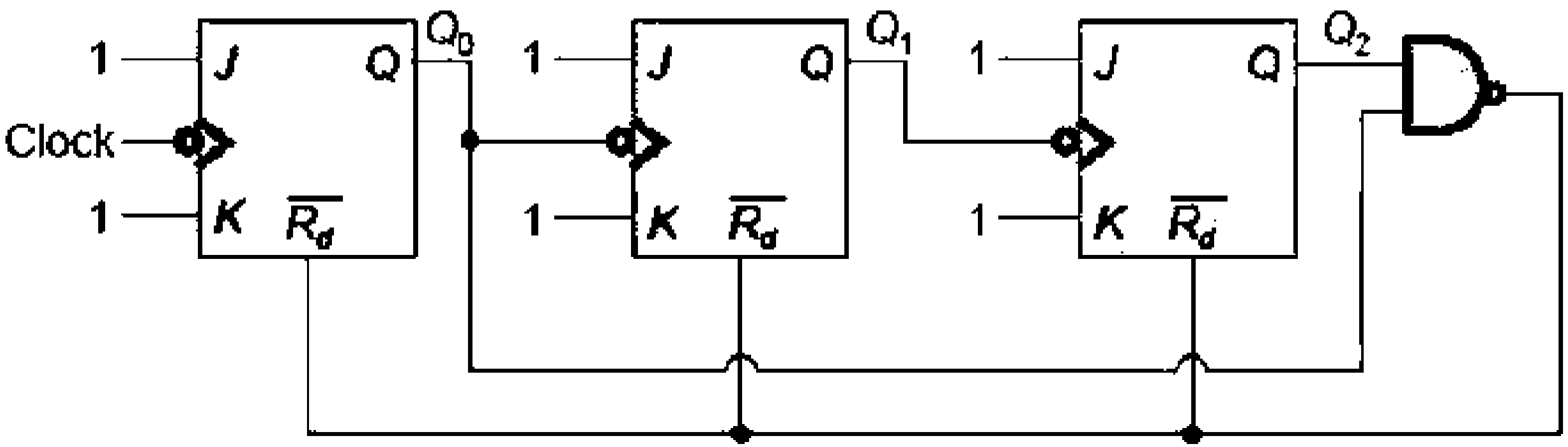


(a) 0110 (b) 1011 (c) 1101 (d) 1111

116. The initial content of a four-bit shift register is 1000. What is the register content after it is shifted four times to the right, with the serial input being 111100? **ESE (2016)**

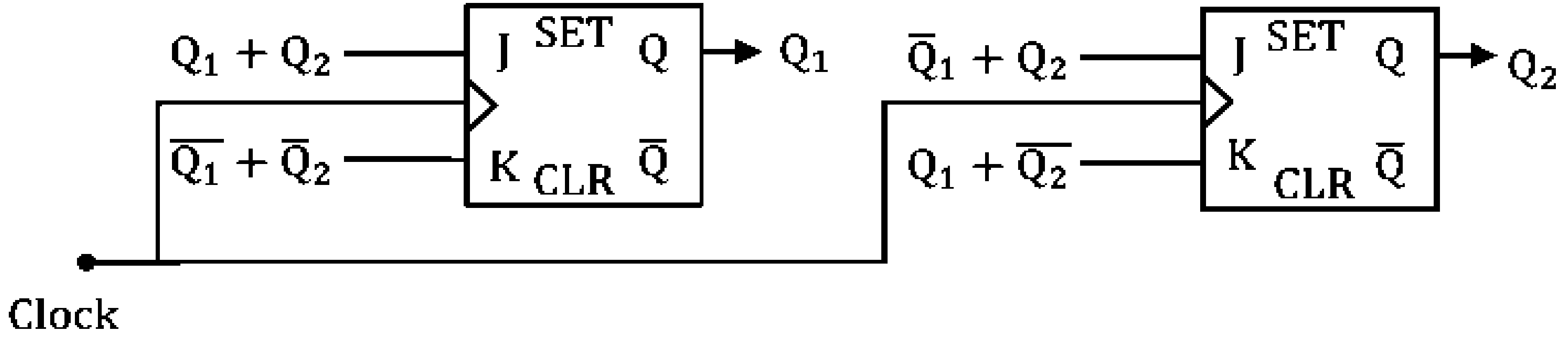
(a) 1111 (b) 1100 (c) 1000 (d) 0011

117. The circuit shown consists of JK flip flops, each with an active LOW asynchronous reset. The counter corresponding to this circuit is ________ GATE (ECE-2015)



(A) Mod – 5 binary UP counter
(B) Mod – 6 binary DOWN counter
(C) Mod – 5 binary DOWN counter
(D) Mod – 6 binary UP counter

118. A 2-bit synchronous counter using two J-K flip flops is shown. The expressions for the inputs to the J-K flip flops are also shown in the figure. The output sequence of the counter starting from $Q_1Q_2 = 00$ is



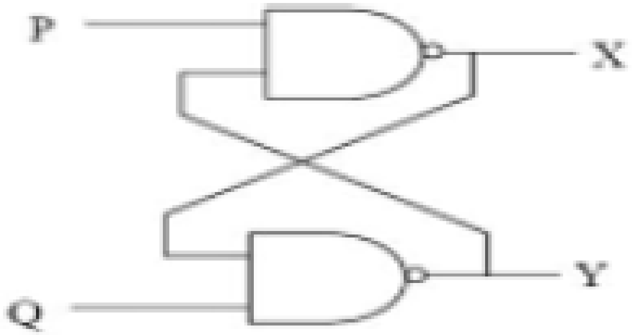
(A) $00 \rightarrow 11 \rightarrow 10 \rightarrow 01 \rightarrow 00 \ldots \ldots$

(B) $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00 \ldots \ldots$

(C) $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00 \ldots \ldots$

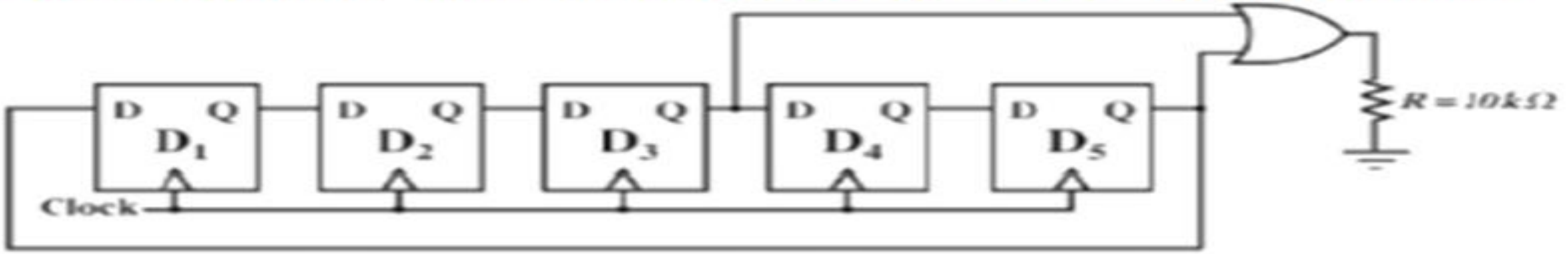
(D) $00 \rightarrow 10 \rightarrow 11 \rightarrow 01 \rightarrow 00 \ldots \ldots$

119. In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delays. The present input condition is: P = Q = '0'. If the input condition is changed simultaneously to P = Q = '1', the outputs X and Y are



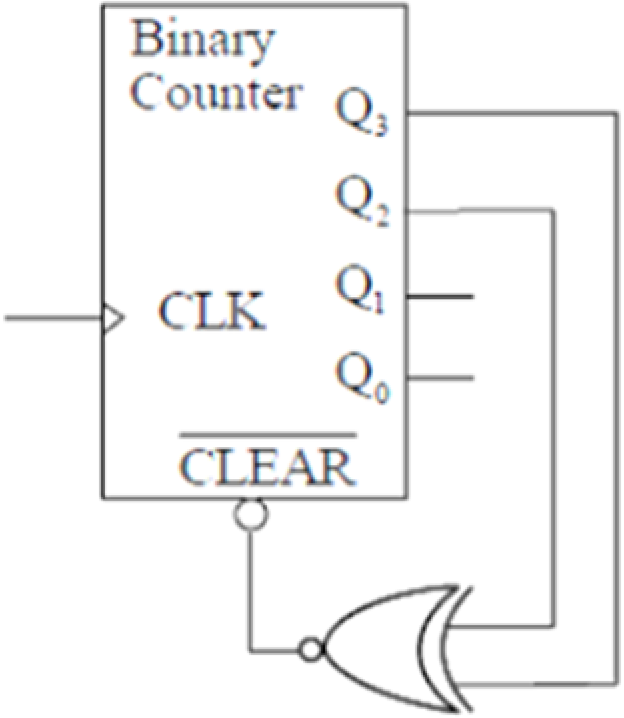
A. X = '1', Y = '1'
B. either X = '1', Y = '0' or X = '0', Y = '1'
C. either X = '1', Y = '1' or X = '0', Y = '0'
D. X = '0', Y = '0'

120. Assume that all the digital gates in the circuit shown in the figure are ideal, the resistor $R = 10k\Omega$ and the supply voltage is 5V. The D flip-flops D_1, D_2, D_3, D_4 and D_5, are initialized with logic values 0,1,0,1 and 0, respectively. The clock has a 30% duty cycle.



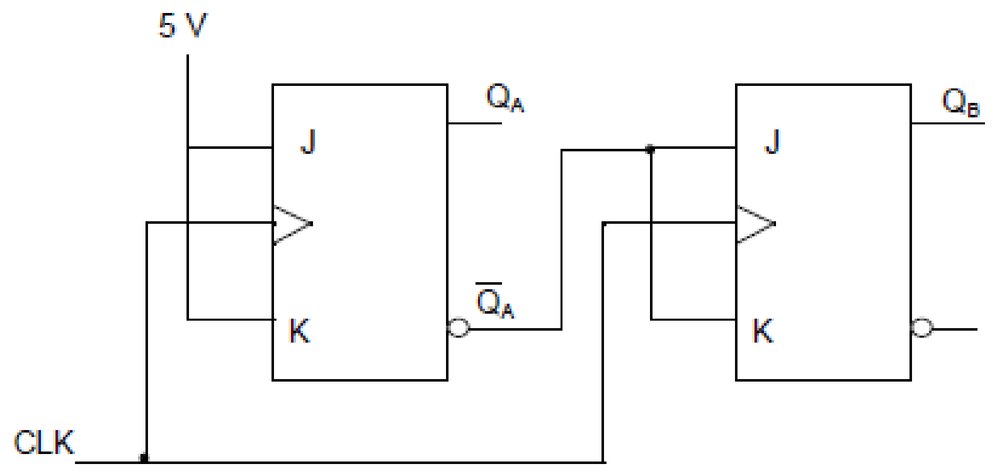
The average power dissipated (in mW) in the resistor R is______

121. The figure shows a binary counter with synchronous clear input. With the decoding log shown, the counter works as a



A. mod-2 counter B. mod-4 counter C. mod-5 counter D. mod-6 counter

122. The current state Q_A Q_B of a two JK flip-flop system is 00. Assume that the clock rise-time is much smaller than the delay of the JK flip-flop. The next state of the system is



(A) 00 (B) 01 (C) 11 (D) 10

123. Which one of the following statements best describes the operation of a negative-edge-triggered D flip-flop ?

(a) The logic level at the D input is transferred to Q on NGT of CLK.

(b) The Q output is always identical to the CLK input if the D input is high.

(c) The Q output is always identical to the D input when CLK = PGT.

(d) The Q output is always identical to the D input.

124. A flip-flop is a

(a) Combinational logic circuit and edge sensitive

(b) Sequential logic circuit and edge sensitive

(c) Combinational logic circuit and level sensitive

(d) Sequential logic circuit and level sensitive

125. *Statement (I) :*

A basic memory unit of a flip-flop is a bistable multivibrator.

Statement (II) :

A flip-flop has two stable states. It remains in one state until it is directed by an input signal to switch over.

126. If the input to a T flip-flop is a 100 MHz signal, the final output of three T flip-flops in a cascade is

(a) 1000 MHz

(b) 520 MHz

(c) 333 MHz

(d) 12·5 MHz

127. In a 4-stage ripple counter, the propagation delay of a flip-flop is 30 ns. If the pulse width of the strobe is 30 ns, the maximum frequency at which the counter operates reliably is nearly

(a) 9·7 MHz

(b) 8·4 MHz

(c) 6·7 MHz

(d) 4·4 MHz

128. For what minimum value of propagation delay in each flip-flop will a 10-bit ripple counter skip a count, when it is clocked at 10 MHz?

(a) 5 ns

(b) 10 ns

(c) 20 ns

(d) 40 ns

129. A cascaded arrangement of flip-flops, where the output of one flip-flop drives the clock input of the following flip-flop, is known as

(a) synchronous counter

(b) ripple counter

(c) ring counter

(d) up counter

130. The number of flip-flops required to construct an 8-bit shift register will be

(a) 32

(b) 16

(c) 8

(d) 4

131. Consider the following statements :

1. Race-around condition occurs in a JK flip-flop when the inputs are 1, 1
2. A flip-flop is used to store one bit of information
3. A transparent latch consists of D-type flip-flops
4. Master-slave configuration is used in a flip-flop to store two bits of information

Which of the above statements are correct ?

(a) 1, 2 and 3 only

(b) 1, 2 and 4 only

(c) 3 and 4 only

(d) 1, 2, 3 and 4

132. Consider the following circuits :

1. Full adder
2. Half adder
3. JK flip-flop
4. Counter

Which of the above circuits are classified as sequential logic circuits ?

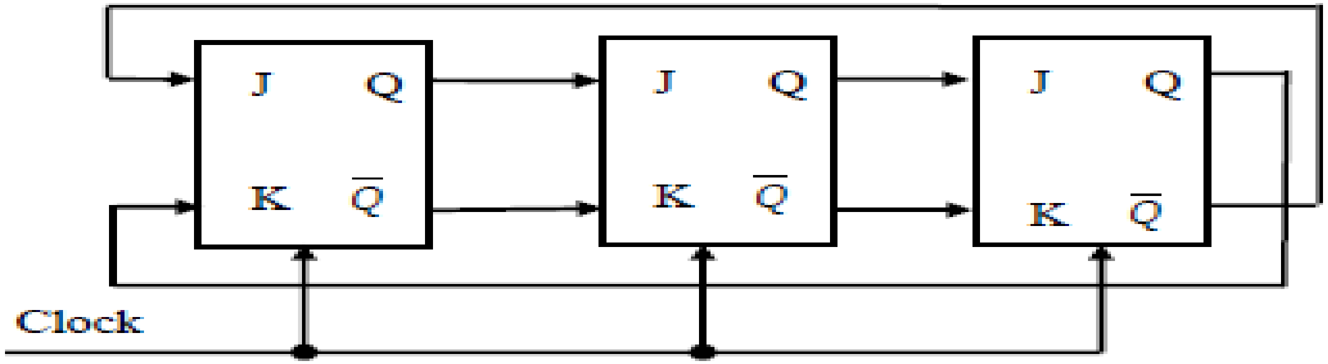
(a) 1 and 2

(b) 3 and 4

(c) 2 and 3

(d) 1 and 4

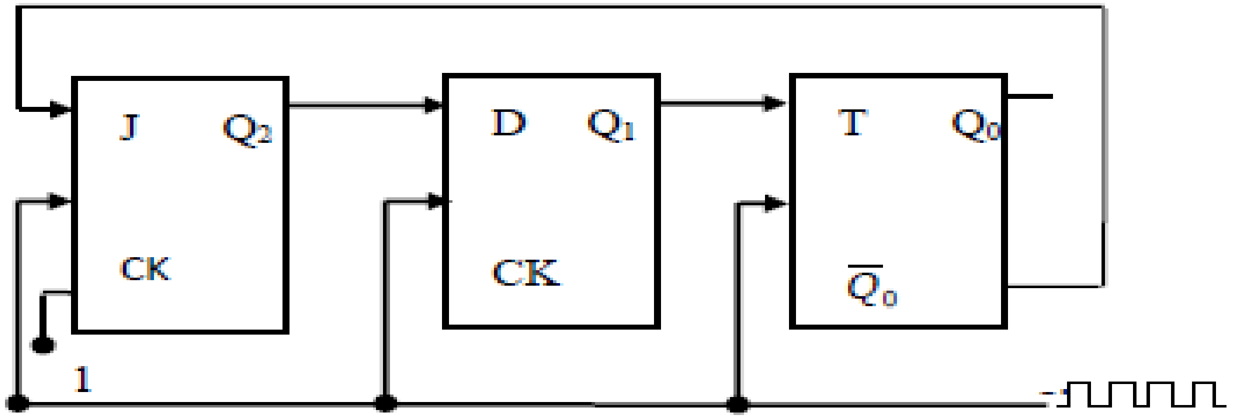
133. For the initial state of 000, the function performed by the arrangement of the J-K flip-flops in figure is:



(A) Shift Register (B) Mod-3 Counter
(C) Mod-6 Counter (D) Mod-2 Counter
(E) None of the above

[1993, 2 Marks]

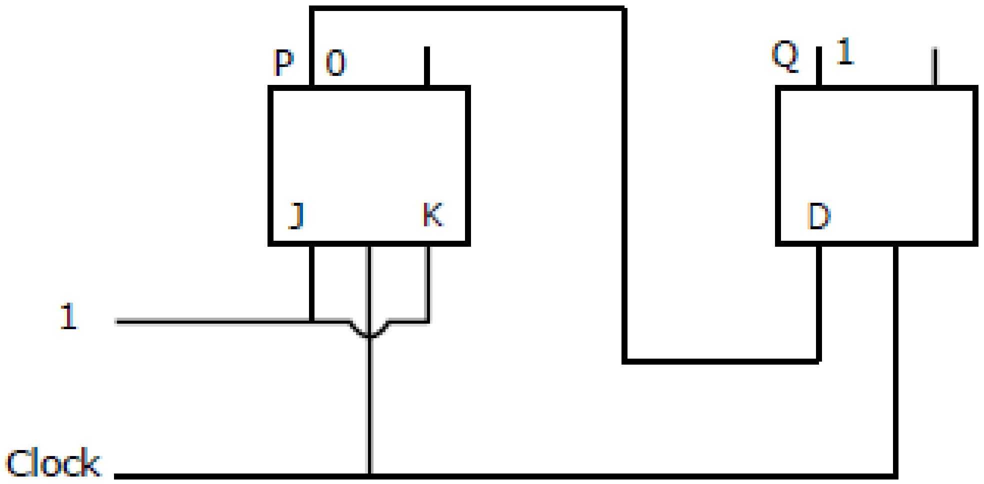
134. Find the contents of the flip-flop Q_2, Q_1 and Q_0 in the circuit of figure, after giving four clock pulses to the clock terminal. Assume $Q_2Q_1Q_0 = 000$ initially.



(Four clock pulses given here)

[1994, 2 Marks]

135. The following arrangement of master-slave flip flops has the initial state of P, Q as 0, 1 (respectively).



After the clock cycles the output state P, Q is (respectively),

(A) 1, 0 (B) 1, 1 (C) 0, 0 (D) 0, 1

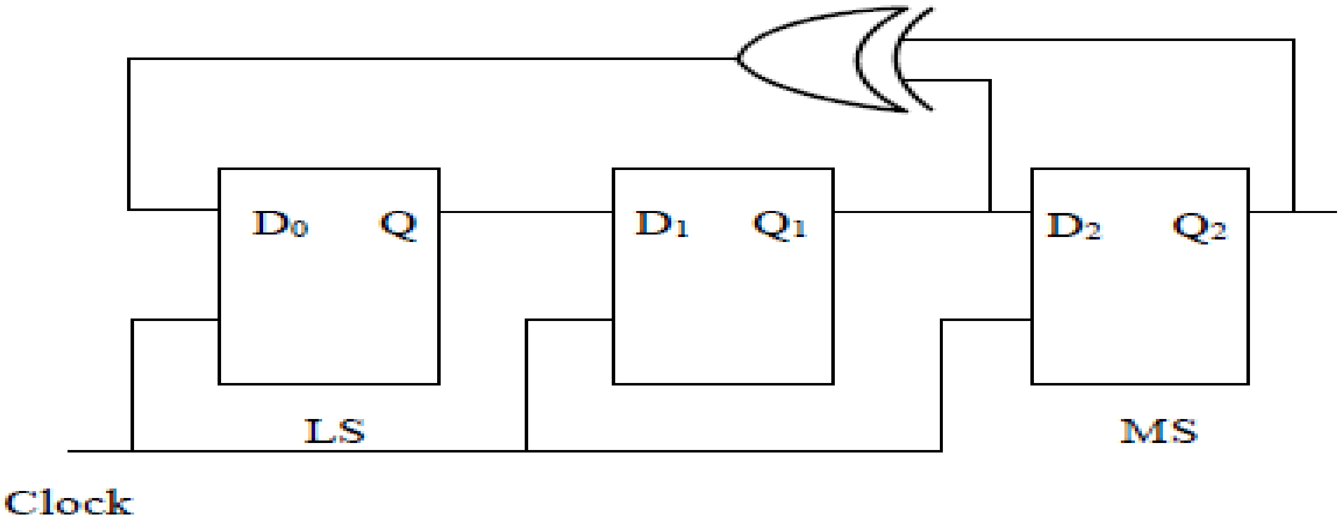
[2000, 2 Marks]

136. The number of flip-flops required to construct a binary modulo N counter is ___________

[1994, 2 Marks]

137. Consider the circuit given above with initial state $Q_0 = 1$, $Q_1 = Q_2 = 0$. The state of the circuit is given by the value $4Q_2 + 2Q_1 + Q_0$.

Which one of the following is the correct state sequence of the circuit?

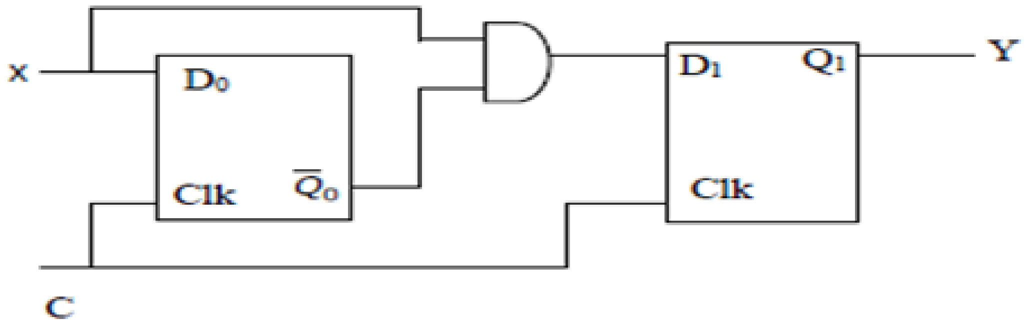


(A) 1,3,4,6,7,5,2 (B) 1,2,5,3,7,6,4

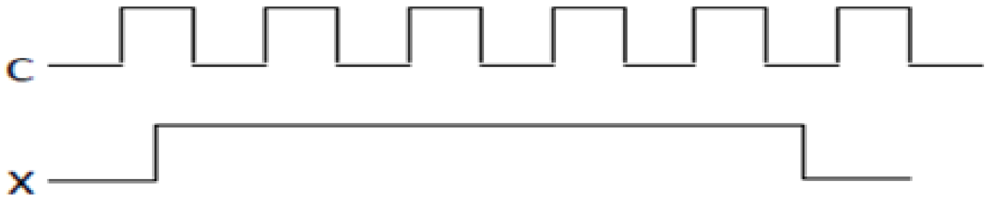
(C) 1,2,7,3,5,6,4 (D) 1,6,5,7,2,3,4

[2001, 2 Marks]

138. Consider the following circuit with initial state $Q_0 = Q_1 = 0$. The D Flip-flops are positive edged triggered and have set up times 20 nanosecond and hold times 0.



Consider the following timing diagrams of X and C; the clock period of $C \geq 40$ nanosecond. Which one is the correct plot of Y?



(A) 

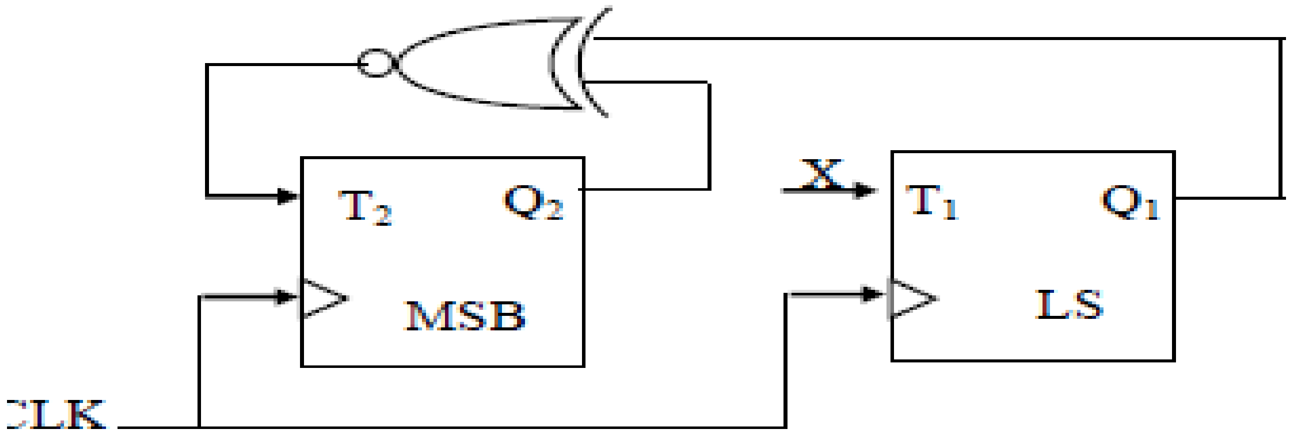
(B) 

(C) 

(D) 

[2001, 2 Marks]

139. Consider the partial implementation of a 2-bit counter using T flip-flops following the sequence 0-2-3-1-0, as shown below.



To complete the circuit, the input X should be

(A) Q_2'

(B) $Q_2 + Q_1$

(C) $(Q_1 \oplus Q_2)'$

(D) $Q_1 \oplus Q_2$

[2004, 2 Marks]

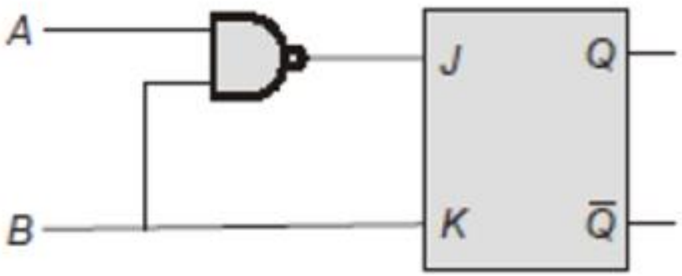
140. In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in

(A) $Q = 0,\ Q' = 1$ (B) $Q = 1, Q' = 0$

(C) $Q = 1, Q' = 1$ (D) Indeterminate states

[2004, 1 Mark]

141. An AB flip flop is constructed from a JK flip flop as shown in the figure. The expression for the next state Q_{n+1} is



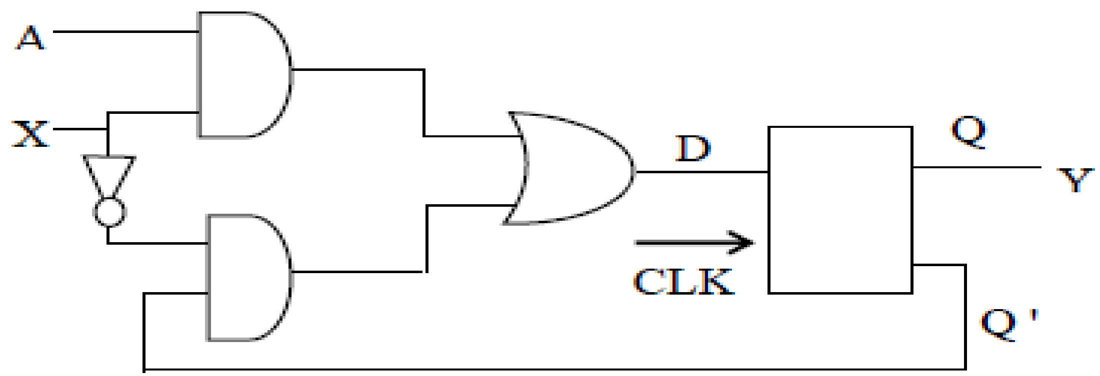
a. $AB + A\bar{B} + \bar{A}B\bar{Q}_n$

b. $\bar{A}\bar{B} + \bar{A}B + \bar{A}B\bar{Q}_n$

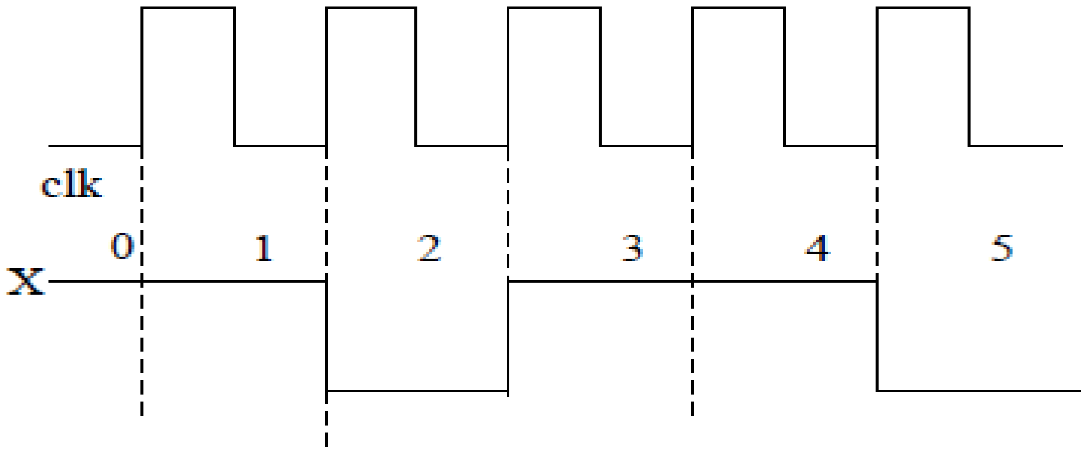
c. $\bar{B} + \bar{A}\bar{Q}_n$

d. $\bar{B} + \overline{AQ_n}$

142. Consider the following circuit involving a positive edge triggered D FF.



Consider the following timing diagram.



Let A_i represent the logic level on the line A in the i-th clock period.

Let A' represent the complement of A. The correct output sequence on Y over the clock periods 1 through 5 is:

(A) $A_0\ A_1\ A_1'\ A_3\ A_4$

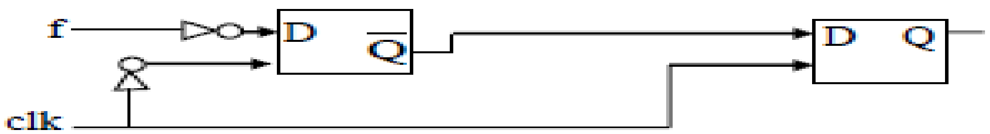
(B) $A_0\ A_1\ A_2'\ A_3\ A_4$

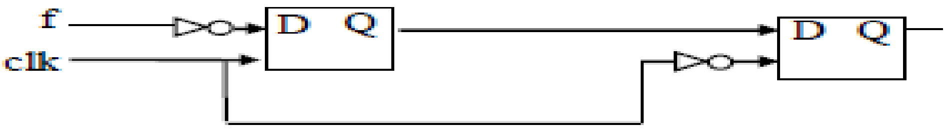
(C) $A_1\ A_2\ A_2'\ A_3\ A_4$

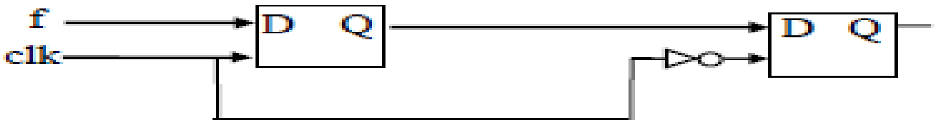
(D) $A_1\ A_2'\ A_3\ A_4\ A_5'$

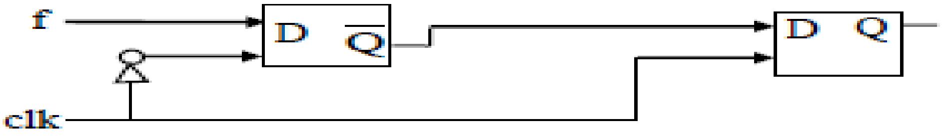
[2005, 2 Marks]

143. You are given a free running clock with a duty cycle of 50% and a digital waveform f which changes only at the negative edge of the clock. Which one of the following circuits (using clocked D flip-flops) will delay the phase of f by 180°?

(A) 

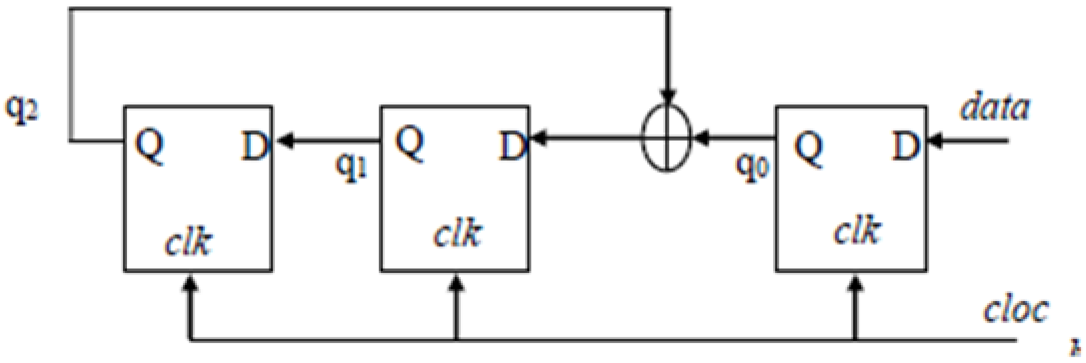
(B) 

(C) 

(D) 

[2006, 1 Mark]

144. Consider the circuit in the diagram. The ⊕ operator represents Ex-OR. The D flip-flops are initialized to zeroes (cleared).



The following data: 100110000 is supplied to the 'data' terminal in nine clock cycles. After that the values of $q_2q_1q_0$ are:

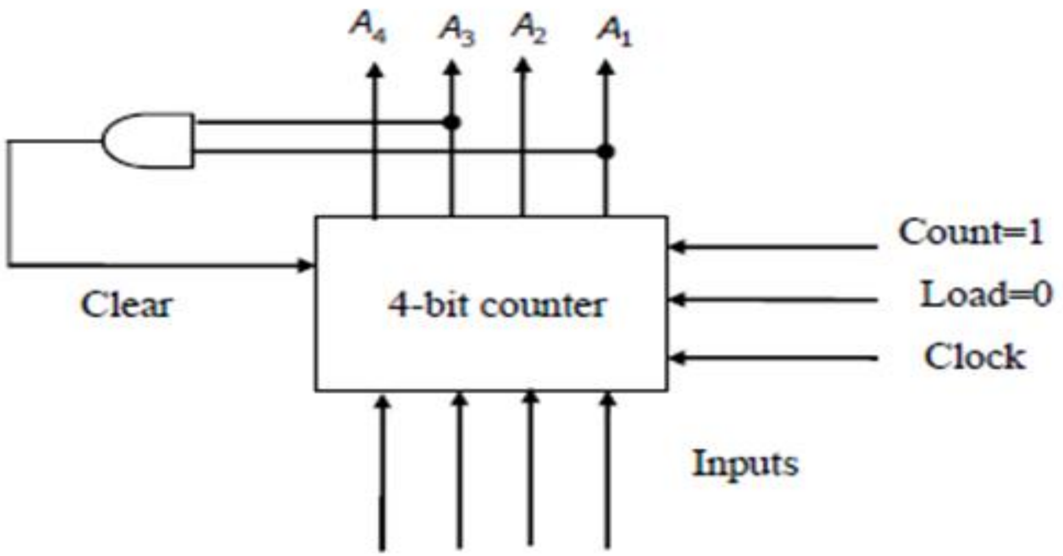
(A) 000 (B) 001 (C) 010 (D) 101

[2006, 2 Marks]

145. The control signal functions of a 4-bit binary counter are given below (where X is "don't care"):

Clear	Clock	Load	Count	Function
1	X	X	X	Clear to 0
0	X	0	0	No change
0	↑	1	X	Load input
0	↑	0	1	Count next

The counter is connected as follows:



Assume that the counter and gate delays are negligible. If the counter starts at 0, then it cycles through the following sequence:

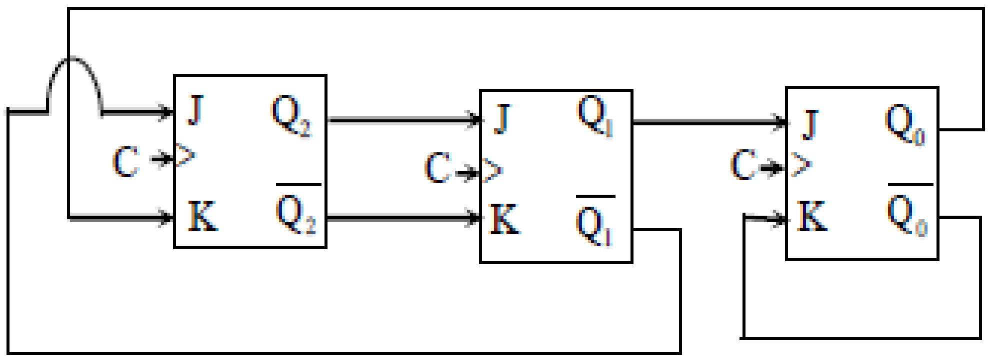
(A) 0, 3, 4
(B) 0, 3, 4, 5
(C) 0, 1, 2, 3, 4
(D) 0, 1, 2, 3, 4, 5

[2007, 2 Marks]

146. We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip-flops required to implement

[2014, 2 Marks]

147. The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is



(A) 001, 010, 011 (B) 111, 110, 101
(C) 100, 110, 111 (D) 100, 011, 001

[2014, 2 Marks]

148. Let $k = 2^n$. A circuit is built by giving the output of an n-bit binary counter as input to an n-to-2^n bit decoder. This circuit is equivalent to a

(A) k-bit binary up counter

(B) k-bit binary down counter

(C) k-bit ring counter

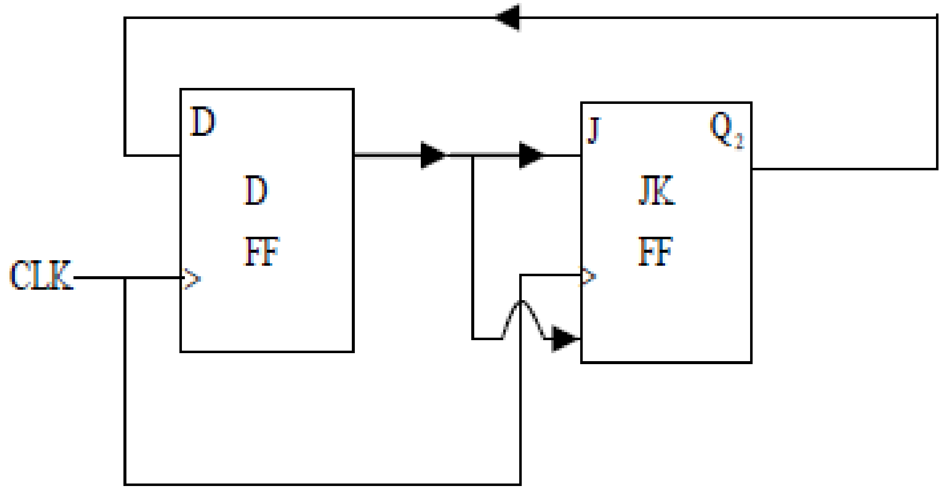
(D) k-bit Johnson counter

[2014, 1 Mark]

149. The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0,0, 1, 1, 2, 2, 3, 3. 0. 0.......) is __________.

[2015, 1 Mark]

150. A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that J = K = 1 is the toggle mode and J = K = 0 is the state-holding mode of the JK flip-flop. Both the flip-flops have non-zero propagation delays.



(A) 0110110...

(B) 0100100..

(C) 011101110...

(D) 01100110

[2015, 2Marks]

151. If the number of unused states in k -bit Johnson counter is 22, then the value of k is _______.

152. The next state table of a 2-bit saturating up-counter is given below.

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuit using T flip-flops. The expression for T_1 and T_0 are

(A) $T_1 = Q_1 Q_0$, $T_0 = \overline{Q}_1 \overline{Q}_0$

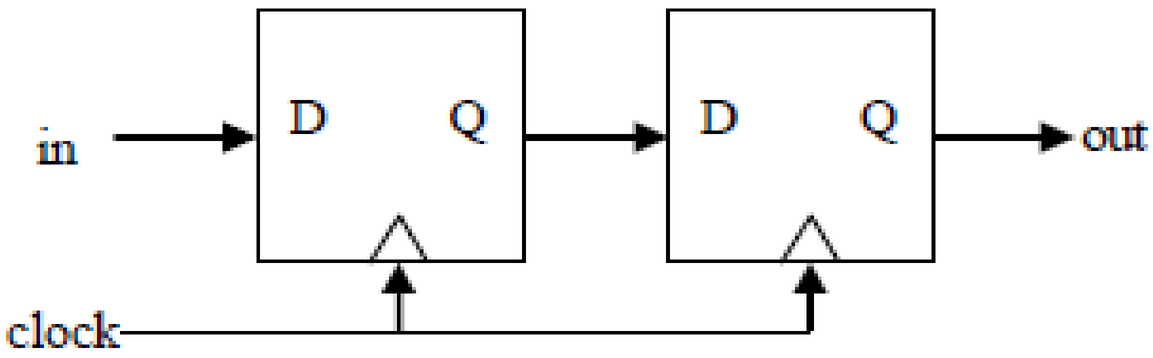
(B) $T_1 = \overline{Q}_1 Q_0$, $T_0 = \overline{Q}_1 + \overline{Q}_0$

(C) $T_1 = Q_1 + Q_0$, $T_0 = \overline{Q}_1 + \overline{Q}_0$

(D) $T_1 = Q_1 Q_0$, $T_0 = \overline{Q}_1 + \overline{Q}_0$

[2017, 2 Marks]

153. Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.

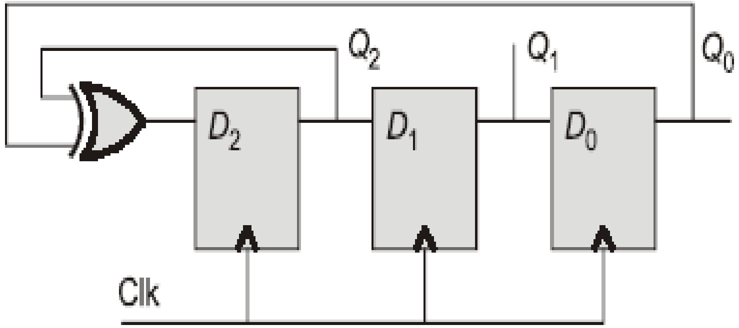


The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is

________.

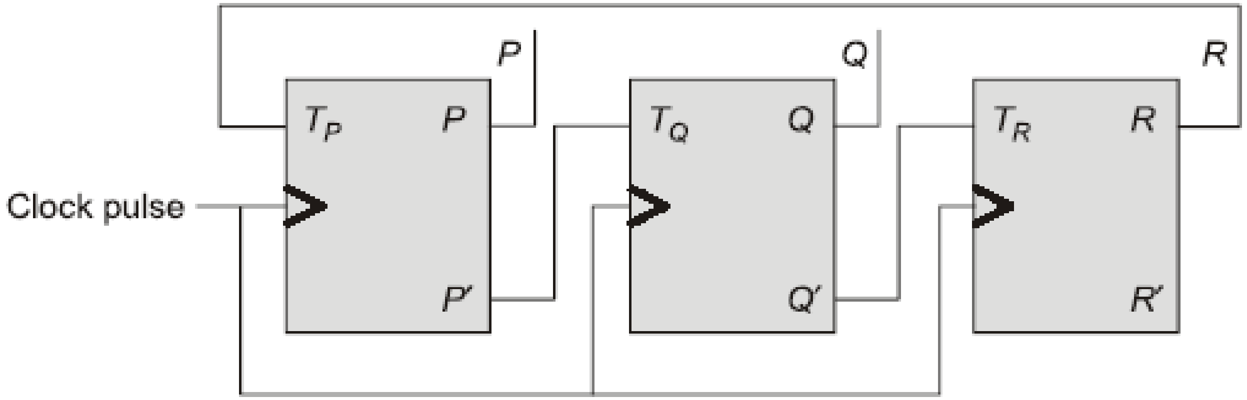
[2018, 1 Mark]

154. The propagation delay of the exclusive-OR (XOR) gate in the circuit in the figure is 3 ns. The propagation delay of all the flip-flops is assumed to be zero. The clock (Clk) frequency provided to the circuit is 500 MHz.



Starting from the initial value of the flip-flop outputs $Q_2Q_1Q_0$ = 111 with $D_2 = 1$, the minimum number of triggering clock edges after which the flip-flop outputs $Q_2Q_1Q_0$ becomes 100 (in integer) is _____.

155. Consider a 3-bit counter, designed using T flip-flops, as shown below:



Assuming the initial state of the counter given by *PQR* as 000. What are the next three states?

(a) 011, 101, 111
(b) 001, 010, 000
(c) 001, 010, 111
(d) 011, 101, 000

156. A counter is constructed with three D flip-flops. The input-output pairs are named (D_0, Q_0), (D_1, Q_1) and (D_2, Q_2), where the subscript 0 denotes the least significant bit. The output sequence is desired to be the Gray-code sequence 000, 001, 011, 010, 110, 111, 101 and 100, repeating periodically. Note that the bits are listed in the $Q_2Q_1Q_0$ format. The combinational logic expression for D_1 is

(a) $Q_2Q_1 + \bar{Q}_2\bar{Q}_1$

(b) $Q_2Q_1Q_0$

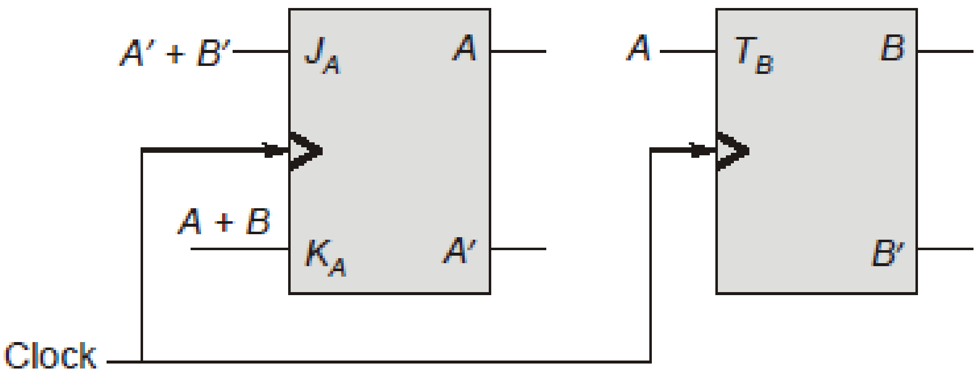
(c) $\bar{Q}_2Q_0 + Q_1\bar{Q}_0$

(d) $Q_2Q_0 + Q_1\bar{Q}_0$

157. A 16-bit synchronous binary up-counter is clocked with a frequency f_{CLK} . The two most significant bits are OR-ed together to form an output Y. Measurements show that Y is periodic, and the duration for which Y remains high in each period is 24 ms. The clock frequency f_{CLK} is ________ MHz.

(Round off to 2 decimal places.)

158. Given below is the diagram of a synchronous sequential circuit with one J-K flip-flop and one T flip-flop with their outputs denoted as A and B respectively, with $J_A = (A' + B')$, $K_A = (A + B)$ and $T_B = A$.



Starting from the initial state (AB = 00), the sequence of states (AB) visited by the circuit is

(a) $00 \rightarrow 10 \rightarrow 01 \rightarrow 11 \rightarrow 00.....$ (b) $00 \rightarrow 01 \rightarrow 11 \rightarrow 00.....$

(c) $00 \rightarrow 10 \rightarrow 11 \rightarrow 01 \rightarrow 00.....$ (d) $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00.....$

159. Match List -I (circuit) with List-II (application) and select the correct answer using the codes given below the lists:

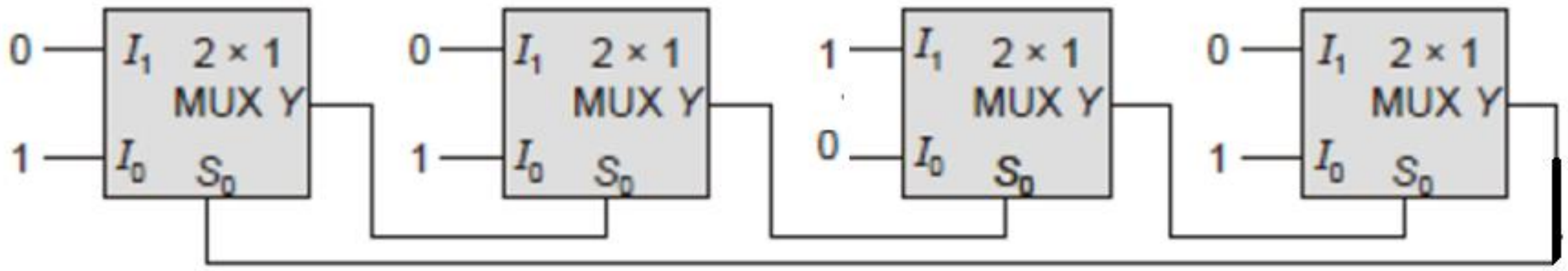
List-I	List-II
A. Left shift register	1. Code conversion
B. Demultiplexer	2. Division
C. ROM	3. Multiplication
D. Right shift register	4. Decoding

Codes:

	A	B	C	D
(a)	2	1	4	3
(b)	3	1	4	2
(c)	2	4	1	3
(d)	3	4	1	2

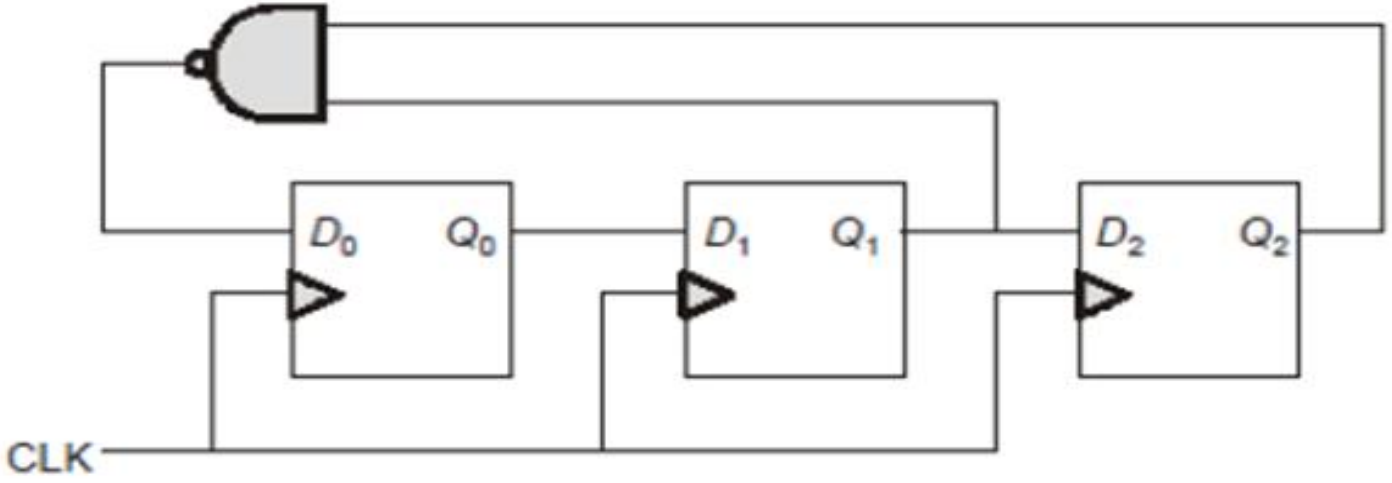
160. A modulo-16 ripple counter uses J-K flip-flops. If the propagation delay of each flip-flop is p ns and the maximum clock frequency that can be used is 5 MHz, then the value of p will be __________.

161. In the multiplexer based circuit shown below, the average propagation delay of each multiplexer is 25 ns. The frequency of the output signal V_0 is



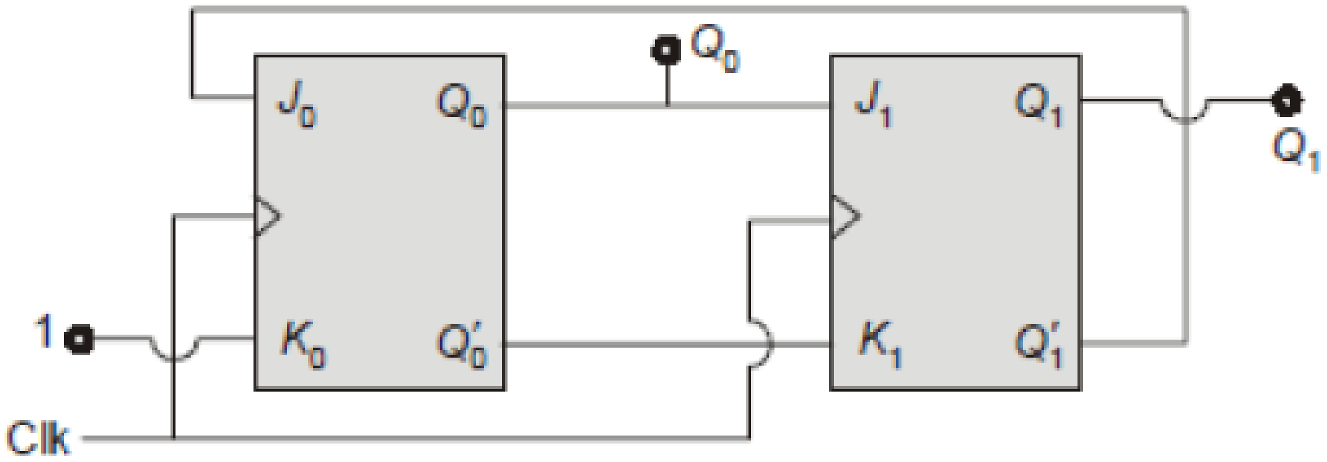
a. 4 MHz
b. 5 MHz
c. 13 MHz
d. 20 MHz

162. In the circuit shown below, initially all flip-flops are reset. The output $Q_2Q_1Q_0$ after four clock pulses is

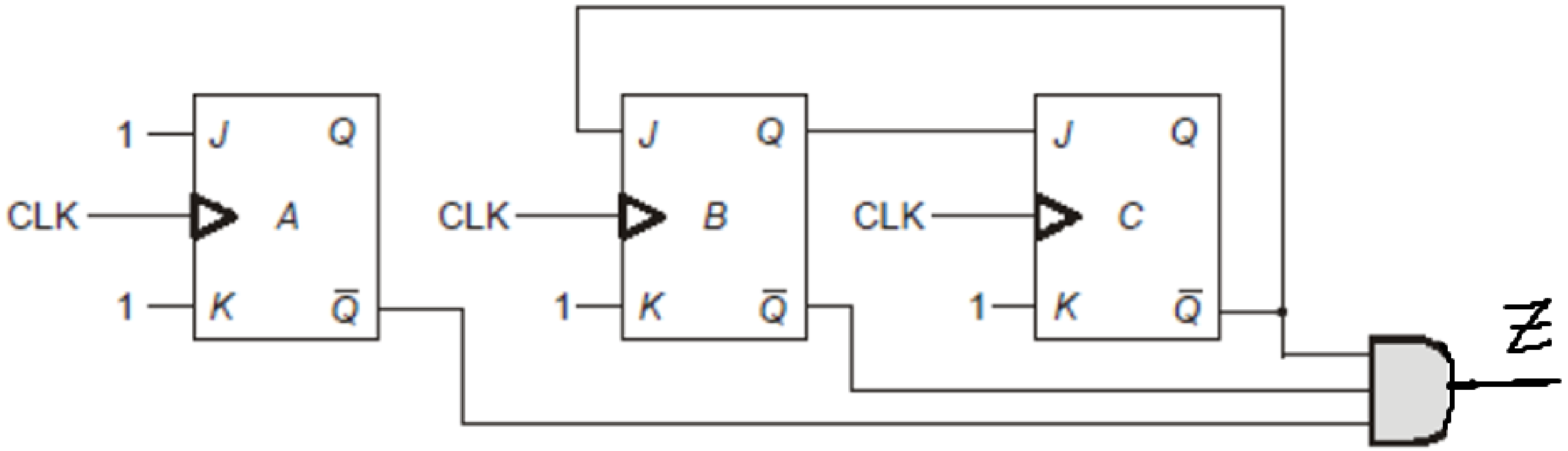


a. 110
b. 111
c. 100
d. 011

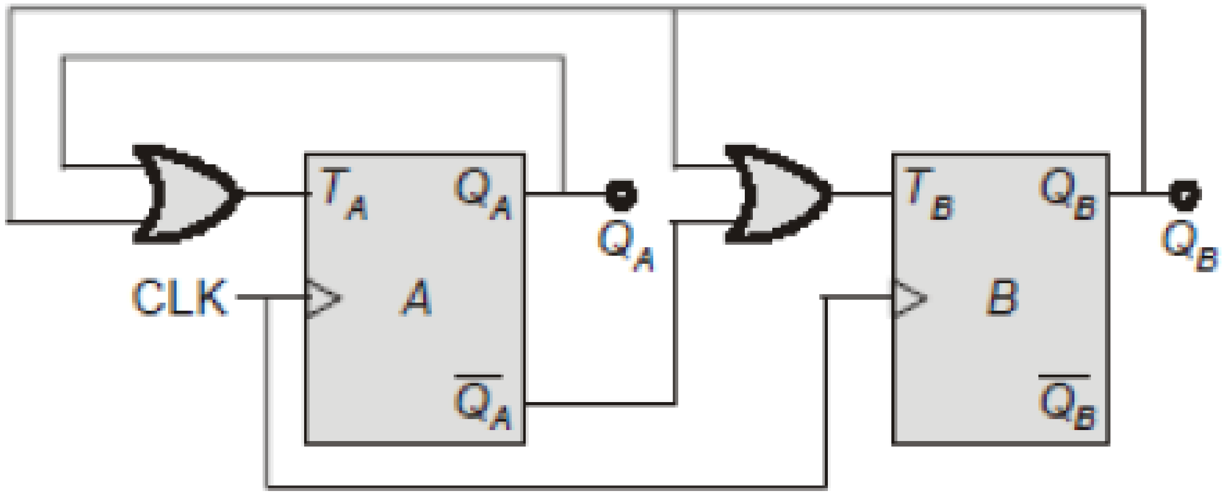
163. If the initial state of the counter ($Q_0 Q_1$) shown below is (00), then the modulus of the counter will be ________.



164. Consider a sequential circuit using three J-K flip-flops and one AND gate as shown in figure. Output (*z*) of the circuit becomes '1' after every *N*-clock cycles. The value of *N* is ___________



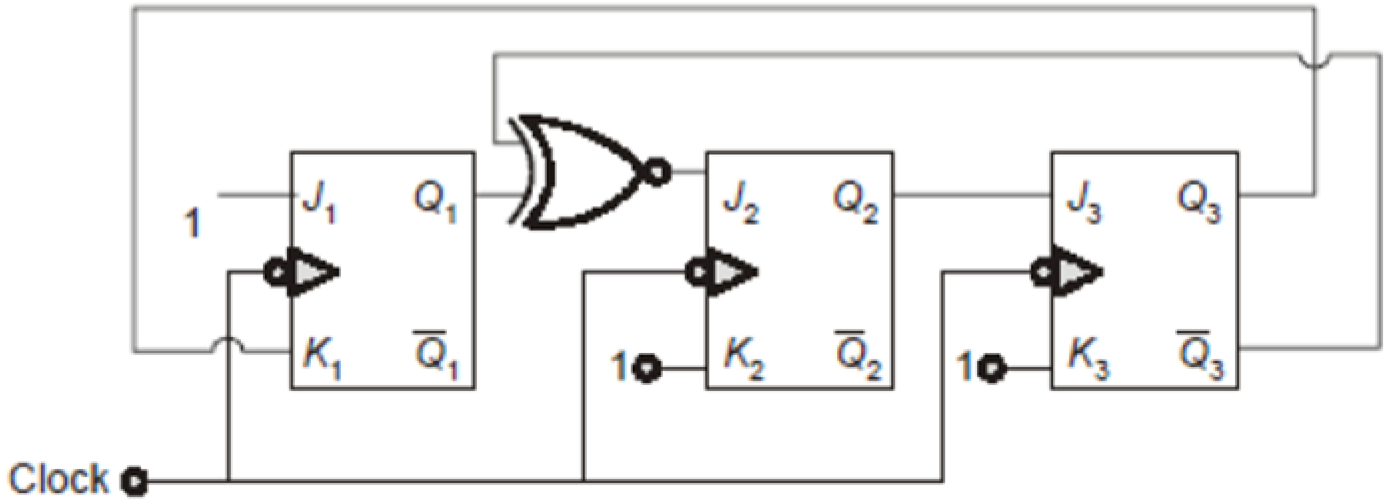
165. Consider the circuit shown below, initially both the flip-flops were cleared.



The modulus of the given counter circuit (i.e. the number of used staes) is

________.

166. Consider the circuit given below. If all the flip-flops are initially cleared then the count (Q_3 Q_2 Q_1) after 111 clock pulses is



a. 000

b. 110

c. 101

d. 011

167. Consider the counter stages shown below

The frequency at point *X* is ____________ Hz.

168. Which of the following is not a synchronous input with reference to Flip-Flops

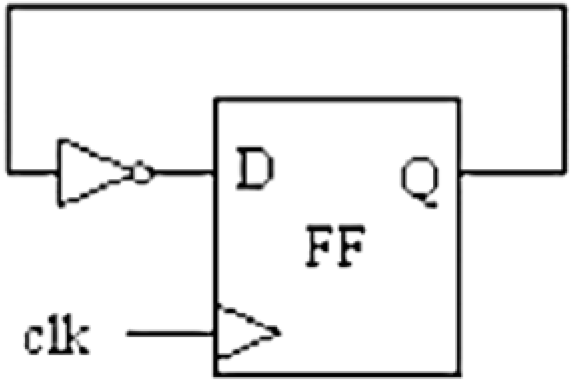
(a)J input of J-K Flip-Flop

(b) S input of R-S Flip-Flop

(c) Preset Input of]-K Flip-Flop

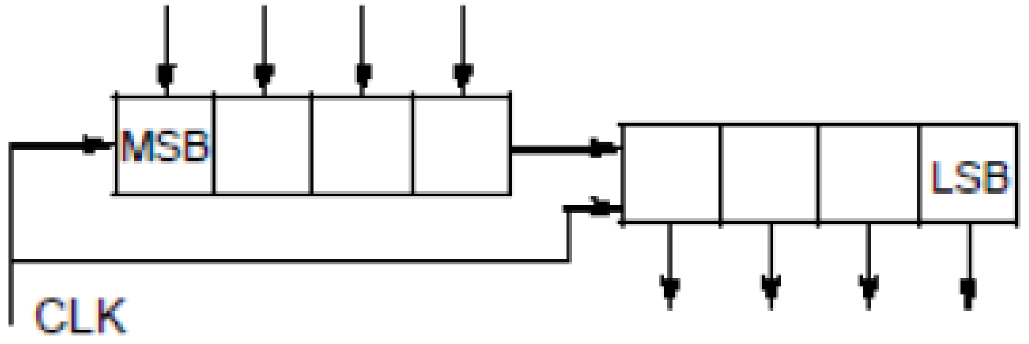
(d) D input of D Flip-Flop

169. The maximum frequency of operation of the following circuit, if propagation delay of Flip-Flop and inverter are 20ns, 5ns respectively is ________ MHz



170. An 8-bit register is made of one 4-bit PISO register [synchronous loading] cascaded with a 4-bit SIPO

register as shown in the figure below.



Total number of clock pulses required to perform write and read operations for one byte is ________.

171. The count of a mod 10 (4 bit) counter is represented as $b_3b_2b_1b_0$, where b_3 is the MSB & b_0 is the LSB. If the counter counts from '0' onwards, then the duty cycle of the bit 'b_2' is

(a) 20 %

(b) 40%

(c) 60%

(d) 50%

172. A counter is needed that will count the number of items passing on a conveyer belt. A photocell and light source combination is used to generate a single pulse each time an item crosses its path. The counter must be able to count as many as one thousand times. The number of flip-flops required are ______

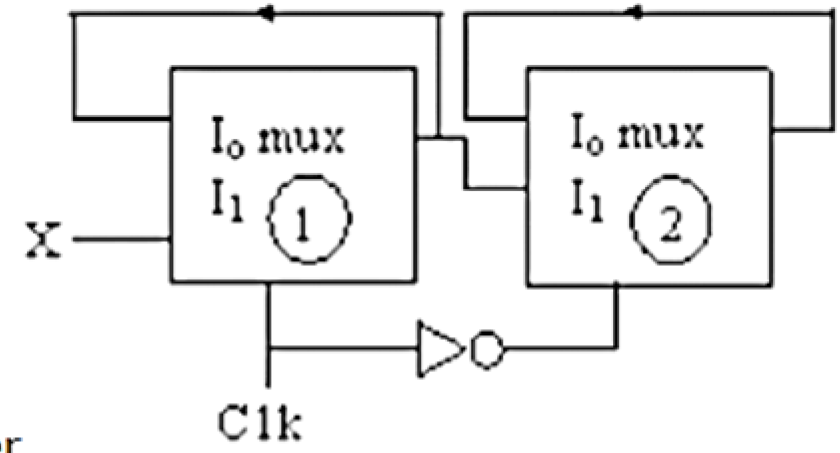
173. Identify the following circuit, constructed using multiplexers

(a) 2-Bit counter

(b) Master-Slave D-FF

(c) D-Latch

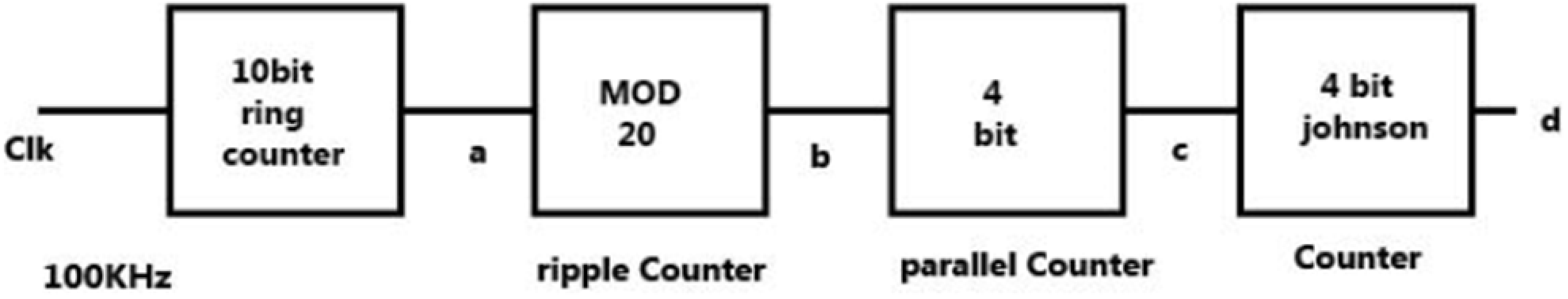
(d) 2-Bit Johnson counter



174. The initial state of the ring counter is 0100010. Determine number of clock pulses required to return to the initial state

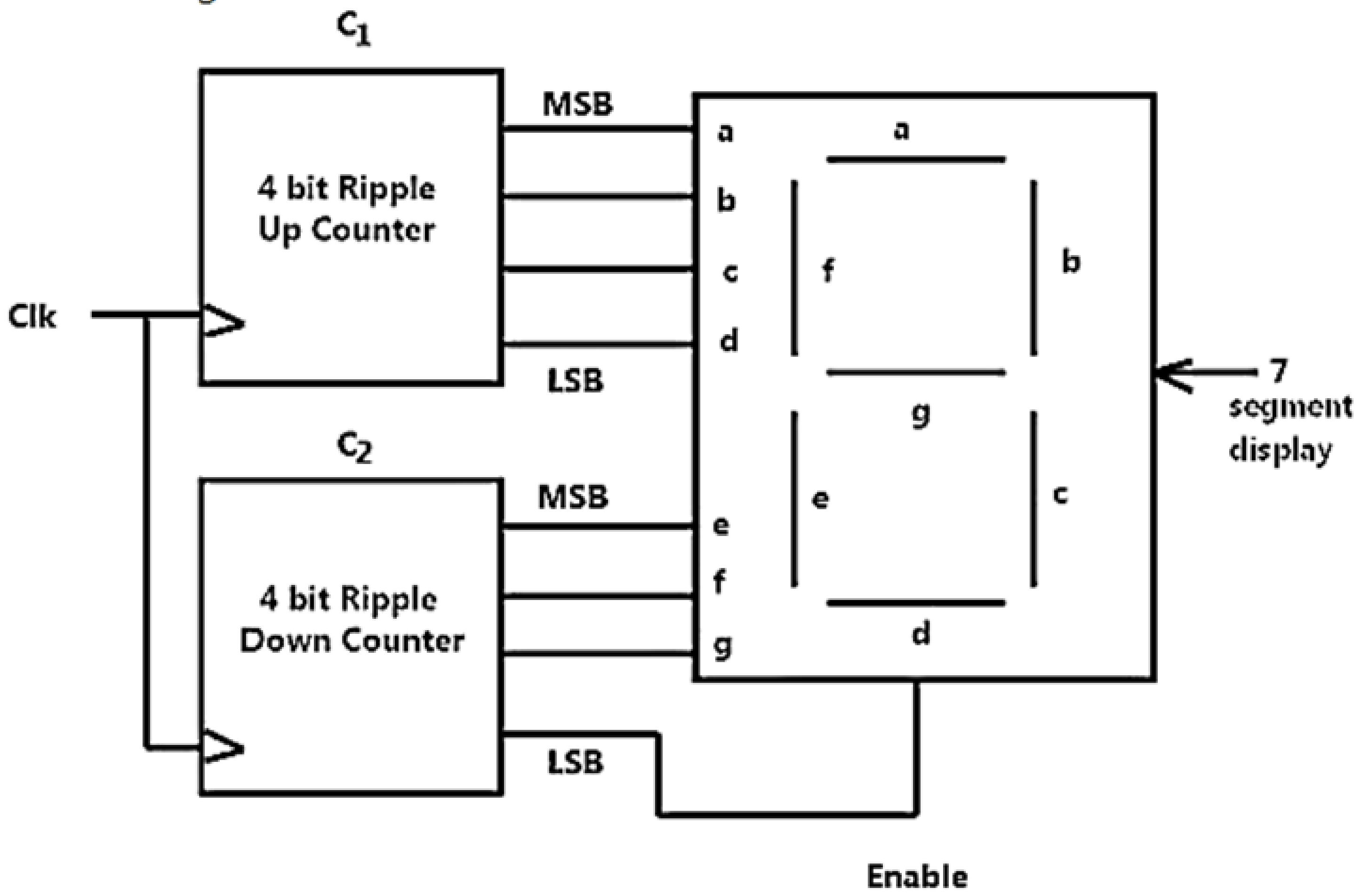
(a) 5
(b) 14
(c) 6
(d) 7

175. The frequency of the pulses at the point d in the following circuit is ____Hz.



176. Consider a mod-1000 ripple up counter. The duty cycle for its MSB is________%.

177. Consider the circuit given below



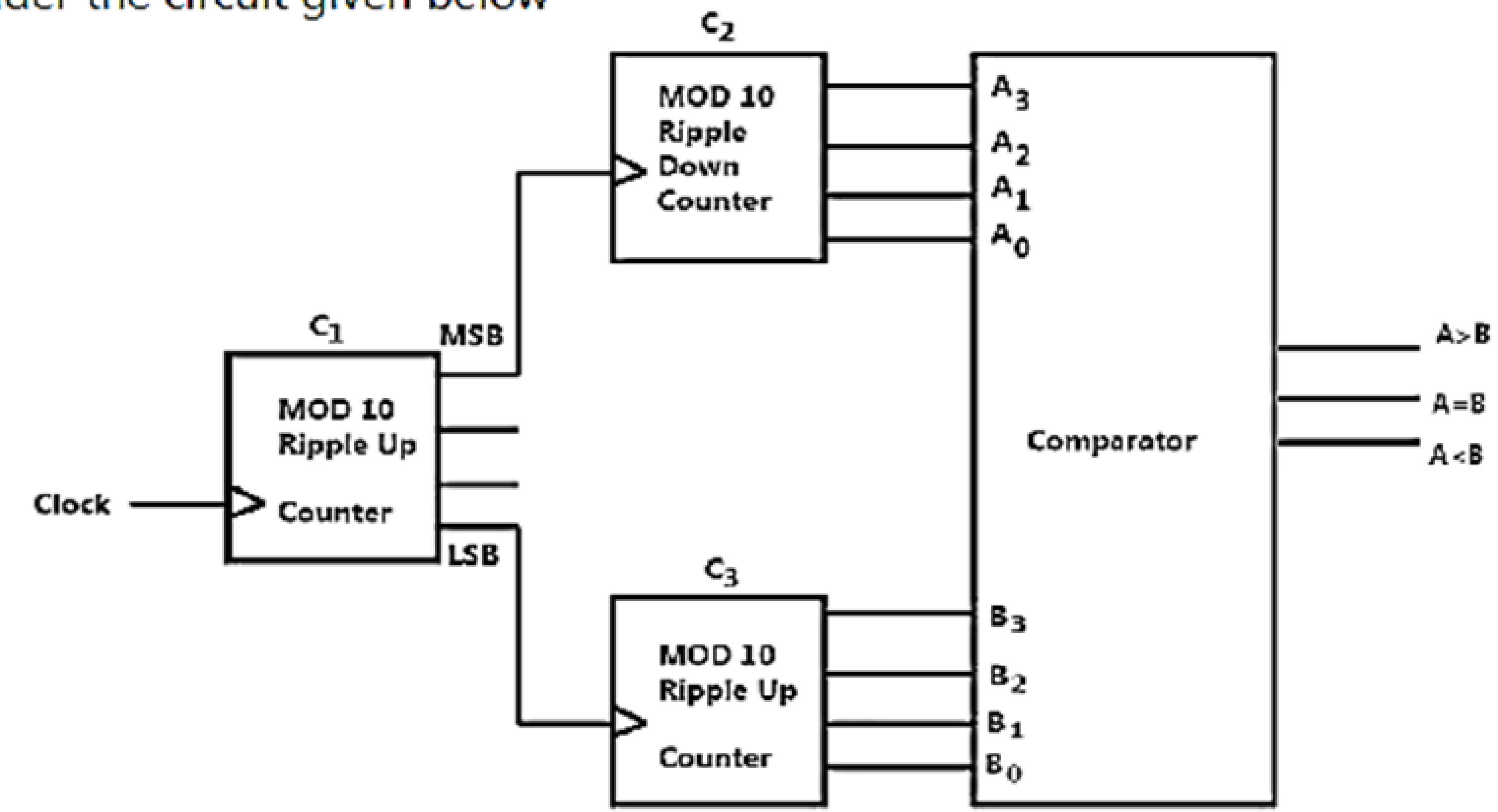
If Enable = 0 ; 7 segment display 11 (b = c = e = f = 1)
Enable = 1 ; 7 segment display data according to Inputs
Initially both the counter were cleared. After 78 clock pulses the data displayed on the 7 segment display is ____.

178. Three 4 bit shift registers are connected in cascade as shown in figure below. Each register is applied with same clock

A 4 bit data 1011 is applied to the shift register 1. The minimum number of Clock pulses required to get same input data at output are with same clock

(a) 11
(b) 12
(c) 9
(d) 7

179. Consider the circuit given below



MSB and LSB of MOD 10 ripple up counter acts as clock to 4 bit ripple down and up counter respectively. Initially all the counter were cleared and output of comparator was A = B. The clock pulse is applied. The minimum number of clock pulses required to make A = B again are